Identifying Galactic Luminous Red Nova Precursors

Harry James Robert Addison

A Thesis presented for the degree of
Master of Physics with Astronomy

Faculty of Engineering and Physical Sciences
Department of Physics
University of Surrey
United Kingdom
January 2022

Abstract

Binary stellar evolution is a topic of great interest in the current age of astronomy due to phenomena such as merging neutron stars and black holes. One of the key phases of binary evolution is the common envelope phase as it is required to bring the components in a binary close together. However, it is also one of the most poorly understood phases due to a lack of direct observations. This may potentially be solved with the relatively new class of transients known as luminous red novae (LRNe), which are currently thought to directly result from the common envelope phase. The LRN precursor phase is associated with the formation of a common envelope, and so observations of these would provide invaluable knowledge of the common envelope phase. However, over the last three decades, there have only been observations of two confirmed LRN precursors. The work presented here aims to increase this number by identifying Galactic LRN precursors. To achieve this the method makes use of the information that yellow supergiants and yellow giants (Hertzsprung gap stars) are likely progenitors of LRNe as they can initiate Roche lobe overflow and hence the formation of a common envelope. Therefore, Hertzsprung gap stars were identified by modelling the density of a Hertzsprung Russell diagram constructed from well behaved Gaia DR2 sources. This resulted in the identification of $\sim 5.4 \times 10^4$ likely progenitors from a sample of $\sim 10^7$ Gaia EDR3 sources. To identify likely precursors from them, the method applies the knowledge that previous precursors exhibited a slow brightening ($\sim 0.27\,\mathrm{mag\,yr^{-1}}$ and $\sim 0.18\,\mathrm{mag\,yr^{-1}}$). As such a slow rising transient detection method was applied to the light curves of the progenitors, which identified 315 apparent slow rising light curves. From these only 21 unique sources were found to be non-false positive detections, and upon further investigation, only three were found to be LRN precursor candidates. Although detailed follow up of these sources is left for future works, these sources could provide invaluable insights into the causes of LRN and potentially the common envelope phase.

Declaration

This dissertation and the work to which it refers are the results of my own efforts. Any ideas, data, images or text resulting from the work of others (whether published or unpublished) are fully identified as such within the work and attributed to their originator in the text, bibliography or in footnotes. This dissertation has not been submitted in whole or in part for any other academic degree or professional qualification. I agree that the University has the right to submit my work to the plagiarism detection service TurnitinUK for originality checks.

Acknowledgements

Firstly, I would like to thank my supervisor Dr. Nadejda Blagorodnova for providing me with the opportunity to work on this very interesting project, as well as providing me with support and guidance throughout. I would also like to thank my second supervisor Prof. Paul Groot for seamlessly fulfilling Nadia's role during her maternity leave.

Thirdly I would like to express my thanks to everyone involved in the Stellar Evolution Group at Radboud University for sharing their wealth of knowledge and expertise in our weekly meetings, as well as providing useful discussions and suggestions on this project. Furthermore, I appreciate the welcoming environment they created despite the virtual nature of all of our meetings.

Finally, I would like to thank family and friends for their support throughout this tough year of remote working.

Contents

List of Figures

List of Tables

CHAPTER 1

Introduction

In the current age of astronomy, Galactic surveys show that over $\sim 60\%$ of star systems are binary or multiple star systems (Gimnez et al., 2006; Moe & Di Stefano, 2017; Sana et al., 2012). Therefore, it is essential to understand the properties of these systems so that we can fully understand the galaxies in which they reside. Furthermore, close binary star systems are of particular interest due to the wide range of phenomena that occur from them such as type Ia supernovae, stripped core-collapse supernovae, the formation and merging of neutron stars and black holes (Postnov & Yungelson, 2014), luminous red novae (Tylenda et al., 2011; Blagorodnova et al., 2017), and many other high energy astrophysical systems. The processes that cause these systems to interact are complex, and despite the efforts to understand them, they are not yet fully understood.

Section 1.1 provides relevant background theory of single and binary stellar evolution. Then Section 1.2 discusses the use of transients in understanding binary evolution. In the final section, Section 1.3, provided is an overview of the project.

1.1 Single and Binary Stellar Evolution Scenarios

1.1.1 Single Stellar Evolution

A single star throughout its lifetime will experience many different phases, beginning as a cloud of dust and gas to ending as a wide variety of astrophysical systems. For the purposes of discussions in future sections and chapters, information is provided only for the relevant phases, which are the main sequence phase and the post main sequence phase up until the red giant branch (RGB).

The main sequence is the phase of stellar evolution where the star's only energy source is the burning of hydrogen in the core to produce helium. During this phase, stars are in a stable equilibrium known as hydrostatic equilibrium, which is the balance of the star's inward gravitational force with its outward gas pressure (Karttunen et al., 2017). Due to their stability, structural changes result from the burning processes, which changes the chemical composition of the core, influencing the rate of fusion. Therefore, the evolution of main sequence stars

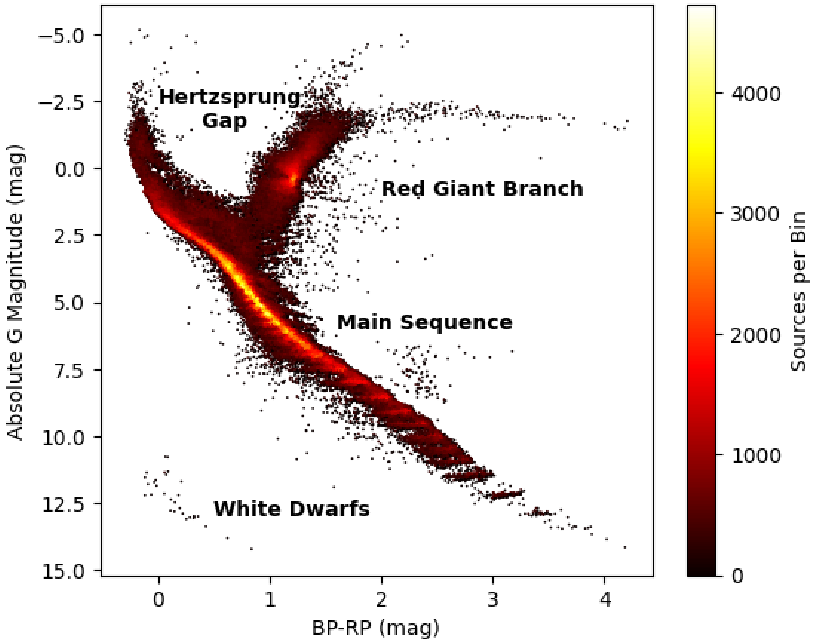


Figure 1.1: Colour-magnitude diagram constructed from a sample of 10^6 low extinction ($E(B-V) < 0.015\,\mathrm{mag}$) sources from Gaia data release 2 (Gaia Collaboration et al., 2016; Brown et al., 2018) using the DPAC (2021a) and the query provided in appendix B of Gaia Collaboration et al. (2018). The colour scale of the data points represents the relative density of stars, with darker regions representing less dense regions.

occurs on the nuclear timescale, which is the time over which the nuclear fusion processes change (Karttunen et al., 2017). The nuclear timescale is the longest of the evolutionary time scales, and so the main sequence is the most prolonged phase of a star's lifetime. Due to this, the main sequence has a large population of stars and is the densest region on a Hertzsprung Russell diagram (HRD), as is shown in the observed Galactic colour-magnitude diagram (CMD) in Figure 1.1.

Main sequence stars enter the post main sequence phase once they exhaust their core hydrogen supply. At this point, there is a decrease in radiation pressure within the core, causing the star to contract (Karttunen et al., 2017). The contraction causes the temperature of the helium core and the surrounding hydrogen shell to increase, resulting in hydrogen burning in the shell (Kippenhahn et al., 2012). Furthermore, in higher mass stars ($M > 2.3\,M_{\odot}$), the contraction of the core can increase its temperature to the point of igniting helium. Both the shell hydrogen burning and core helium burning increase the star's luminosity, which causes the star's envelope to expand in order to re-establish hydrostatic equilibrium. This evolution of stars off of the main sequence causes them to move towards the RGB, and for stars with masses greater than $2.3\,M_{\odot}$ this evolution is rapid, occurring on the thermal timescale (Kippenhahn et al., 2012). This timescale is defined as the time taken for a star to radiate all of its thermal energy if its nuclear reactions suddenly stopped (Karttunen et al., 2017). Due to the fast evolution of more massive stars, the chance of observing a star in this evolutionary phase is unlikely, meaning that the population of these stars is small. Therefore, in a HRD, this forms a sparsely population region between the main sequence and RGB known as the Hertzsprung gap, as is shown on the CMD in Figure 1.1.

1.1.2 Binary Stellar Evolution

Binary evolution can lead to a wide range of processes that result in events such as neutron star and black hole mergers (Postnov & Yungelson, 2014), and type Ia supernovae. Presented in this section is relevant information regarding the evolution of binary systems containing main sequence and early post main sequence stars.

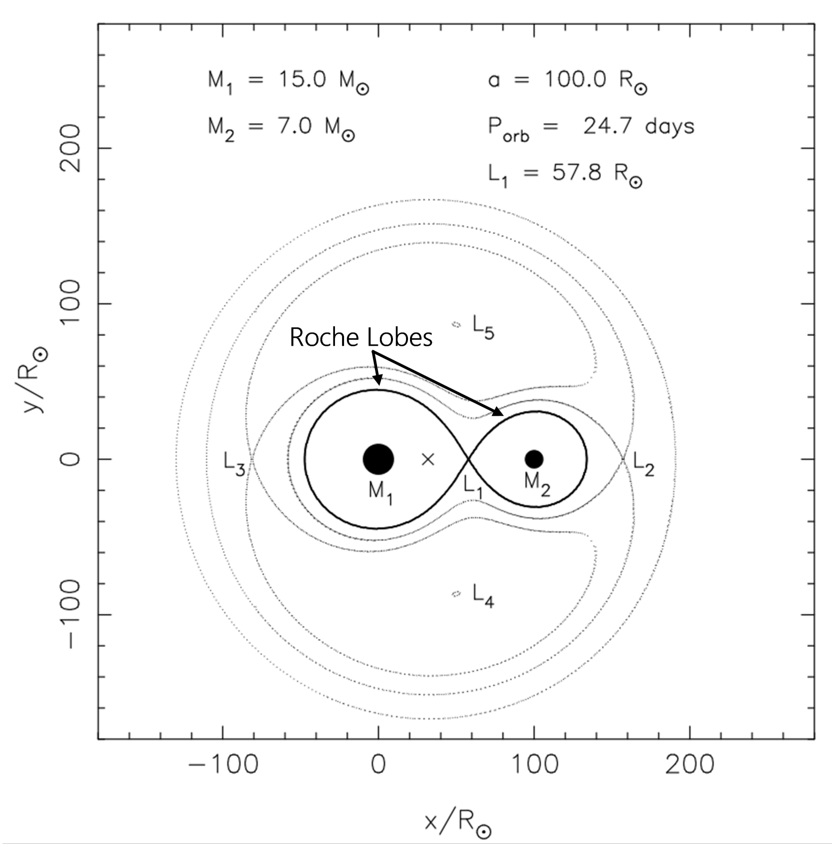


Figure 1.2: Gravitational equipotential lines in the orbital plane of a binary system with parameters noted at the top of the figure, after Tauris & van den Heuvel (2006). The Roche lobes of the binary system are defined by the innermost equipotential that encloses both components. Also indicated on this figure are the five Lagrange points denoted as L_{1-5}.

Binary systems consisting of main sequence stars are generally detached binaries as the two stars are in hydrostatic equilibrium and usually confined within their Roche lobes. The Roche lobe is the region around a star in a binary system defined by the innermost gravitational equipotential that encompasses both components. It also passes through the first Lagrange point (L_1), the point between the two stars where the gravitational forces and centrifugal forces are balanced. A detached binary system has many different evolutionary paths that it can follow, although only one path, the formation of a common envelope (CE) (Paczynski, 1976), is relevant for future discussions.

In the case of a binary consisting of two main sequence stars a CE is the scenario when the binary's secondary component (lower mass zero age main sequence star) is engulfed by the primary component (higher mass zero age main sequence star), thus forming a system where the two cores of the stars exist within one envelope (Paczynski, 1976). Currently, there are

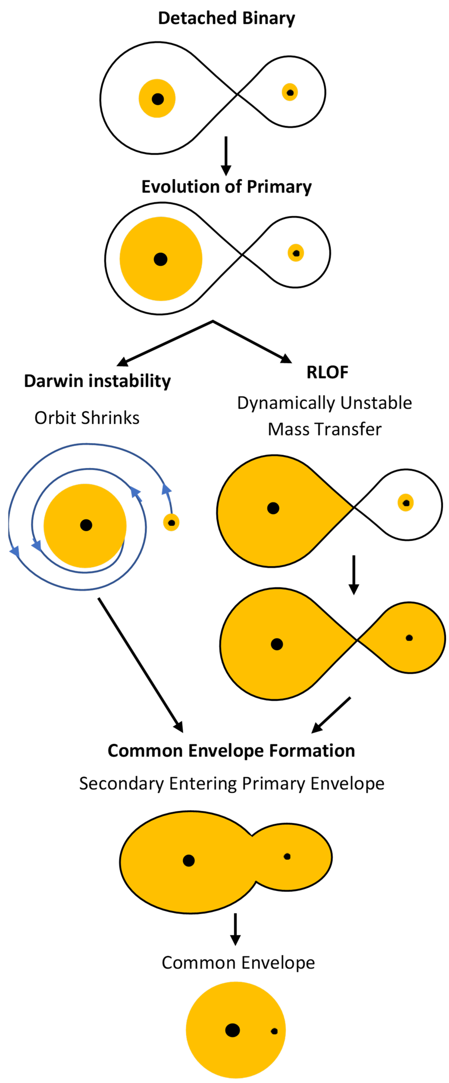


Figure 1.3: Summary of the formation of a common envelope though the mechanisms of Darwin instability and Roche lobe overflow (RLOF), after Tauris & van den Heuvel (2006). Throughout the figure the cores of the two stars are represented by the black circles, while their envelopes are represented in yellow. The black tear drop shaped lines represent the Roche lobe of the binary system.

two mechanisms thought to cause the formation of a CE: Darwin instability (Darwin, 1879), and dynamically unstable mass transfer (Paczynski, 1976; Ivanova et al., 2013a; Livio & Soker, 1988; Pejcha, 2014). These two channels are summarised in Figure 1.3 and are explained in detail below.

The first mechanism of CE formation, Darwin instability, is an instability of the binary system caused by the synchronisation of the orbit and the spin of the star (Darwin, 1879). This instability can arise through the evolution of the primary component from the main sequence to the post main sequence phase, resulting in the expansion of the star, and due to the conservation of angular momentum, there is a decrease of the star's angular velocity. Consequently, this desynchronises the star's spin angular momentum and the orbital angular momentum, therefore causing a transfer of angular momentum between the star and the orbit in order to re-establish tidal synchronisation. Synchronisation is achievable if the primary's spin angular momentum is less than a third of the orbital angular momentum. However, if the primary's spin angular momentum is more than one-third of the orbital angular momentum, then synchronisation of them is no longer possible (Darwin, 1879; Eggleton, 2006). The reason behind this is a runaway transfer of angular momentum between the orbit and the primary component. The transfer of angular momentum from the orbit to the star causes the orbital radius to decrease and its frequency to increase. Thus, the star needs more angular momentum to synchronise with the orbit, resulting in the transfer of more angular momentum, which causes the orbit to shrink further. The star can never receive enough angular momentum to synchronise with the orbit, meaning that the angular momentum transfer is continuous. Eventually, this leads to the orbital radius of the binary shrinking such that the secondary component's core enters the envelope of the primary, forming a CE.

The second mechanism for CE formation is mass transfer from the primary to the secondary in a runaway scenario in the case where the binary consists of two main sequence stars. Like Darwin instability, this mechanism begins with the primary star evolving from the main sequence to the post main sequence phase. However, unlike before, the expansion of the primary causes it to fill its Roche lobe, which initiates a mass transfer from the primary to the secondary through the first Lagrange point (L_1) in a process known as Roche lobe overflow (RLOF). The outcome of RLOF on the binary system is dependent on the mass-transfer rate, and hence its stability. The stability of the mass transfer has three scenarios, which are: stable mass transfer, thermal-timescale mass transfer, and dynamically unstable mass transfer. Only dynamically unstable mass transfer is responsible for CE formation, which occurs when the primary component cannot recover from the disturbance of its hydrostatic equilibrium on a dynamical timescale, which is defined as the timescale on which a star counteracts a perturbation made to its hydrostatic equilibrium. As the primary star cannot re-establish hydrostatic equilibrium, RLOF continues with an ever-increasing mass-transfer rate, thus leading to the formation of a CE.

The CE phase of binary evolution is one of the least understood processes in binary stellar evolution. It is also arguably one of the most important phases due to its proposed involvement in the formation of many different types of systems such as close binaries with a carbon-oxygen white dwarf (Izzard et al., 2011), double neutron stars, and AM CVn systems (Nelemans, 2005) to name a few. One of the main reasons for our poor understanding of the CE phase is the lack of direct observations which is due to its relatively short evolutionary time scale of a few years. The next section discusses the events that may allow us to directly observe the CE phase.

1.2 The Role of Transients in Understanding Binary Stellar Evolution

The evolution of binary systems is very complex due to many interactions that can occur between the components. In particular the CE is one of the most important, but yet most poorly understood phases. This is predominately due to the lack of direct observations of the phase despite the plethora of post CE evidence in the form of close binaries (Izzard et al., 2011). As is discussed in this section, this lack of observations may be solved with the relatively new class of transients known as luminous red novae (LRNe), which have been proposed to be an outcome of the CE phase (Blagorodnova et al., 2017; Tylenda et al., 2011; Pejcha, 2014).

LRNe are a class of intermediate luminosity transients due to having a peak brightness greater than that of novae (absolte magnitude of -6 to -8 mag), but less than that of supernovae (absolute magnitude < -14 mag) (Blagorodnova et al., 2017). The first observation of a LRN, although not known at the time, was made by Rich et al. (1989) in 1988, and since then there have been observations of at least 18 more LRNe (Blagorodnova et al., 2021). Of these events, M101 OT2015-1 (Blagorodnova et al., 2017; Goranskij et al., 2016) and V1309 Scorpii (V1309 Sco) (Nakano et al., 2008; Mason et al., 2010; Tylenda et al., 2011) are of particular interest due to the archival data available for them, which has allowed for investigations into their progenitors and thus likely causes. Provided here is a summary of the studies of these two events, with more detailed discussions presented in sections III.A and III.C of Appendix A.

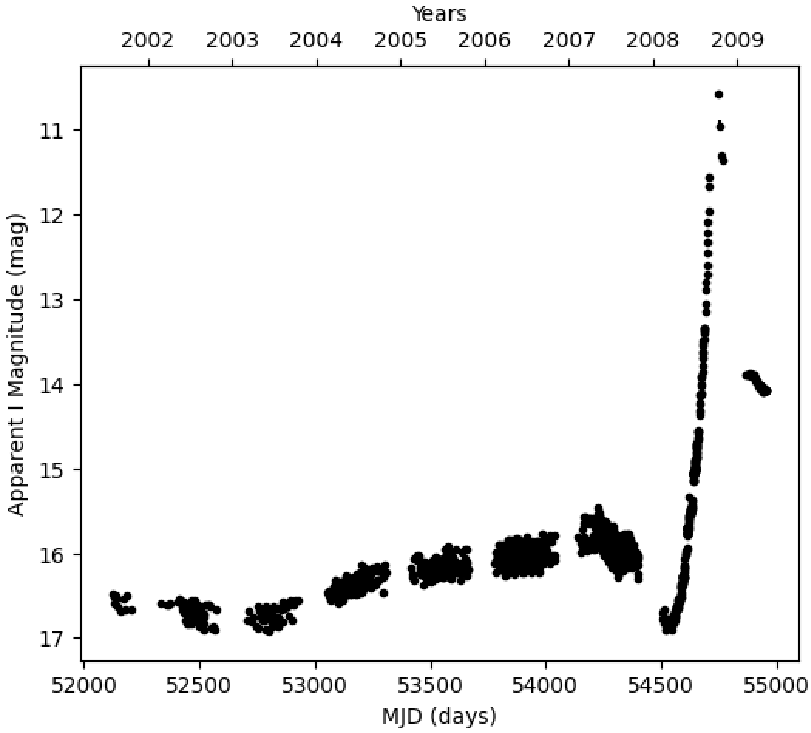


Figure 1.4: Light curve of V1309 Sco's precursor and outburst produced from the I band data used by Tylenda et al. (2011), which was obtained from the Optical Gravitational Lensing Experiment (OGLE, 2021).

The LRN V1309 Sco was first discovered in 2008 by Nakano et al. (2008) during its outburst. V1309 Sco's precursor phase and outburst were observed by the Optical Gravitational Lensing Experiment (OGLE) (Udalski et al., 2003; OGLE, 2021). Tylenda et al. (2011) used this data, shown in Figure 1.4, to investigate the nature of the progenitor system and causes of the outburst. They began by exploring the ~ 0.5 mag scatter in the precursor phase of the light curve (~ 2500 days to ~ 200 days before the outburst) by searching for a periodicity. They showed that in 2002 it exhibited a period of ~ 1.4 days and that over time it decreased exponentially. Based on the short period and the phase folded light curves, Tylenda et al. (2011) concluded that the system was a contact binary, a binary where the two components are in contact with each other. Furthermore, they suggested that the decreasing period was caused by an instability in the system due to either Darwin instability or mass loss from the second Lagrange point (L_2) due to RLOF. Pejcha (2014) favoured the idea of an L_2 outflow of mass, as the timescale of the instability was much shorter than the timescale of Darwin instability. Tylenda et al. (2011) also used the L_2 outflow to explain the dip in brightness at MJD $\sim$ 54300. They suggested that the outflow formed an optically thick excretion disc that obscured the system, which Pejcha et al. (2016) confirmed could be possible through modelling of the L_2 outflow. Finally, Tylenda et al. (2011) suggested that the rapid brightening seen at MJD $\sim$ 54600, was an indication of the two components merging as a result of the formation of a CE. Pejcha (2014) also agreed with this idea, however, they also proposed that the rapid brightening could be caused by a shock wave passing through the L_2 outflow.

Unlike V1309 Sco, M101 OT2015-1 was detected during its precursor phase by the intermediate Palomar Transient Factory survey, which identified the source at the time as slowly rising in brightness (Blagorodnova et al., 2017; Cao et al., 2015). Blagorodnova et al. (2017) and Goranskij et al. (2016) conducted studies of M101 OT2015-1's progenitor and outburst using the plethora of available archival and follow up data. Both studies determined the size of the source during the progenitor phase, and again during the peak of the event. Blagorodnova et al. (2017) estimated the size of the photosphere to be $230 \pm 13\,R_\odot$ six years before the outburst, and $6500\,R_\odot$ during the outburst. Similarly, Goranskij et al. (2016) estimated the radius to be $400\,R_\odot$ at 240 days before the outburst, and $3300\,R_\odot$ during the outburst. Although both sets of estimates do not fully agree with one another, they both show that the pre-outburst photosphere radius increased by at least a factor of eight. Goranskij et al. (2016) suggested that the increase in radius was caused by the formation and ejection of a CE. This idea is also supported by the spectroscopy of M101 OT2015-1, as well as the post-outburst colour evolution of the source, which occurred over a few hundred days (Blagorodnova et al., 2017). This also provides evidence for the formation of optically thick dust, which could have resulted from the ejection of the CE, and hence supports the CE theory. Furthermore, both studies concluded that the progenitor source was a post main sequence star that was evolving towards the RGB, and so if it was a part of a binary system, then the formation of a CE would occur.

The aforementioned works support the idea that LRNe are caused by the formation of a CE through RLOF. However, this may not be the case as other causes were suggested for V1309 Sco (Tylenda et al., 2011; Pejcha, 2014), and there is a lack of LRNe events with observations of the progenitor and precursor phase from which conclusions about the actual causes can be drawn from. Additionally, the data available of LRNe precursors is very limited as it is generally archival data, which means that the analysis of the events is restricted. Despite the lack of LRNe precursor observations thus far, they are good candidates for events that could provide a means of directly studying the CE phase of binary stellar evolution.

1.3 Project Overview

The main aim of this project is to fill the gap in the current literature of LRNe, which is that there are very few observed LRN precursors. Furthermore, the observations of these previous precursors were generally from sky surveys and were coincidental, meaning that the available data was limited and did not provide all of the necessary data to conduct in depth analyses of the events. The project outlined here aims to solve these two issues by identifying Galactic LRN precursors. This will not only provide us with more LRNe events to explore, but it will allow us to take real time observations with all of the available techniques. Thus, this will allow us to increase our knowledge of LRNe, as well as the CE phase of binary stellar evolution.

As of current it is believed that there has been no development of a method to identify

LRN precursors. One reason for this is due to the lack of large observational surveys until recently. These relatively new large surveys such as Gaia (Gaia Collaboration et al., 2016) and the Zwicky Transient Facility (ZTF) (Masci et al., 2018; Bellm et al., 2018) allow for easy access to time-domain observations of a large number of sources. This is important when looking for LRN precursors as they are uncommon with a Galactic rate of $\lesssim 0.5$ per year (Kochanek et al., 2014). So by having a large sample of observed Galactic sources, as is provided by the relatively new surveys, the chance of identifying a LRN precursor is increased. The method presented here makes use of data from the Gaia and ZTF surveys to identify Galactic LRN precursors. This method can be split into two main steps which are, likely progenitor selection, and likely precursors selection. Presented in Chapters 2 and 3 are the details, results, and discussions regarding these two steps, which are very briefly outlined below.

The first step involves the identification of sources from a set of Galactic sources that are likely LRN progenitors. Such sources are those that are capable of causing the formation of a CE. As is discussed in more detail in Section 2.1 the sources of interest are yellow super giants (YSG) and yellow giants, which are types of post main sequence star evolving to the red giant phase and thus reside in the Hertzsprung gap. To select sources within the gap, it was defined by using a density model of the observed Galactic CMD. Stars in the low density region of the model were then selected as likely LRN progenitors. See Chapter 2 for the full details of the method, as well as the results and discussions.

The second step in identifying Galactic LRN involves collecting the light curves of the likely progenitors and selecting those that exhibit characteristics of a LRN precursor. The light curves of the likely progenitors were collected from the ZTF (Masci et al., 2018; Bellm et al., 2018), which is a photometric survey that scans the Northern Hemisphere's night sky (Bellm et al., 2018). To identify likely precursors from the light curves, a method was applied to them to see if they exhibited a slow rate of increase in brightness. The selected brightening light curves were then manually looked over, removing any false positives and leaving any likely precursors. Presented in Chapter 3 is a detailed explanation of the method, along with the results and discussions.

CHAPTER 2

Likely Progenitor Selection

This chapter presents the details and discussions of the method used to select likely progenitor systems of LRNe. First, a brief description of this method is provided in Section 2.1, followed by Sections 2.2 and 2.3 that contain further details of the input data and the method. The final two sections, 2.4 and 2.5 contain the results and discussion of the method.

2.1 Method Overview

LRN progenitor systems are thought to commonly contain a YSG component, as their fast expansion after leaving the main sequence can result in the formation of a CE, as described in Section 1.1. The ejection of this CE is the most likely explanation for astrophysical transients associated with LRNe (Blagorodnova et al., 2017). Consequently, the method presented here focuses on identifying the Galactic population of YSG as progenitor systems of LRNe.

The YSG phase is a relatively short lived phase of a star's lifetime, being around 1% of the time span of the main sequence phase (Drout et al., 2009; Kippenhahn et al., 2012), which implies that the population of YSG is small, residing in a sparsely populated region of the HRD known as the Hertzsprung gap, which is the area between the main sequence and the RGB on the HRD, as is seen in Figure 1.1. The method presented here exploits the knowledge that YSG reside in the Hertzsprung gap in order to identify them as likely LRN progenitors. Determining the Hertzsprung gap can be achieved by modelling the density of the HRD, as the model will represent the main sequence and RGB as dense regions, while the Hertzsprung gap will have a relatively lower density. This model can then be used to identify likely progenitors as those that lie within the low density region which is the Hertzsprung gap.

Among different density modelling techniques I selected to use the Gaussian mixture model (Pearson, 1894; McLachlan et al., 2019), which models a training data set by fitting several two dimensional Gaussian distributions to it. They thus form a larger distribution which represents the density of the training data, from which the training data set could have been drawn. One of the main reasons to have chosen Gaussian mixture model over other techniques is its ability to make inferences about a different data set based on the model. These inferences include the likelihoods of given individual data points to belong to the model. In this project, these likelihoods are used to form a sample of the likely progenitor sources. Discussed in Section 2.2 are the details of the training data used to form the model and the data set that the likely progenitors are selected from. Then provided in Section 2.3 are the specifics of the Gaussian mixture model and how it is used to select likely progenitors.

2.2 Training Data and Fitting Data

The first step in producing the model of the HRD is to form a data set on which the Gaussian mixture model will be trained. As the aim of this project is to identify Galactic LRNe, the training data set was produced from data collected by the Gaia mission (Gaia Collaboration et al., 2016), as it has produced some of the most complete catalogues of Galactic sources. More specifically the training data set was selected from Gaia DR2 (Brown et al., 2018). This version was chosen over the newer, larger data release, Gaia EDR3 (Brown et al., 2021), as it contained extinction values for individual sources which are required to obtain an extinction-free HRD.

The chosen parameter space used for the training data set, and hence all subsequent data sets, was absolute G magnitude against a colour of $G - RP$. I chose to avoid using the BP photometry as it has a known systematic inconsistency (Maíz Apellániz, J. & Weiler, M., 2018; Weiler, Michael, 2018). The training data set consisted of the columns; apparent G magnitude, $G - RP$, parallax, and $A(G)$ extinction, and was formed using the online Gaia Archive (DPAC, 2021a) with the constraints and ADQL queries presented in Table 2.1.

Quantity	Constraint	ADQL Query
Parallax / Error	≥ 20	gaiadr2.gaia_source.parallax_over_error ≥ 20
Flux / Error	≥ 70	gaiadr2.gaia_source.phot_g_mean_flux_over_error ≥ 70
		gaiadr2.gaia_source.phot_rp_mean_flux_over_error ≥ 70
RUWE	≥ 0.95	gaiadr2.ruwe.ruwe ≥ 0.95
	≤ 1.05	gaiadr2.ruwe.ruwe ≤ 1.05
Extinction	≤ 0.75	gaiadr2.gaia_source.a_g_val ≤ 0.75
Absolute G Magnitude	≤ 1.5	(gaiadr2.gaia_source.phot_g_mean_mag - gaiadr2.gaia_source.a_g_val - 10 + 5 * log10(gaiadr2.gaia_source.parallax)) ≤ 1.5

Table 2.1: Constraints and ADQL queries used to obtain the training data set from Gaia DR2 (Brown et al., 2018). The constraints placed on the parallax over error, flux over error RUWE, and extinction are used to provide a "clean" sample of Galactic sources. The final constraint on the absolute G magnitude is to limit the data to the region of the Hertzsprung Russell diagram that contains the Hertzsprung gap and hence likely progenitors.

In order to produce a clean sample of Galactic sources in the region of the Hertzsprung gap, I applied the constraints in Table 2.1. A clean sample in this case meaning that the sources have relatively low parallax errors ($\leq$ 5%), photometric errors ($\leq$ 1.43%), and extinction values ($\leq$ 0.75 mag). Furthermore, a constraint was placed on the renormalised unit weight error (RUWE), as it is a measurement of the quality of the astrometry for a source, with values close to one representing good astrometry. A tightly constrained training data set is needed as large errors and uncertainties in the training data will result in a model that poorly resembles the true underlying distribution of the data. However, it is also important to not constrain the data too much, as this will result in a data set with too little data that will also not represent the true underlying distribution. The constraints and values in Table 2.1 provided a data set that is balanced between being clean and complete. This data set is shown in the form of a HRD in Figure 2.1, which also shows the stellar evolution tracks from Modules for Experiments in Stellar Astrophysics (MESA; Paxton et al., 2011, 2013, 2015, 2018) Isochrones and Stellar Tracks (MIST: Dotter, 2016; Choi et al., 2016). The shown tracks are of the main sequence to core helium burning phase for stars from 2 to 10 $M_\odot$ with a solar metallicity.

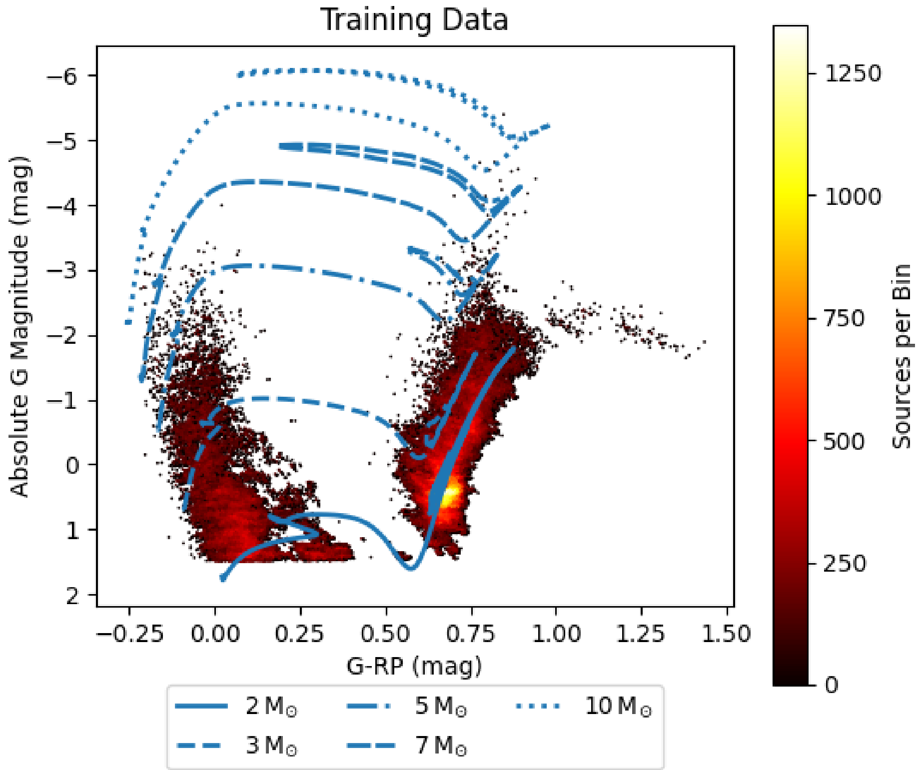


Figure 2.1: Training data collected from Gaia DR2 (Gaia Collaboration et al., 2016; Brown et al., 2018) presented as a Hertzsprung Russell diagram with the MIST stellar evolution tracks (Dotter, 2016; Choi et al., 2016; Paxton et al., 2011, 2013, 2015, 2018) from the main sequence to core helium burning phase for stars with masses 2, 3, 5, 7, and 10 $M_\odot$. The parameter space of the training data is limited to absolute G magnitudes above 1.5 mag, as this parameter space contains the Hertzsprung gap, seen between the main sequence and the red giant branch.

After creating the model for the clean training data set, details of which are provided in Section 2.3, the likely progenitors were selected from a second, more extensive "fitting" data set. As it was fitted to the model it consisted of the same parameter space as the training data set, apparent G magnitude against $G - RP$. However, unlike the training data set, the fitting data set did not need to be as tightly constrained as it was not used to form a model. Therefore, there was no need to constrain the RUWE or extinction values, and the parallax and photometric error constraints were loosened. Another difference to the training data set is that the fitting data set was formed from Gaia EDR3 (Brown et al., 2021) instead of from Gaia DR2 (Brown et al., 2018). This choice was made due to Gaia EDR3 containing approximately 10^8 more sources than Gaia DR2 (Brown et al., 2018, 2021), which would potentially provide more likely progenitor sources. Another important difference to the training data set is that the sources are all above a declination of $-30°$, which is a limitation imposed by the declination range of the ZTF survey (Masci et al., 2018; Bellm et al., 2018) used when identifying precursors from the likely progenitor systems (see Chapter 3). The final difference between the two data sets is that the fitting data set does not contain the parallax measurements of sources but instead contains the geometric distances from Bailer-Jones et al. (2021). These distances were used instead of directly inverting the parallaxes, as the parallax errors in the fitting data set were as large as 20%, meaning that the inversions of the parallaxes would result in inaccurate distance estimations. The fitting data set was constructed using the online Gaia archive (DPAC, 2021a) using the constraints and ADQL queries presented in Table 2.2. The resulting data set consisted of the following columns: apparent G magnitude, $G - RP$, right ascension (RA), declination (Dec), and geometric distance from Bailer-Jones et al. (2021).

Quantity	Constraint	ADQL Query
Parallax / Error	≥ 5	gaiaedr3.gaia_source.parallax_over_error ≥ 5
Flux / Error	≥ 10	gaiaedr3.gaia_source.phot_g_mean_flux_over_error ≥ 10
		gaiaedr3.gaia_source.phot_rp_mean_flux_over_error ≥ 10
Absolute G Magnitude	≤ 3	(gaiaedr3.gaia_source.phot_g_mean_mag - 5 *
		log10(external.gaiaedr3_distance.r_med_geo / 10)) ≤ 3
$G - RP$	≤ 1.22	gaiaedr3.gaia_source.g_rp ≤ 1.22

Table 2.2: Constraints and ADQL queries used to obtain the fitting data set from Gaia DR3 (Brown et al., 2021). The final two constraints on the absolute G magnitude and $G - RP$ is to limit the data to the region of the Hertzsprung Russell diagram that contains the Hertzsprung gap and hence likely progenitors.

The constraints presented in Table 2.2 were used to form the fitting data set. The use of constraints on the parallax and flux errors was to limit the uncertainty in the positioning of the sources on the HRD, reducing the contamination from sources outside of the Hertzsprung gap. By constraining the errors, the positioning of sources in the HRD is more certain and so the likely progenitor sample contained less false positives. The other two constraints on the absolute G magnitude and $G - RP$ were used to select sources more massive than $\sim 2\,M_{\odot}$ and to avoid highly extincted stars as the aim is to select stars within the Hertzsprung gap.

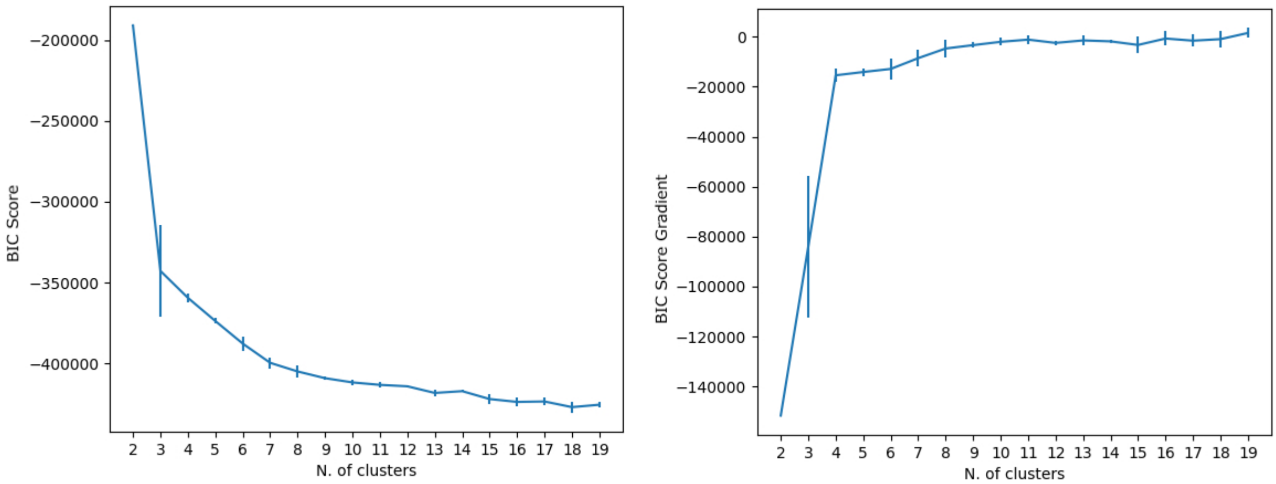
2.3 Method

To produce the model of the HRD, the Gaussian mixture model was trained using the training data set that was described in Section 2.2. Training the model entailed finding the best fit of a set number of two dimensional Gaussian distributions (components) to the training data set, which was accomplished computationally by using the `Python` module `sklearn.mixture.GaussianMixture` (Pedregosa et al., 2011). To select the optimum number of components to be fitted to the training data, three tests were used: Bayesian information criterion (BIC) (Schwarz, 1978), distance (Connor et al., 2013), and silhouette (Rousseeuw, 1987).

The first of the tests, BIC (Schwarz, 1978), was used to measure how well a set of components described the training data set, with a lower BIC score representing a better fit. This test also took into consideration the complexity of the model, penalising models that were more complex, as they can result in overfitting. When using BIC in selecting the number of components to be used for the model, I created models of the training data with 2 to 19 components and computed the BIC for each one of them. As the modelling method is statistical in nature, the test was repeated 20 times in order to calculate averages with errors of the BIC scores for each of the models. The results from this analysis are shown in Figure 2.2 along with the gradient of the BIC results.

The second of the tests, the distance test, is a test that looks at the reproducibility of a particular model. It does this by splitting the training data set into two random samples and then forming a model for each of them. Then they are compared using the Jensen-Shannon distance (Connor et al., 2013) which is a measure of how much the two models differ, with lower values representing models that differ less and thus are more reproducible. Like with the BIC tests, the distance test was applied to models with 2 to 19 components and repeated 20 times to calculate average scores. The results of this test are shown in Figure 2.3.

The final test, the silhouette test, provides an evaluation of the validity of the model by looking at the average silhouette width, which is based on the tightness and separation of the components of the model (Rousseeuw, 1987). A model with a silhouette score closer to one is considered to have a more valid fit of the components to the data, and is thus a better model. As with the previous two tests, the silhouette test was applied to models with 2 to 19 components and repeated 20 times to provide average scores. The results of the silhouette test are shown in Figure 2.4



(a) BIC scores for models with 2 to 19 components (clusters).

(b) Gradient of the BIC scores for models with 2 to 19 components (clusters).

Figure 2.2: Results of the BIC test on models with 2 to 19 components (clusters).
a) BIC scores: resembles how well the model fits the data, with better fitting models having lower scores. Based on the BIC scores the model with 18 components fits the data better.
b) Gradient of the BIC scores: shows if there is a significant change in the fit of the model to the data. There is no significant benefit in increasing the number of components past 4.

From the results of the BIC test shown in Figure 2.2, the BIC scores suggest that a model with 18 components is the best fit to the data. However, this is a complex model and the gradient of the BIC scores shows that there is very little benefit in increasing the number of components above 4. It should also be noted that the improvements between the models with

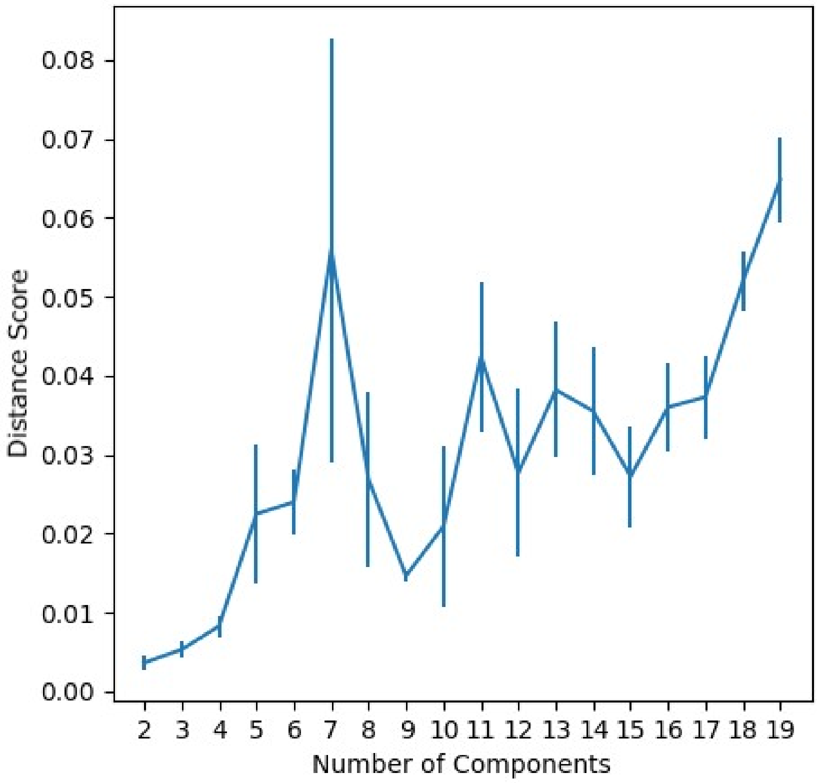


Figure 2.3: Distance scores for the models with 2 to 19 components. Models with a lower distance score are more reproducible. The most reproducible model contains only two components, and the reproducibility generally decreases as the number of components increases.

2 to 4 components are significant, as is indicated by the large ($\sim 6 \times 10^4$) changes in the BIC score gradients. The improvements between models with more than 4 components are not as significant due to the relatively small ($\lesssim 5 \times 10^3$) changes in the BIC score gradients. From the BIC scores and gradients, the best suited number of components to use in the model is four, however, the other test results must also be considered.

The results of the distance test in Figure 2.3 suggest that a model with two components is the most reproducible due to it having the lowest distance score and a small error. Other comparably reproducible models include the ones with 3 and 4 components, as their distance scores differ by less than ~ 0.005 considering the errors. Furthermore, these three models are the only ones that have a distance score of less than 0.01, with the next lowest score being ~ 0.015, which corresponds to the 9 component model. It is also worth mentioning that the errors of the scores of models with 2, 3, 4, and 9 components are less than ~ 0.002, which is significantly less than the errors of the other model's scores which are all greater than ~ 0.01. As with the BIC test, the best number of components for the model cannot be drawn from the results of this test alone.

Finally, the results from the silhouette test presented in Figure 2.4 show that models with 2 to 5 components have the most valid clustering due to their relatively high scores. Their scores are above 0.45 while the other models all have scores lower than 0.4, and thus have less valid clustering. Although the silhouette test shows that models with 2 to 5 components are the more valid clustered models in order to decide on the best number of components to use,

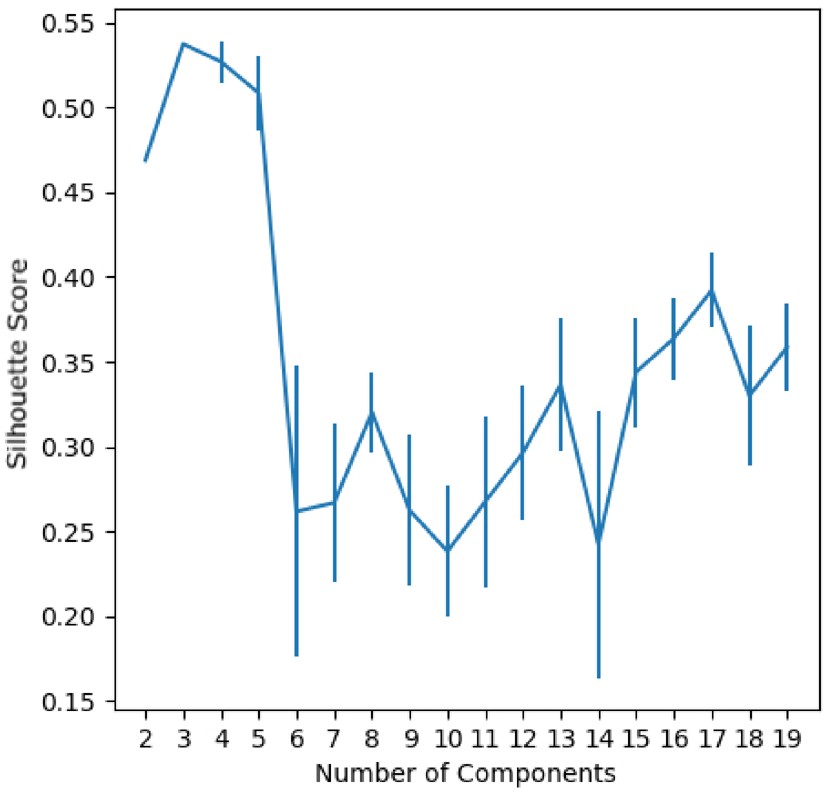


Figure 2.4: Silhouette scores for models with 2 to 19 components. The higher the silhouette score the more valid the clustering is and thus the better the model. Models with 2 to 4 components have the highest silhouette scores and are therefore suggested to be the best models for the data.

the results from the previous two tests also need to be considered.

By considering the results of all three tests, the best model consists of 4 components, but upon visual inspection of the model it appeared that it did not represent the training data very well and that a more complex model was required. To select the next best suited model is difficult, as the three tests do not unanimously suggest one particular model. The BIC test suggests that the models with more than 4 components are similarly good, while the distance test suggests that the next best model contains 9 components. The results from the silhouette test suggests that 17 components is the next best model, although the scores for models with more than 6 components are quite similar with a minimum difference in the scores of ~ 0.1. It was decided that 9 components should be used in the model due to its reproducibility and simplicity over the other similarly matched models. The model of the training data set consisting of 9 components is shown in Figure 2.5, where the components are represented by ellipses. More specifically, each ellipse corresponds to one standard deviation of the Gaussian distributions centred on the white dots, while the colour represents the superposed Gaussian distributions which corresponds to the density of the data.

With the model created, next the fitting data set was fitted to it and the likelihood values of each data point belonging to the model were calculated using the Python module `sklearn.mixture.GaussianMixture` (Pedregosa et al., 2011). The resulting log likelihood values for the fitting data set are shown in Figure 2.6a.

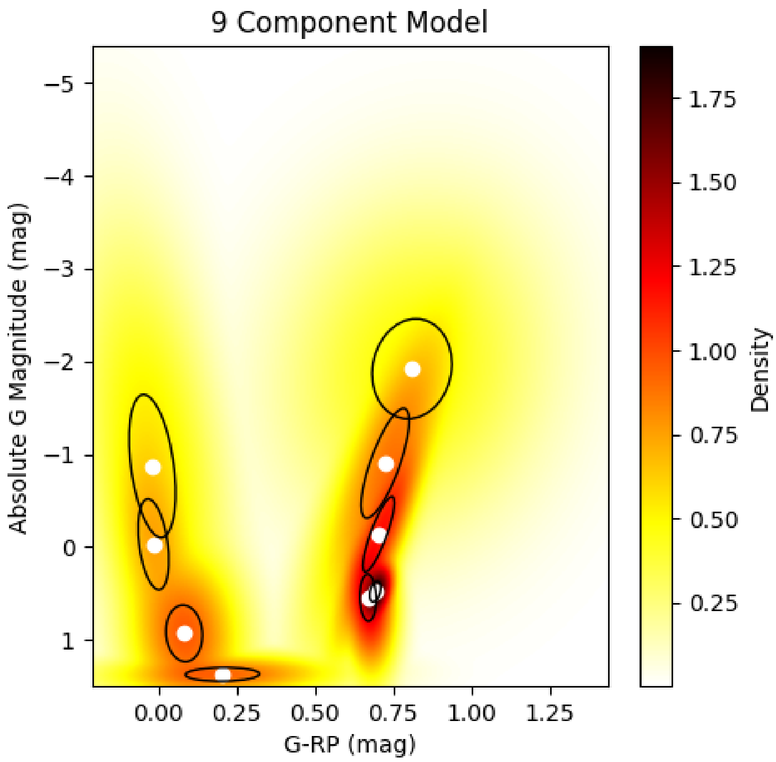
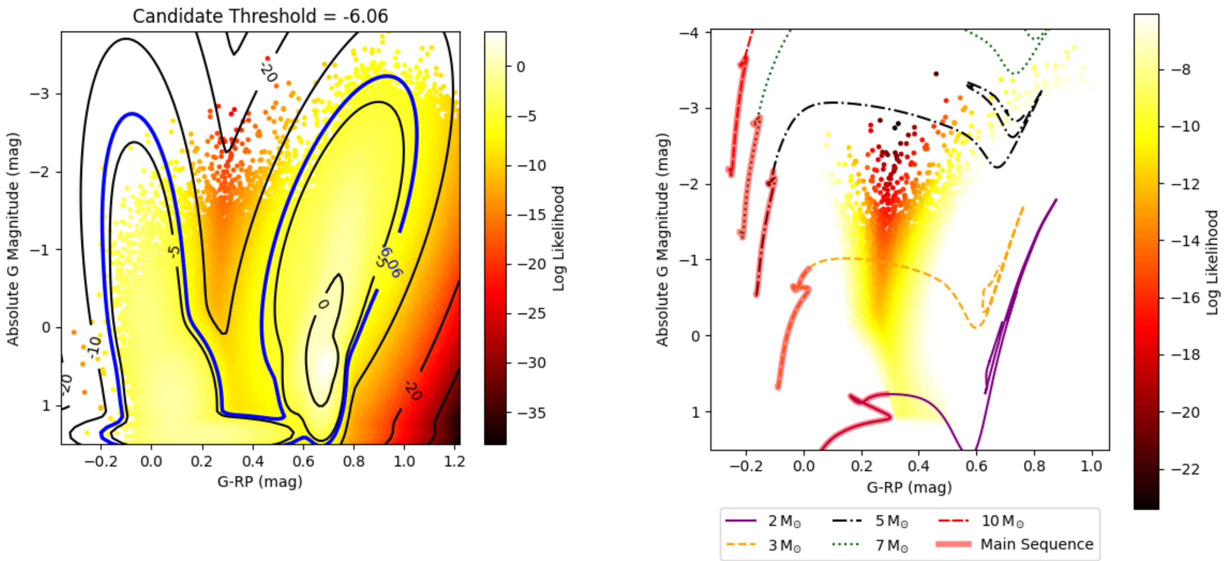


Figure 2.5: Resulting nine component Gaussian mixture model of the training data set. The components of the model are represented by the ellipses which correspond to one standard deviation of each of the Gaussian distributions. The collective distribution of all nine Gaussian distributions corresponds to the density distribution of the data and is shown with colour, with darker areas representing more dense regions.

To select likely LRN progenitor sources from the fitting data set, first the likelihood values were used to determine the sources that did not belong to the model, i.e. stars that are not main sequence or RGB. Sources excluded from the model reside in the low density regions of the HRD, which includes the Hertzsprung gap as well as some unwanted regions. To use the likelihood values to determine if a data point belonged to the model or not, I calculated a threshold value that corresponded to the Gaussian model's 3 standard deviations, which was achieved by using a cross-validation data set. This involved randomly sampling the training data set into two samples, one with 90% of the data and the other with the remaining 10%. A Gaussian mixture model with nine components was then formed using the 90% sample and the 10% sample was then fitted to the model. For each of the data points in the sample fitted to the model, their likelihood values of belonging to the model were calculated. The threshold log likelihood value was then chosen as the value in which 99.7% of the fitted data was enclosed by. As the Gaussian mixture model is statistical in nature, this process was repeated 100 times to produce 100 different threshold values, which were then averaged to give a threshold log likelihood value of $\log \mathcal{L} = -6.06$.

With the threshold $\log \mathcal{L} = -6.06$, sources with a log likelihood of less than this were selected as residing in the low density regions of the HRD. As can be seen in Figure 2.6a, not all of the sources with log likelihoods less than the threshold reside in the Hertzsprung gap.



(a) Hertzsprung Russell diagram of the fitting data.

(b) Hertzsprung Russell diagram of the likely progenitors.

Figure 2.6: a) Hertzsprung Russell diagram of the fitting data set with the log likelihood values of the data belonging to the Gaussian mixture model. The contour lines represent lines of constant log likelihood values, and the blue contour line with a value of -6.1 represents the log likelihood threshold used for selection of likely progenitor candidates.
b) Hertzsprung Russell diagram of the selected likely progenitors from the fitting data, along with their log likelihood values of fitting the model. Over plotted, as a reference to the main sequence and red giant branch, are the MIST (Dotter, 2016; Choi et al., 2016; Paxton et al., 2011, 2013, 2015, 2018) stellar evolution tracks from main sequence to core helium burning phase of stars with masses 2, 3, 5, 7, and 10 $M_{\odot}$.

For example, the region in the lower right ($G - RP \geq 0.8$, $G \geq 0$) of Figure 2.6a is below the RGB and is thus not a part of the Hertzsprung gap, however it contains sources with log likelihood values lower than the threshold value. Due to there being regions of low density not corresponding the the Hertzsprung gap, cuts to the absolute G magnitude and $G - RP$, provided in Table 2.3 with a description, were made to the sources with log likelihood values below the threshold. The applied cuts to the parameter space removed the sources that were to the left of the main sequence and to the right of the RGB. Therefore, the remaining sources form a sample that lie in the Hertzsprung gap and are thus likely LRN progenitors.

After having selected a sample of likely progenitors, one final condition was applied to the sources. This condition was that the likely precursors could not have a nearby source within $2''$. This was required due to the limitation of the astrometric resolution of the ZTF time domain survey (Masci et al., 2018; Bellm et al., 2018) from which light curves were obtained from, as is described in Chapter 2. Due to the seeing conditions at the telescope's site, the ZTF's image quality is only $2''.0$ FWHM in the r-band (Masci et al., 2018; Bellm et al., 2018), which means that sources are only resolved if they have a separation of more than $\sim 2''$. Due to this I decided to discard candidates that have another source within $2''$, as the collected light curves

Cut	Parameter Space Removed
$G \geq 1.25\,\text{mag}$	Data below main sequence and red giant branch.
$G - RP \geq 1\,\text{mag}$	Data to the right of the right most point of the red giant branch.
$G \leq -2\,\text{mag}$ and $G - RP \geq 0.6\,\text{mag}$	Data below the red giant branch.
$G - RP \leq G - RP$ of MIST main sequence ends	Data to the left of the end of the main sequence from the MIST stellar evolution tracks (1).

Table 2.3: Parameter space cuts made to the sources in the fitting data sets with log likelihoods less than the threshold value (-6.06). A description of the parameter space removed by the cuts is provided in terms of the main sequence and red giant branch.
(1) Dotter (2016); Choi et al. (2016); Paxton et al. (2011, 2013, 2015, 2018)

would likely be a blend of multiple sources. To apply this condition, $2''$ cone searches centred on the coordinates of the likely progenitors were performed on Gaia EDR3 (Brown et al., 2021). Those likely progenitors with only one source within the cone were selected as "single" sources and were further analyzed for variability.

2.4 Results

The progenitor selection method described in this chapter allowed us to identify a total of 54347 unique sources in the Hertzsprung gap without a resolved companion. Table 2.4 shows the number of sources at the different stages of the likely progenitor selection process, which are also presented in the form of density sky maps in Figure 2.7. As expected, the sky map in Figure 2.7a shows that the relative density of the fitting data set is highest in the Galactic plane. Figure 2.7b shows that the density of the sources in the Hertzsprung gap is also higher in the Galactic plane, although the maximum density has decreased by a factor of ~ 75. Furthermore, the relative density outside of the plane is generally less than 100 with some regions containing no sources at all. The final sky map in Figure 2.7c shows that the maximum relative density of sources identified as single source likely progenitors is very similar to that of the sources in the Hertzsprung gap.

Stage of Selection	Number of Sources
Fitting Data Set	9697315
Likely Progenitors (Hertzsprung Gap)	56593
Single Source Likely Progenitors	54347

Table 2.4: Numbers of sources at different stages of the likely progenitor selection process. The different stages are the full fitting data set, likely progenitors which are sources within the Hertzsprung gap, and finally single source progenitors which are sources in the Hertzsprung gap that have no other sources within $2''$. The last two stages have been split into the three different extinction corrections that were applied.

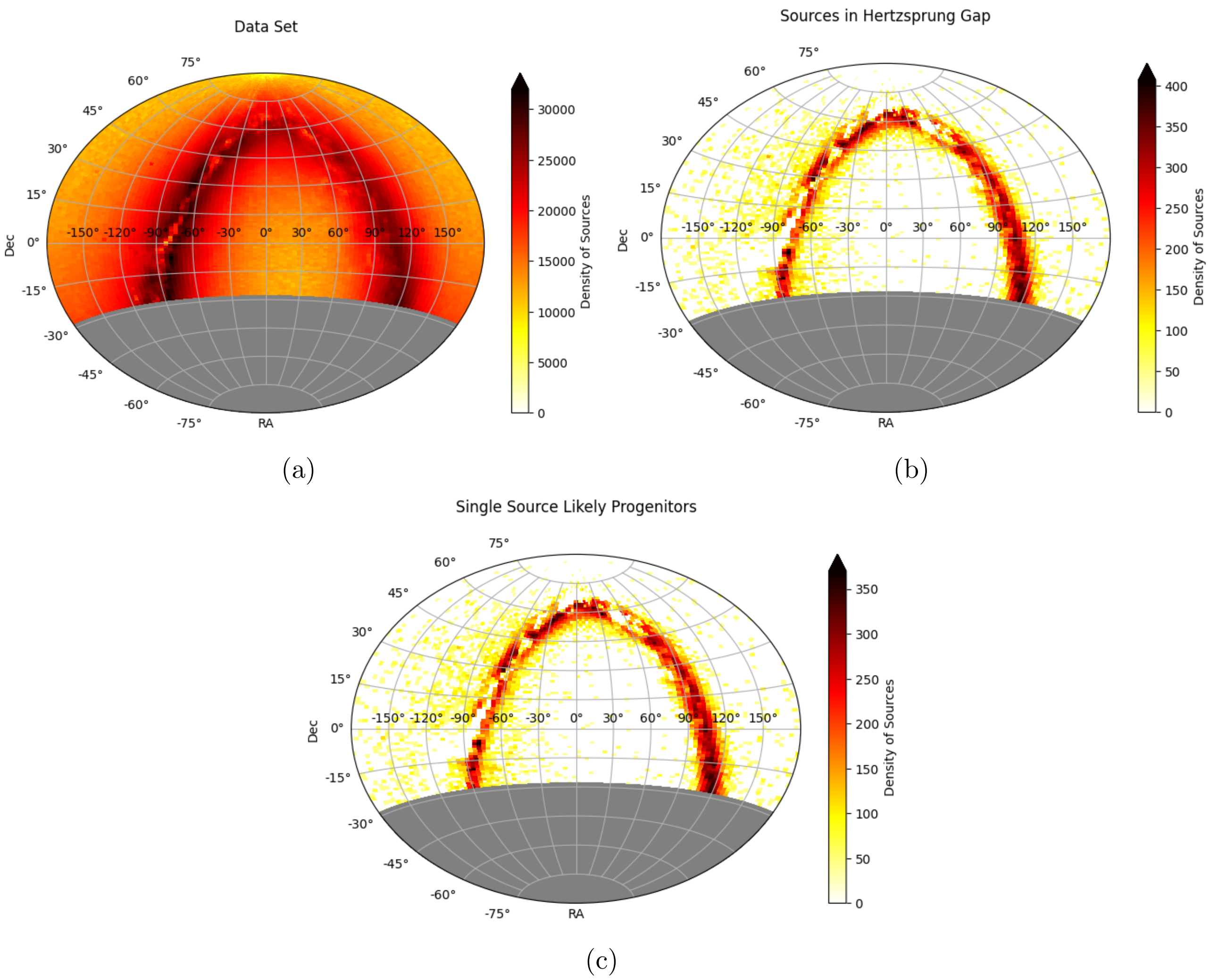


Figure 2.7: Density sky maps of the sources at the three different stages of progenitor selection: (a) full fitting data set, (b) likely progenitors which are sources within the Hertzsprung gap, (c) and likely progenitors without another source within 2″. The sky maps are in terms of celestial coordinates, right ascension (RA) and declination (Dec). The grey region represents the region of the sky where no data was collect due to the declination range of the ZTF survey being above a declination of $\sim -30°$ (Bellm et al., 2018).

2.5 Discussion

In total 56593 sources were identified as likely progenitors, with 54347 of them having no other sources within 2″. These single source likely progenitors provide a reasonably sized sample for the precursor analysis, described in Chapter 3. Next I discuss the limitations and improvements that could be made to both the data used and the method.

The first of the changes that could be made to the method would be to relax the parallax and flux error constraints on the fitting data set, as this would increase the number of sources within the data set. Therefore, there would potentially be more sources that will reside in the Hertzsprung gap and do not have another source nearby, thus increasing the sample of single source likely progenitors that the precursors are identified from. One negative consequence of reducing these constraints is that the sources would have larger positional uncertainties in the

HRD, which could lead to an increase in false detections of Hertzsprung gap sources.

The next improvement comes with the upcoming release of Gaia DR3 in quarter 2 of 2022 (DPAC, 2021b). The data within Gaia DR3 will build upon Gaia EDR3 and will include extinction values for independent sources. As a result, the method could be improved by forming an extinction corrected fitting data set from Gaia DR3. This would therefore provide more accurate absolute magnitudes and colours for the sources in the fitting data set than those used in the current method, which does not apply extinction corrections. This more accurate data set could result in the identification of more sources in the Hertzsprung gap as the uncorrected sources previously not in the gap may have a large enough extinction that would actually place them in the gap.

When considering the limitations of the method, the first one is that it focuses on identifying yellow giants and YSG as likely LRN progenitors. This excludes other possible progenitor binary systems where neither of the components is a YSG or a yellow giant. However, due to the small sample of observed progenitors there is some uncertainty in the true nature of the components of the progenitors of LRNe. As at least three progenitor systems have been observed to contain yellow giants or YSG (MacLeod et al., 2017; Blagorodnova et al., 2017; Blagorodnova et al., 2021), it is reasonable to start the search for progenitor systems by focusing on only yellow giants and YSG. As more progenitor systems are discovered, the method of identifying LRN precursors could be adapted accordingly.

The final limitation of the method is that the fitting data set was limited to declinations above $-30°$ due to the declination range of the ZTF survey (Bellm et al., 2018), which is used in the method presented in Chapter 3. Due to this imposed declination range, it is possible that likely progenitor sources and hence precursors were missed in this analysis. Therefore one improvement would be to extend the search for likely progenitor systems to all declinations. To achieve this the method discussed in Chapter 3 would need to makes use of data from additional time domain surveys that cover declinations below $-30°$.

CHAPTER 3

Likely Precursors Selection

This chapter provides the details and discussions of how likely LRN precursors are identified from the candidate progenitor systems selected in Chapter 2. Provided first is an outline of the method, which is then explained and explored in more detail in the sections that follow. Presented in Section 3.4 are the results from the method which are then discussed in Section 3.5.

3.1 Method Overview

In order to identify LRN precursors, the method presented here utilises a feature their light curves, which is the slow rise in brightness before the outburst. This brightening was seen in the precursors of V1309 Sco and M101 OT2015-1, where the slow magnitude increases were ~ 0.8 mag over ~ 1600 days and ~ 1.5 mag over ~ 2000 days respectively. For both events, the slow increase in brightness was suggested to be caused by the formation of a common envelope (Tylenda et al., 2011; Goranskij et al., 2016), which is currently thought to be one of the main causes of LRNe (Ivanova et al., 2013b; Blagorodnova et al., 2017). Therefore, it is expected that the light curves of other precursors will also exhibit a slow increase in brightness. As such, the method presented here to identify likely precursors from the candidate progenitor systems makes use of this information.

The method to identify likely precursors began with collecting light curve data of the candidate progenitor systems selected in Chapter 2. This process is discussed in detail in Section 3.2. The next step in identifying likely precursors was to apply a slow rising transient detection method to the obtained light curves, which selects sources that slowly increase in brightness among other types of variability. In Section 3.3 are the details and discussions regarding the slow transient detection methods explored. The final step in the method was to manually select sources exhibiting a clear slow brightening from the resulting light curves of the slow transient detection method. The results are presented in Section 3.4 and discussed in Section 3.5 along with the limitations and improvements to the method.

3.2 Light Curve Data

Obtaining the light curve data of the likely progenitor sources identified in Chapter 2 is the first step in identifying the precursors from them. In this section are the details of how and where the light curve data was obtained from, as well as the analysis carried out on the light curves.

In this method, the light curve data was obtained from the ZTF time domain survey (Masci et al., 2018; Bellm et al., 2018). This was accomplished by using the `Python` package `ztfquery` by Rigault (2018), which provides an automated means of performing a cone search of a given coordinate, radius, and light curve band to collect. For each of the likely progenitor sources, a cone search with a $2''$ radius was used to collect the light curves in the g, i, and r bands. The collected light curves were then cleaned and filtered by applying quality constraints on certain statistical properties of the individual data points and the overall light curve.

The first quality constraint applied was to the individual observations using their CATFLAG values, which are values that represent if the observation is unusable, bad, usable, or clean (ZTF team, 2021). An observation is unusable or bad if it is effect by clouds or moon-contamination, and these observations are marked with CATFLAG values greater than 32768 (ZTF team, 2021). Observations with CATFLAG values of less than 32768 are considered to be mostly usable, and those with a value of zero are perfectly clean observations (ZTF team, 2021). It was decided that only clean observations were to be kept and so any observations with a CATFLAG value different from zero were excluded from further use.

The second constraint applied to the light curves was that the time span of each of the light curves had to be longer than one year. This was applied as the aim is to identify LRN precursors which, based on the events V1309 Sco and M101 OT2015-1, only exhibit a magnitude increase of about one magnitude over approximately 1600 days. Therefore, if a light curve of a LRN precursor had a time span of less than a year, the expected magnitude increase would be less than approximately 0.22 mag. With such a small expected magnitude increase and time span it would be difficult to distinguish between what might be a LRN precursor and other sources such as variable stars. Hence, the minimum time span of one year was set in order to have a higher certainty of if the light curves exhibited a slow brightening rather than a periodic source such as a variable star.

The final constraint enforced on the light curves was that each one had to contain more than 25 clean observations. This constraint was applied to reduce the number of false detections of likely LRN precursors, as light curves with too few data points will not accurately resemble the true underlying distribution. Thus, these light curves could coincidentally resemble a brightening source when in fact this is not the case. As the number of points in the light curve increases, the more they will resemble the true underlying distributions, thus reducing the number of false positive detections of slow rising transients. However, in the case of ZTF light curves, increasing the required number of data points also reduces the number of light curves that satisfy the condition. Therefore, a balance is needed between reducing the number of false positives and rejecting too many light curves. As such, the value of 25 data points was chosen, as the number of light curves with 20 or more observations in the ZTF DR7 is about 50.9% of the total number of light curves (ZTF team, 2021). Therefore, if the constraint was increased further, say to 50 or 100 data points, the number of light curves in ZTF DR7 that would satisfy this one constraint would be less than half the total light curves.

The constraints discussed above were applied to all of the light curves obtained of the likely progenitor sources. If a light curve did not meet all of the constraints then it was discarded from further use in the method to select likely precursors. The remaining light curves that satisfied all of the above mentioned constraints were then tested to be slow rising transients using the methods explored in Section 3.3.

3.3 Slow Transient Detection Methods

To identify LRN precursors from the light curves that were obtained using the method described in Section 3.2, applied to them was a slow transient detection method. Presented and discussed in this section are the two methods that were explored for detecting slow transients. The first method presented is that by Kostrzewa-Rutkowska et al. (2018) which uses the von Neumann statistic (von Neumann, 1941) and skewness. The second method explored is one that makes use of the gradient of the light curve's general trend. These two methods are described, explored, and discussed in Sections 3.3.1 and 3.3.2 respectively.

3.3.1 Von Neumann Statistic and Skewness Method

Method

The method of using the von Neumann statistic and skewness of the light curve as a means of detecting transients has previously been used by Kostrzewa-Rutkowska et al. (2018) to detect nuclear transients in the Gaia data, and Wevers et al. (2017) to detect fast transients in the Gaia data. Here the method presented by Kostrzewa-Rutkowska et al. (2018) is explored for the use of detecting slow rising transients. The method first calculates the von Neumann statistic (von Neumann, 1941), described as the ratio of the successive mean square difference to the variance, and the skewness of the light curves using the formulae 3.1 and 3.2 respectively.

$$\eta = \frac{\frac{1}{n-1}\sum_{i=1}^{n-1}(x_{i+1} - x_i)^2}{s^2} \tag{3.1}$$

$$\gamma = \frac{\frac{1}{n}\sum_{i=1}^{n-1}(x_i - \bar{x})^3}{s^3} \tag{3.2}$$

Where η is the von Neumann statistic, γ is the skewness, x are the magnitude measurements, n is the number of measurements, and s is the variance of the light curve.

The skewness of the light curve is used to infer if there is a magnitude increase or not. A positive value suggests that the light curve is decreasing in magnitude or it exhibits a dip in magnitude, while a negative value suggests that the magnitude is increasing (Kostrzewa-Rutkowska et al., 2018). The von Neumann statisic is then used as a measure of the smoothness of the variation, with smoother variations having a lower von Neumann statistic. To select transients based on the two statistics, Kostrzewa-Rutkowska et al. (2018) applied two cuts to the skewness and reciprocal of the von Neumann statistic parameter space. These cuts are $\gamma < 0$, and $\gamma < (\log(1/\eta)/\log(4)) - 1$, and are depicted as the blue dashed lines on Figure 3.1, showing the γ vs. $1/\eta$ parameter space for different variable stars and LRN precursors. Kostrzewa-Rutkowska et al. (2018) identify transients as the sources below the applied cuts.

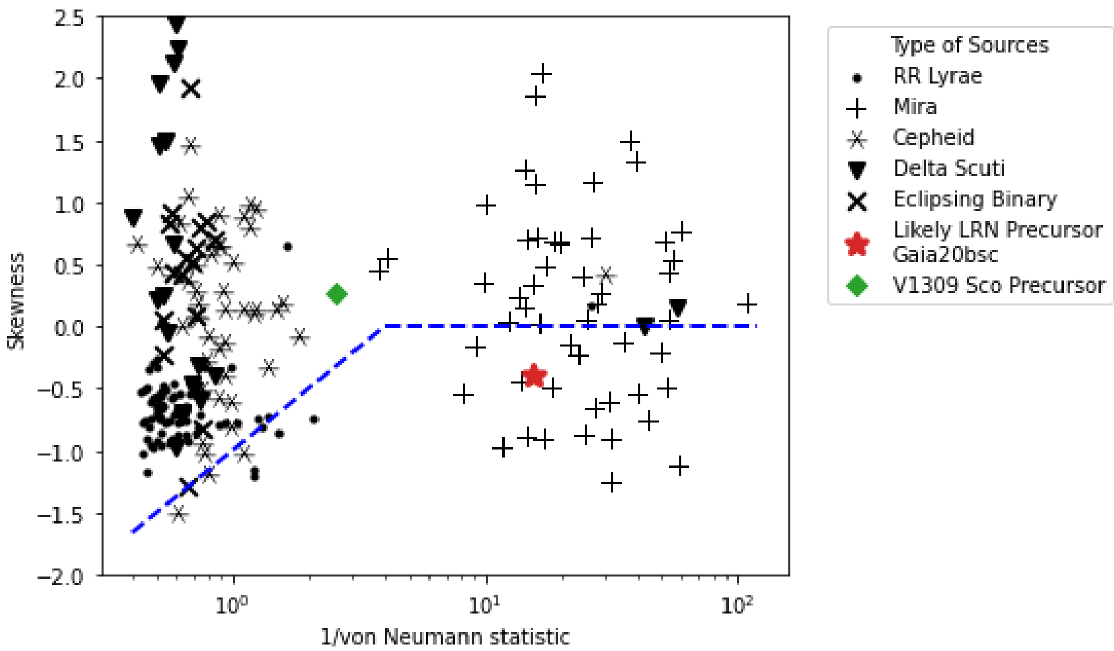


Figure 3.1: Skewness and the reciprocal von Neumann statistic parameter space populated with the values from light curves of varying sources. The sources present in the Figure are different types of variables, LRN precursors, and a possible LRN precursor. The blue dashed lines represent the cuts applied by Kostrzewa-Rutkowska et al. (2018) to select transients which lie within the space below the cuts ($\gamma < 0$, and $\gamma < (\log(1/\eta)/\log(4)) - 1$). The majority of the different variables are not picked as transients, with the exception of Mira variables. The possible LRN precursor Gaia20bsc is identified as being a transient, however the precursor of the confirmed LRN V1309 Sco is not.

Exploration of the Method

To explore the use of the von Neumann statistic and skewness method for the identification of LRN precursors, the method was used on different types of variable sources, the precursor of V1309 Sco, and an unconfirmed but likely LRN precursor Gaia20bsc (Alerts, 2020). The primary aim of this was to determine if the method was capable of detecting LRN precursors as transients, and the secondary aim was to determine if different types of variable sources were falsely identified as transients.

In this exploration of the method the following types of variable sources were used: RR Lyrae, Mira variables, Cepheid variables, Delta Scuti variables, and eclipsing binaries. The light curve data for these variable sources was obtained from the ZTF database (Masci et al., 2018; Bellm et al., 2018), and the data for the likely precursor Gaia20bsc was obtained from the Gaia alert of this source (Alerts, 2020). The data used for V1309 Sco is that used by Tylenda et al. (2011) which was originally obtained from OGLE (Udalski, 2003; OGLE, 2021). The results of the calculations of the von Neumann statistic and skewness for the sources mentioned are plotted in Figure 3.1, which also shows the cuts (blue dashed lines) used to select transients.

From the results of the von Neumann statistic and skewness shown in Figure 3.1, it can be seen that the majority of the variables sources are above the cuts (blue dashed lines), and are therefore correctly not selected as transients. However, the Mira variables are spread evenly across the applied cuts at higher values of the reciprocal von Neumann statistic, meaning that those below the cuts are falsely identified as being transients. It is also worth mentioning that the Mira variables are distinguishable from the other types of variable sources based on the reciprocal von Neumann Statistic, as the Mira variables reside at larger values while the other types generally reside at lower values. This suggests the Mira variable light curves consist of smoother variations compared to the other types of variables investigated, which is the case here.

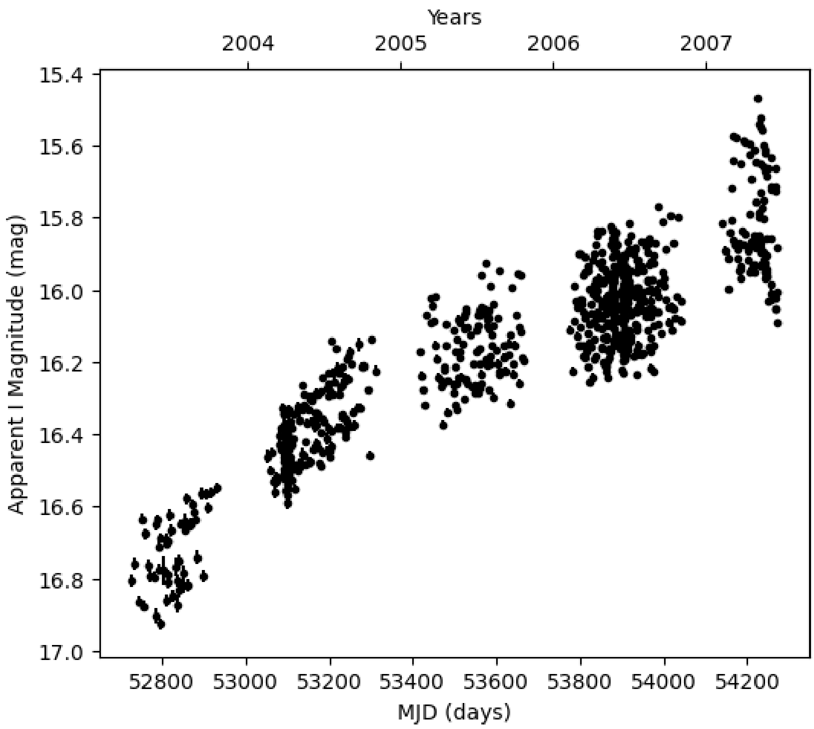


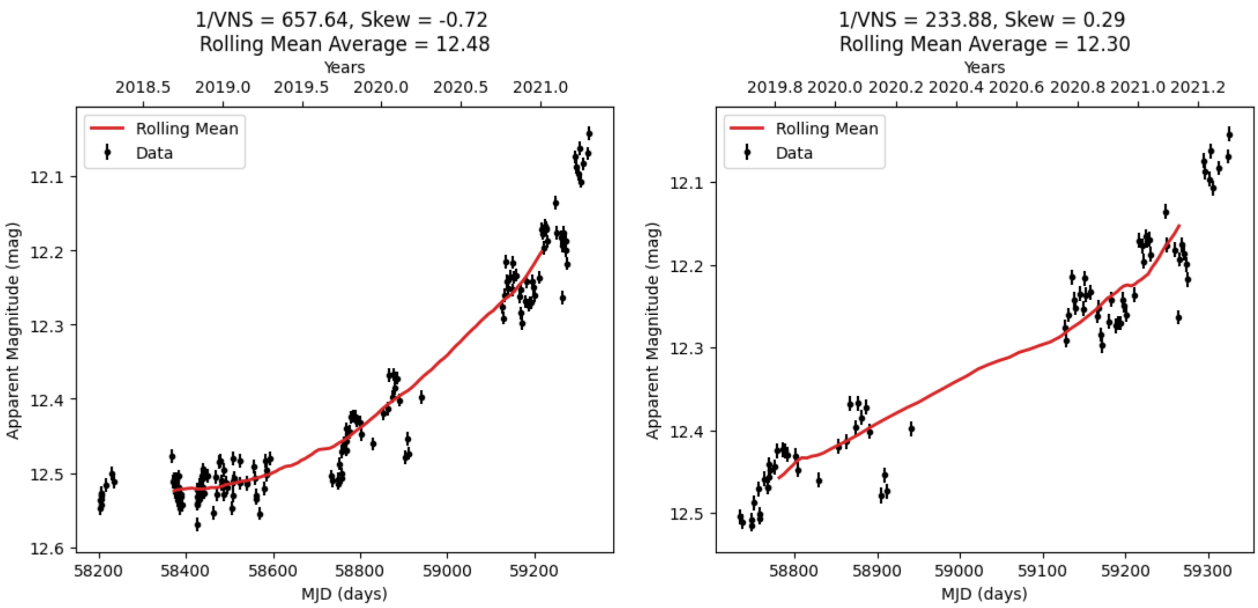
Figure 3.2: Light curve of V1309 Sco's precursor used in the exploration of using the von Neumann Statistic and skewness as a slow rising transient detection method. The light curve was produced from the *I* band data used by Tylenda et al. (2011), which was obtained from the Optical Gravitational Lensing Experiment (Udalski, 2003; OGLE, 2021).

Another interesting result from the plot in Figure 3.1, is that the likely LRN precursor Gaia20bsc satisfies the transient selection cuts, however, the precursor of V1309 Sco at ~ 1 years before its outburst (see Figure 3.2) does not satisfy them. In fact, the skewness of V1309 Sco's precursor light curve is positive with a value of approximately 0.25, suggesting that the light curve either experiences a dip in brightness or is decreasing in brightness. This is however not the case as the light curve of V1309 Sco's precursor, shown in Figure 3.2, is overall increasing in brightness albeit at a decreasing rate.

Discussion of the Method

One possible reason behind V1309 Sco's positive skewness is the eclipsing binary variation that is present in the precursor light curve. To test if this was the case, the eclipsing binary variation was removed from the general trend of the light curve by taking the moving average of the data. The von Neumann statistic and skewness were then calculated for the general magnitude trend (moving averages) of the light curve, with the resulting values being 0.0001 and 0.9455 respectively. By using the general trend of the magnitude the skewness had decreased but it was still positive, falsely suggesting that the light curve was decreasing in brightness.

Another possible reason as to why V1309 Sco's precursor light curve was not detected as a transient, due to its positive skewness, could be that the skewness of a light curve is not a good indicator of a long period increase in magnitude. In the studies by Kostrzewa-Rutkowska et al. (2018) and Wevers et al. (2017) they used the von Neumann statistic and skewness as a means of detecting nuclear and fast transients respectively. The light curves of these types of transients exhibit relatively fast brightness increases, and so the brightening period is relatively small compared to the total time span of the light curve. This means that the mean used in calculating the skewness is relatively close to the average pre-brightening magnitude, and so the skewness is effectively measuring the variation about this magnitude. However, in the case of V1309 Sco's precursor the brightening period is equal to the total time span of the light curve, and so the mean of the light curve lies somewhere between the maximum and minimum magnitudes. This means that the skewness is not measuring the variation of the data about a pre-brightening magnitude, thus giving no indication of if the light curve is increasing or decreasing in brightness relative to a pre-brightening magnitude.



(a) Full available light curve of the brightening source.

(b) Brightening phase (MJD > 58600) of the light curve.

Figure 3.3: Light curves of a brightening source obtained from the ZTF light curve database (Masci et al., 2018; Bellm et al., 2018). a) Full available light curve of the source. b) Brightening phase ($MJD > 58600$) of the full light curve. The von Neumann statisitic (VNS) and skewness were calculated for the rolling mean values of the light curve which are represented by the red line.

The effect on the skewness described above is also shown by the light curves presented in Figure 3.3, which are light curves of a brightening source taken from the ZTF database (Masci et al., 2018; Bellm et al., 2018). Plot 3.3a shows the full available light curve, which starts to brighten after approximately 58600 MJD, and plot 3.3b shows the brightening phase (MJD

> 58600) of the light curve. Calculated for both of these light curves were the von Neumann statistic and skewness using the moving average values, which are represented by the red lines in the plots in Figure 3.3. From this figure it can be seen that the skewness of the full light curve is negative, while the brightening phase light curve has a positive skewness. This shows that the skewness is not a good indicator of an increasing trend when the brightening period is equal to the total time span of the light curve, and hence this method is not sufficient for detecting slow rising transients that occur over many years.

3.3.2 Sum of Differences Method

Method Description and Development

The sum of differences method was developed as a means of detecting slow rising transients and hence LRN precursors. The main principle behind the method is that the magnitude trend of a slow rising transient will exhibit an overall net increase in magnitude. As such, the sum of differences method first finds the general trend of a light curve by calculating the rolling average, which is then used to calculate the net magnitude change by summing the differences between consecutive rolling average values. This was done using the following formulae:

$$\Theta = \sum_{i=1}^{N-1} (\xi_{i+1} - \xi_i) \tag{3.3}$$

$$\xi_i = \frac{\sum_{j=0}^{n_{bin}-1} (x_{i+j})}{n_{bin}} \tag{3.4}$$

$$n_{bin} = \frac{1}{3} \times n_{total} \tag{3.5}$$

Where Θ is the sum of differences, ξ_i is the value of the i^{th} rolling average bin, N is the total number of rolling average bins, n_{bin} is the number of data points in each rolling average bin, n_{total} is the total number of data points in the light curve, and x are the magnitude measurements. A negative sum means that the light curve exhibits a brightening trend, a positive sum means that the trend of the light curve is dimming, and a sum of zero means that there is no change in the brightness trend of the light curve.

Although slow rising transients will have a negative sum of differences value, a negative value on its own is not enough to identify slow rising transients for several different reasons. The first reason is that the data used is observational and is not ideal. This means that a light curve with a relatively constant magnitude trend that should ideally result in a sum of differences value of zero, may actually result in a slightly negative or positive value due to the errors and uneven time sampling. Shown in Figure 3.4 is an example of a light curve where this is the case. As is seen in the Figure, the sum of differences value is negative (-0.01 mag), meaning that the light curve is increasing in brightness, which would falsely suggest that it is a slow rising transient.

The second reason why a negative sum of differences value is not enough to identify a slow rising transient is that a negative value alone does not constrain the time span of the magnitude increase. This means that sources with a negative sum of differences value may have a brightness increase over any time span, and so are not limited to only slow rising transients.

To solve the first issue mentioned, a threshold could be applied to the sum of differences

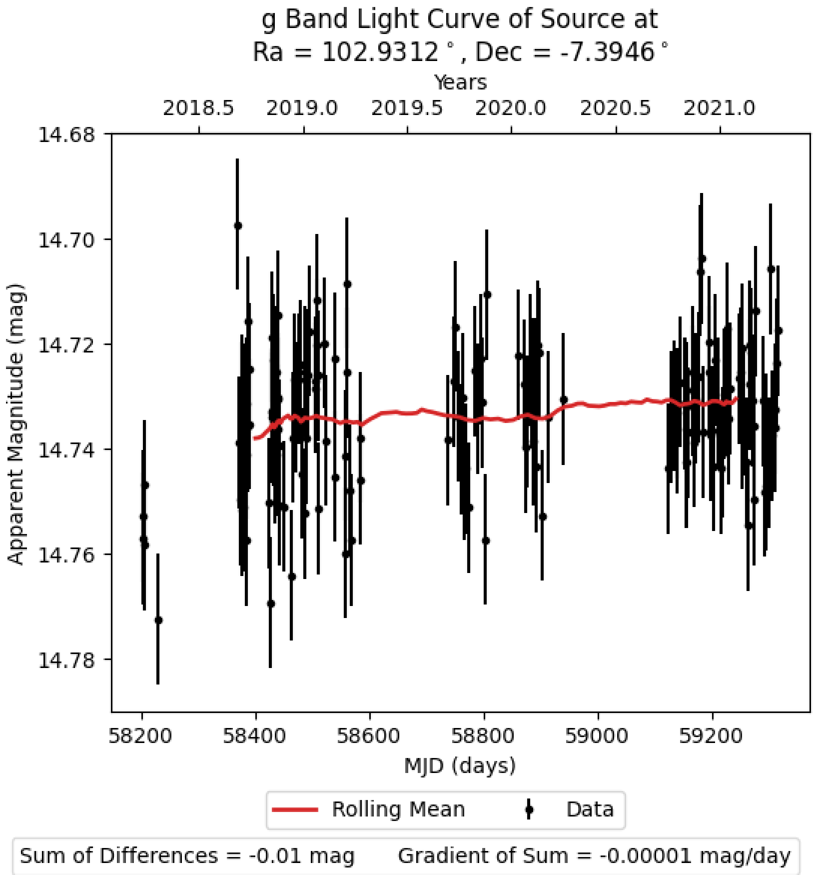


Figure 3.4: Light curve of a source obtain from the ZTF light curve database (Masci et al., 2018; Bellm et al., 2018). The light curve has relatively constant magnitude observations and magnitude trend as shown by the red line. The resulting sum of the differences of the consecutive rolling mean values is -0.01 mag.

values, and so light curves with a value less than the threshold value are identified as slow rising transients. This would limit the number of false positive detections of slow rising transients, as it would remove the light curves that do not exhibit an increase in brightness that would be expected of a precursor. However, applying a threshold to the sum of differences would not solve the second issue mentioned, as it would not consider the time span of the increase, and so the threshold could be applied to the gradient of the sum of differences. By applying a threshold to the gradient of the sum of differences, it limits the number of false detections as well as considering the rate of brightening.

To select a suitable threshold value for the gradient of the sum of differences, first a lower limit was calculated based on the two known LRN precursors V1309 Sco and M101 OT-2015-1, which experienced magnitude changes of ~ -0.8 mag over ~ 1600 days and ~ -1.5 mag over ~ 2000 days respectively. The rate of change of magnitude for V1309 Sco's precursor is therefore -5×10^{-4} mag/day, and for M101 OT2015-1's precursor it is -7.5×10^{-4} mag/day. Therefore, the lower bound of the threshold value is set by V1039 Sco's precursor with a value of -5×10^{-4} mag/day.

To further select a suitable threshold value, the sum of differences method was used on a small sample of light curves and the threshold value used to select slow rising transients was

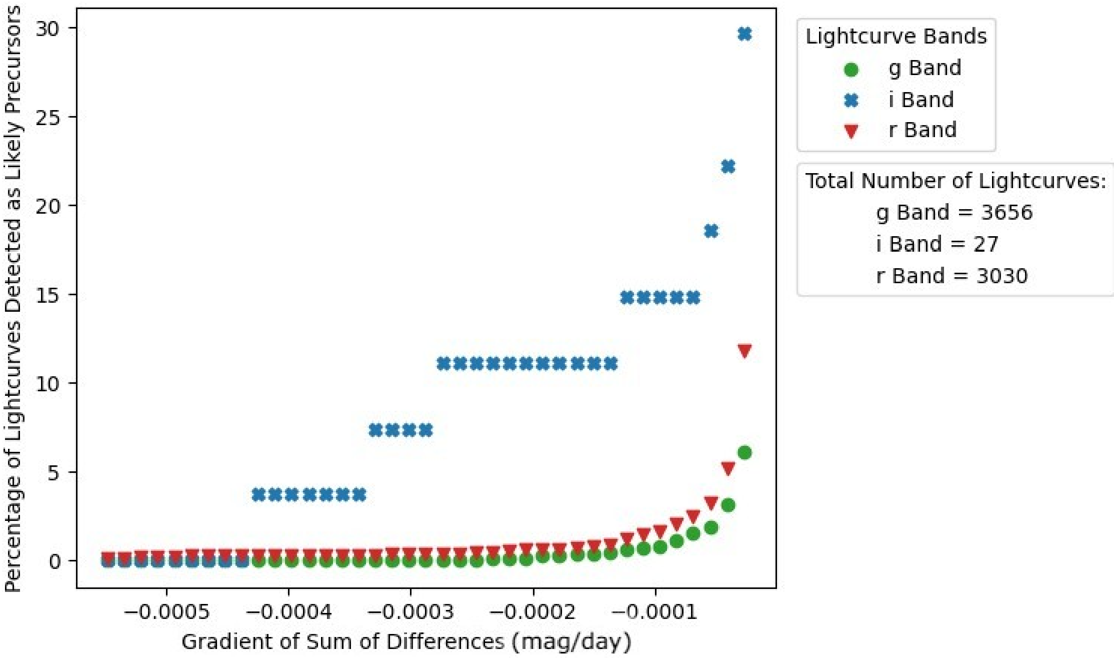


Figure 3.5: Percentage of g, i, and r band light curves from the ZTF database (Masci et al., 2018; Bellm et al., 2018) identified as likely precursors by the sum of differences method as a function of the threshold value used for the gradient of the sum of differences. The total number of light curves in each filter band are shown on the right. It should be noted that none of the light curves used are of likely precursors, and so all identifications are false positives.

varied from -5.5×10^{-4} mag/day to -3×10^{-5} mag/day. The small sample of light curves was formed from the light curves collected of 10% ($\sim$ 5400) of the candidate progenitor sources. In total 3656 g-band light curves, 27 i-band light curves, and 3030 r-band light curves were obtained from the ZTF light curve database (Masci et al., 2018; Bellm et al., 2018). The purpose of testing the method with different threshold values was to find out how it affected the number of light curves falsely detected as rising transients. The results from the test are presented in Figure 3.5, which shows for each of the filter bands, g, i and r, the percentage of light curves detected as likely precursors as a function of the gradient threshold value.

From the results in Figure 3.5 it can be seen that the percentage of g and r band light curves detected as likely precursors remains fairly constant at gradient threshold values of less than approximately -2×10^{-4} mag/day, after which there is an exponential-like increase in the percentage. This trend is, however, not seen in the i-band light curves, where the percentage of light curves detected as likely precursors increases in steps. This is due to the small number of i-band light curves that were obtained from the ZTF database, which is due to the lack of i-band light curves in this specific band. It should be noted that the sample of light curves contained no plausible precursors, and so the likely precursor detections recorded in Figure 3.5 are all false positives.

Based on the lower limit and the results from Figure 3.5, it was decided that a gradient threshold value of -2×10^{-4} mag/day was to be used as it significantly reduces the number of false positive detections to $\lesssim 1\%$ in the g and r bands and $\sim 11\%$ in the i-band. Furthermore, this value also considers the uncertainty in LRN precursor brightening rates due to there currently only being two observed LRN precursors.

Exploration of the Method

To explore the limitations of the sum of differences method for identifying LRN precursors, the method was applied to simulated light curves that were based on V1309 Sco's precursor. The simulations were formed with different parameters such as the ratio of the eclipsing binary amplitude, $\Delta m_{\rm eclipse}$, to the secular magnitude increase, $\Delta m_{\rm sec}$, (amplitude ratio=$\Delta m_{\rm eclipse}/\Delta m_{\rm sec}$), the ratio of the eclipsing binary period, $P_{\rm eclipse}$, to the the time span of the secular magnitude increase, Δt, (period ratio=$P_{\rm eclipse}/\Delta t$), and the number of data points in the light curve. The range of the two ratios used were $0 \leq \Delta m_{\rm eclipse}/\Delta m_{\rm sec} \leq 20$ and $0.0002 \leq P_{\rm eclipse}/\Delta t \leq 1.8$, and were based on the following amplitude and period ranges: $0\,{\rm mag} \leq \Delta {\rm m}_{\rm eclipse} \leq 2\,{\rm mag}$, $0.037\,{\rm mag} \leq \Delta {\rm m}_{\rm sec} \leq 1.5\,{\rm mag}$, $0.19\,{\rm days} \leq {\rm P}_{\rm eclipse} \leq 650\,{\rm days}$, and $1\,{\rm year} \leq \Delta {\rm t} \leq 3\,{\rm years}$.

To form the simulated light curves, first a 3rd order polynomial was fitted to V1309 Sco's precursor light curve. The resulting polynomial was then superposed with an eclipsing binary variation, which was simplified to a sinusoidal function with a phase of zero. Due to this simplification, for every binary orbital period, there are two sinusoidal periods which make up the eclipsing binary variation. The amplitude and period of this eclipsing binary variation was determined by the relevant ratios and using the constant values -0.8 mag and 1600 days for the secular magnitude increase and time span respectively, which are based on V1309 Sco's precursor. Next, data points were formed from the sinusoidal superposed polynomial at a rate of 1000 points per day of the light curve time span, and a photometric error was added to each data point. The error values were selected at random from a Gaussian distribution with a standard deviation of 0.05 mag, which was based on ZTF's photometric errors of an 18.7 magnitude source. Finally random samples of 25, 50, 75, 100, and 500 data points were selected to form the simulated light curves. It should be noted that this sampling is simplistic and is unlike the sampling seen in the light curves from ZTF, where there is generally a cadence of 2 observations per night every 2-3 nights.

The different simulated light curves were analyzed using the sum of differences method. The resulting sum of differences and gradient values were plotted as a function of the amplitude ratio, period ratio, and number of data points in the light curve. Four of the resulting plots are shown in Figures 3.6 and 3.7, while the complete set of plots can be found in Appendix B. The plots in Figures 3.6 and 3.7 show, from two different perspectives, the sum of difference results of the simulated light curves with 25 and 500 data points respectively.

From the results in Figures 3.6 and 3.7 one of the most noticeable differences between them is the scatter in the sum of difference values. The results of the simulated light curves with 25 data points, shown in Figure 3.6, appear to have larger fluctuations between the neighbouring sum of difference values, whereas the values of the simulated light curves with 500 data points, shown in Figure 3.7, appear to change more gradually. From examining the full set of results it appears that, the larger number of points in the light curve, the smoother is the change in the sum of differences.

Another interesting result from Figures 3.6 and 3.7 is that, for constant period ratios the sum of differences values decrease as the amplitude ratio increases. This is caused by the eclipsing binary variation amplitude becoming dominant over the secular magnitude increase when the amplitude ratio is greater than one. At amplitude ratio values less than one, the secular magnitude increase is dominant. Provided that this value is constant in all of the simulated light curves, the sum of differences values does not vary much. This implies that the method can produce reliable and reproducible results regardless of the random distribution of the light curve data.

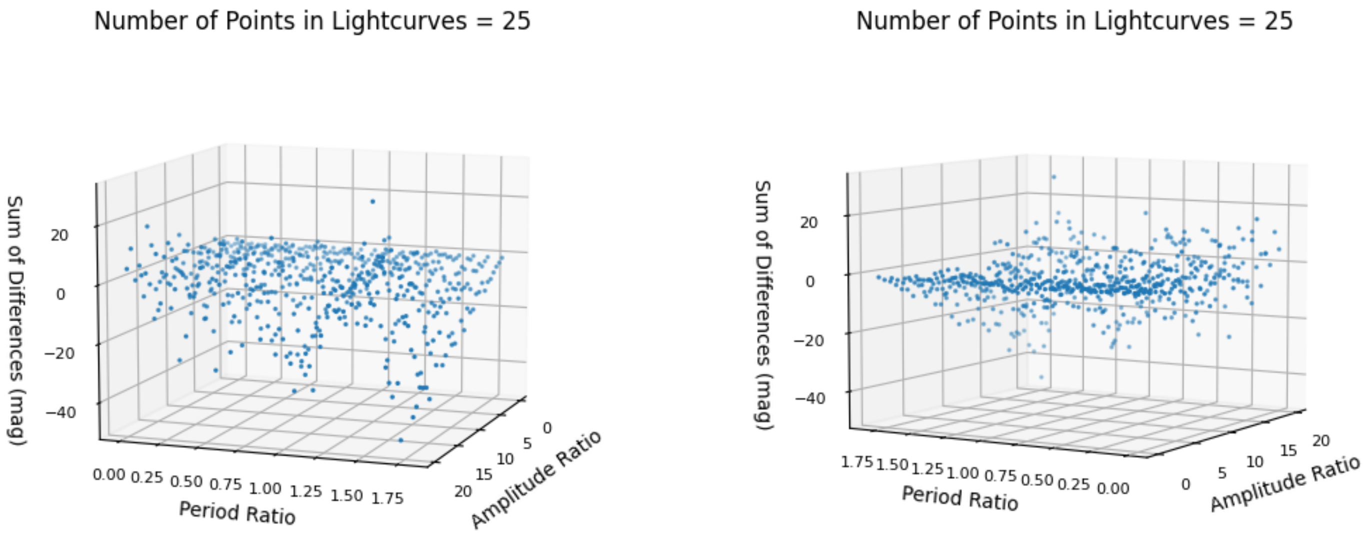


Figure 3.6: Calculated sum of differences values for the simulated light curves with 25 data points, and varying period and amplitude ratios. The two plots show the same results but from different viewing angles.

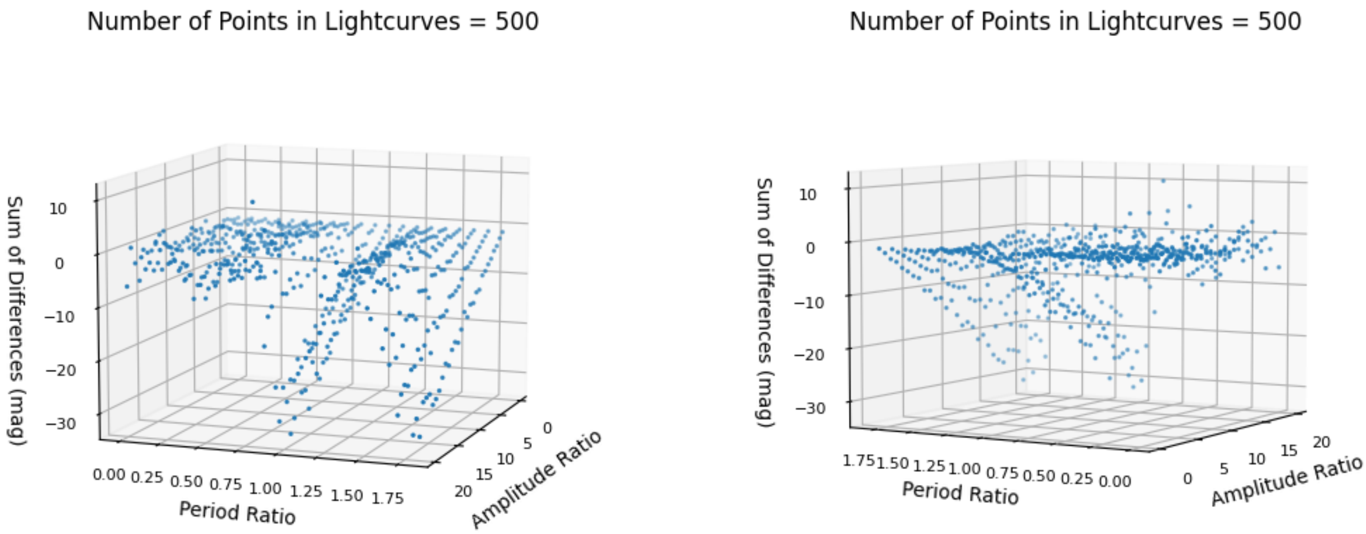


Figure 3.7: Calculated sum of differences values for the simulated light curves with 500 data points, and varying period and amplitude ratios. The two plots show the same results but from different viewing angles.

Lastly, it can be seen that the sum of differences at higher constant amplitude ratios varies in a cyclical fashion as a function of the period ratios, which is more clearly seen in Figure 3.7

than Figure 3.6. This is caused by the binary amplitude variation becoming dominant over the secular magnitude increase at amplitude ratios above one. Consequently, the sum of differences effectively only measures the net change in cyclical magnitude of the eclipsing binary variation.

The results of the gradient of the sum of differences, which are presented in Appendix B, show the same features as described for the sum of differences results.

Discussion of the Method

The testing of the method revealed several limitations for the identifications of lrn precursors. The first of the limitations is the existence of large variations between the resulting sum of difference and gradient values of the simulated light curves with less data points. The higher variations for the poorer sampled light curves suggests that the true underlying magnitude increase of them may not actually be represented by the sum of differences. This limitation was minimised by using light curves with a minimum of 25 data points, however, it could be further minimised by requiring a greater minimum number of data points. One disadvantage to this, as previously mentioned, would be that the number of light curves obtained would decrease as you increase the required minimum number of data points, thus reducing the chance of identifying LRN precursors.

A second limitation of the sum of differences method is that the gradient of the sum of differences represents the average gradient of the trend of the light curve. This means that, for light curves that start to increase half way through the light curve's time span, the average gradient of the increase is effectively halved. Therefore, a precursor that begins half way through the light curve's time span may not be selected as a slow rising transient as it does not meet the threshold value. Although this is an issue, the selected threshold value of -2×10^{-4} mag/day partly accounts for this, as a precursor with an average gradient similar to that of V1309 Sco (-5×10^{-4} mag/day) could begin over half way through the light curve and still be selected as a slow rising transient by the method. Furthermore, the light curves collected from the ZTF have a maximum time span of approximately three years and so precursors starting within the last third of them would be near undetectable due to the small magnitude increase that would occur in that time. Thus, the gradient of the sum of difference issue is not a major problem for precursors that begin in the last third of the light curve's time span. However, if the sum of differences method was to be applied to light curves with longer time spans, then this may become a major limitation of the method. However, this could be avoided by taking 3 year samples of longer time span light curves.

3.4 Results

Presented in Table 3.1 are the resulting number of light curves obtained from the ZTF database (Masci et al., 2018; Bellm et al., 2018) for each of the filter bands. Also shown in this table are the number of light curves that satisfy all of the requirements described in Section 3.2. From Table 3.1 it can be seen that of the 5.4×10^4 likely progenitors only 65.3% have g-band light curves, 17.0% have i-band light curves, and 56.6% have r-band light curves available in the ZTF light curve database. Also from this table it is clear to see that the majority of the r and g band light curves collected satisfied the requirements discussed in Section 3.2, with 96.2% g-band and 96.0% r-band light curves satisfying the requirements. This is not the case for the i-band light curves where only 17.3% of those collected satisfy the requirements.

Band	Sources	With Light Curves	Accepted Light Curves	Passed Test	In ATLAS
g	54347	35514 (65.3%)	34164 (96.2%)	70 (0.2%)	56 (80.0%)
i	54347	9212 (17.0%)	1593 (17.3%)	80 (5.0%)	79 (98.8%)
r	54347	30752 (56.6%)	29531 (96.0%)	165 (0.6%)	131 (79.4%)

Table 3.1: Number of sources and light curves at the different stages of analysis: progenitor sources with ZTF light curves (Masci et al., 2018; Bellm et al., 2018), accepted light curves that satisfy the quality constraints, light curves that pass the sum of differences test, and the number of sources in the ATLAS variable star catalogue (Heinze et al., 2018). Also noted are the percentages of sources and light curves that are present from the previous step in the analysis.

By using the sum of differences method on the light curves that satisfied the requirements discussed in Section 3.2, a total of 315 light curves were identified as being slow rising transients. The sources of these light curves were then cross-matched to the ATLAS variable star catalogue (Heinze et al., 2018) to obtain their possible classifications, which are summarised in Table 3.2. It should be noted that not all of the sources were present in the ATLAS variable star catalogue, hence the difference in the total number of classifications and the number of light curves identified as slow rising transients.

By using the ATLAS variable star catalogue, sources with classifications of STOCH, PULSE, NSINE, MSINE, and MPUSLE (see Table 3.2 for classification descriptions) were directly discarded as false positives and the remaining source's light curves were visually inspected for false positives. In total 292 light curves were identified as false positives, leaving only 23 light curves that showed a clear magnitude increase. These 23 light curves corresponded to 21 unique sources, and presented in Figure 3.8 are their r and g band light curves obtained from ZTF (Masci et al., 2018; Bellm et al., 2018) along with the V band light curves from ASAS-SN (Shappee et al., 2014; Jayasinghe et al., 2019). ASAS-SN light curves cover time periods between $\sim$ 2015 and $\sim$ 2019, which is before the time period covered by the ZTF DR7 (March 2015 - July 2021) (ZTF team, 2021). Combining both the ZTF and ASAS-SN light curves provides information of the variability of the sources over a longer time period, thus giving a further insight into the nature of the sources.

In addition to the light curves of the 21 sources, Table 3.3 provides additional information such as their coordinates, magnitudes, and parallaxes. Furthermore, literature and catalogues were searched to see if any of the 21 sources had been previously observed and analysed. These results are presented in Table 3.4, and shows information such as if there are spectra available, if they are Hα emitters, if they are an X-ray source, and their classifications from Simbad (Wenger et al., 2000) and the ATLAS variable star catalogue (Heinze et al., 2018).

Finally, the location of the 21 sources of interest are shown on a celestial coordinate sky map presented in Figure 3.9. Also shown in this figure is the density sky map produced from the fitting data set discussed in Section 2.2. The Galactic plane in this figure is represented by the region of the density sky map where the density is $\gtrsim$ 20000. As such, it can be seen that the sources of interest reside in the Galactic plane and more specifically the outer Galactic disc. This is likely to have been a consequence of not correcting for extinction of sources, as sources towards the Galactic centre are more extinct than those in the outer disc and not in the disc.

Class	Description	Number of Sources		
		g	i	r
CBF	Close binary, full period	0	2	5
CBH	Close binary, half period	0	0	2
DBF	Distant binary, full period	0	0	11
DBH	Distant binary, half period	2	0	12
dubious	Star might not be a real variable	6 (2)	3	11 (1)
IRR	Irregular: catch-all for difficult short-period cases	7 (2)	11	9 (1)
LPV	Long-period variable: catch-all for difficult cases	4 (3)	1	4 (2)
MPULSE	Modulated pulse: likely multimodal pulsator	7	7	14
MSINE	Modulated sine: multiple cycles of sine wave were fit	1	0	2
NSINE	Noisy sine: pure sine was fit, but residuals are large or non-random	0	0	2
PULSE	Pulsating variable	28	51	57
STOCH	Stochastic: certainly variable, yet more incoherent than even IRR	3	4	2

Table 3.2: Number and type of ATLAS variable star catalogue (Heinze et al., 2018) classifications of the identified slow rising transient sources. The bracketed numbers represent the number of sources of interest for the respective classification and band. The total number of classifications is less than the total number of slow rising transient detections due to the absence of some sources from the catalogue.

Source No.	RA (deg)	Dec (deg)	l (deg)	b (deg)	Plx (mas)	Δ Plx (mas)	G (mag)	$G - Rp$ (mag)	$E(B-V)$ (mag)
1	110.31019	−21.3153	235.2184	−3.3386	0.15	0.01	12.43	0.31	0.61
2	104.46141	−13.2202	225.4437	−4.5921	0.15	0.02	13.97	0.46	0.65
3	103.80181	−1.4815	214.6801	0.1629	0.24	0.02	12.47	0.35	0.60
4	95.98428	6.2353	204.2375	−3.2322	0.17	0.02	12.61	0.33	0.52
5	88.97008	20.7680	188.2093	−2.1820	0.25	0.02	12.27	0.45	0.89
6	87.82820	28.9491	180.6315	1.0736	0.19	0.02	13.43	0.44	0.64
7	66.59266	44.0087	158.4535	−3.5461	0.39	0.02	12.21	0.43	0.61
8	298.78552	42.6502	77.4310	7.4304	0.11	0.01	12.38	0.19	0.35
9	316.54535	47.7675	88.9251	0.3345	0.29	0.01	12.73	0.46	2.35
10	28.47645	55.7053	131.7065	−6.0986	0.39	0.02	12.06	0.44	0.35
11	39.56234	57.4849	136.9226	−2.4564	0.44	0.01	12.20	0.47	0.81
12	34.35626	57.7478	134.2272	−3.2259	0.26	0.01	12.72	0.48	0.62
13	7.99930	63.5143	120.7663	0.7244	0.33	0.01	11.99	0.47	1.30
14	3.11473	62.9781	118.5214	0.4407	0.39	0.01	12.80	0.36	0.80
15	358.91366	63.8671	116.8359	1.6576	0.48	0.01	12.06	0.43	1.13
16	359.73548	63.9497	117.2067	1.6630	0.20	0.01	13.71	0.45	0.91
17	30.43455	62.0812	131.0598	0.3240	0.32	0.02	11.72	0.49	1.20
18	27.39134	63.9136	129.2696	1.7594	0.36	0.02	12.97	0.45	1.35
19	14.92932	60.1786	123.9622	−2.6766	0.30	0.02	12.89	0.28	0.43
20	3.09651	60.8852	118.1973	−1.6271	0.34	0.01	12.79	0.45	0.97
21	34.45508	64.0326	132.2165	2.7291	0.14	0.02	13.85	0.39	0.78

Table 3.3: Information of the sources of interest provided by Gaia EDR3 (Brown et al., 2021) with the line of sight extinctions ($E(B-V)$) provided by Schlafly & Finkbeiner (2011).

Source No.	Spectrum	Hα Emitter	X-ray Source	Simbad Classification (13) (Secondary Classification)	ATLAS Classification (14)
1				Star (Infrared Source)	Irregular Variable
2				Star	Irregular Variable
3	Yes (1)	Yes (2)	Yes (3, 4)	Long Period Variable (Emission-line Star, Infrared Source)	Long Period Variable
4	Yes (5)	No (5)		–	–
5		Yes (6, 12)		Emission-Line Star	Long Period Variable
6	Yes (5)	Yes (5)		–	Dubious: Probably not a variable
7	Yes (5)	Yes (5, 6, 12)		Emission-Line Star (Infrared Source)	–
8				Variable Star (Infrared Source)	Long Period Variable
9				–	–
10		Yes (6, 7, 8, 9)		Emission-Line Star (Infrared Source)	Dubious: Probably not a variable
11		Yes (6, 11)		Emission-Line Star (Infrared Source)	–
12				–	–
13		Yes (6)		Emission-Line Star (Infrared Source)	–
14		Yes (6, 10)		Emission-Line Star (Infrared Source)	–
15				Star (Infrared Source)	–
16				–	–
17				Star (Infrared Source)	–
18				–	–
19				–	–
20				–	–
21				–	–

Table 3.4: Results from the literature search of the 21 sources of interest. Empty inputs represent cases where no literature was found. Inputs consisting of "–" represent cases where the source was not present in the specified catalogue.
References: (1) Follow up by Mogawana (2021), (2) Robertson & Jordan (1989), (3) Pavlinsky et al. (2021), (4) Evans et al. (2020), (5) (Luo et al., 2015), (6) Kohoutek, L. & Wehmeyer, R. (1999), (7) MacConnell & Coyne (1983), (8) Coyne et al. (1978), (9) Stephenson & Sanduleak (1977), (10) González & González (1956a), (11) Coyne & MacConnell (1983), (12) González & González (1956b),(13) Wenger et al. (2000), (14) Heinze et al. (2018).

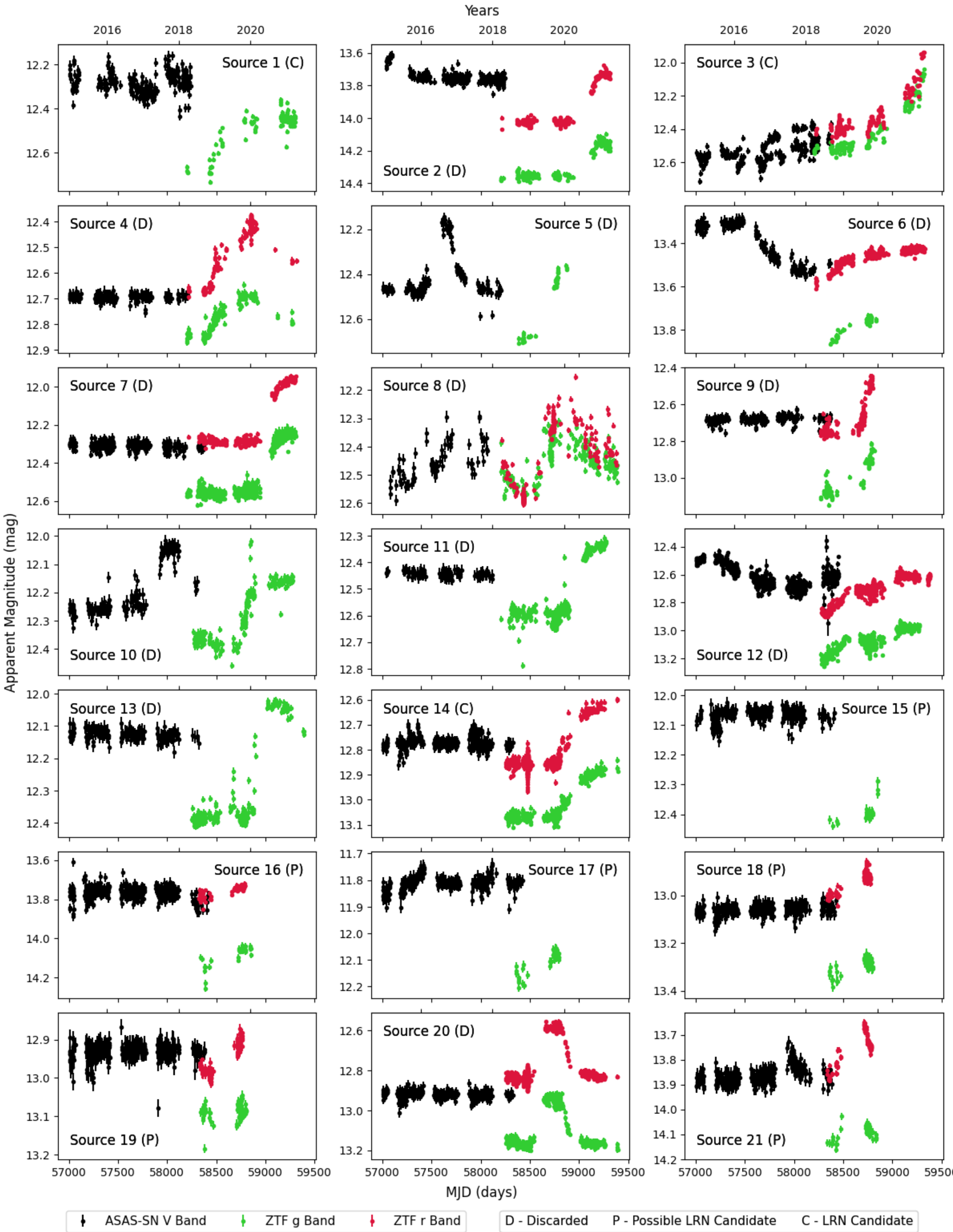


Figure 3.8: Light curves of the 21 detected slow rising transients formed from the g and r band data from ZTF (Masci et al., 2018; Bellm et al., 2018) and the V band data from ASAS-SN (Shappee et al., 2014; Jayasinghe et al., 2019).

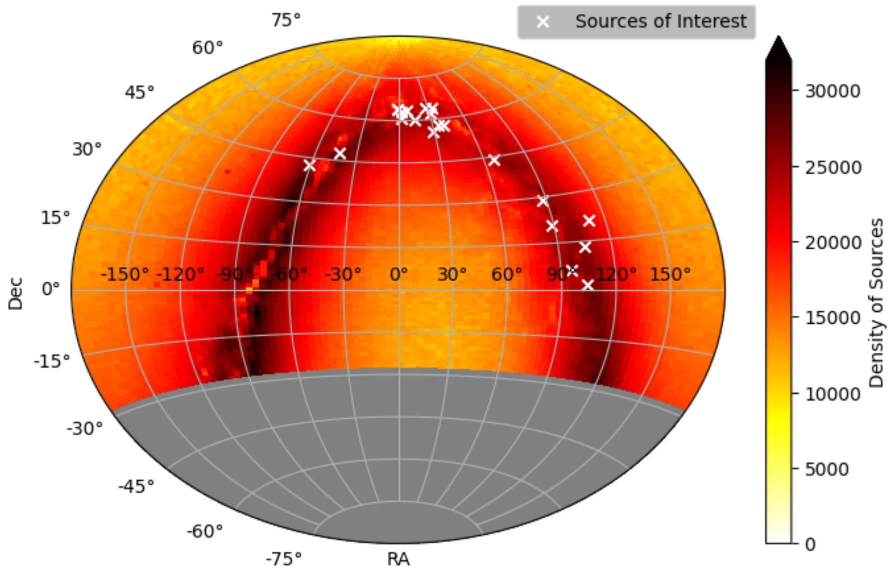


Figure 3.9: Coordinates of the 21 sources of interest identified as slow rising transients plotted on a sky map in terms of celestial coordinates, right ascension (RA) and declination (Dec). They are over plotted onto a density sky map constructed from the fitting data set. The grey region represents the region of the sky where no data was collect due to the declination range of the ZTF survey being above a declination of $\sim -30°$ (Bellm et al., 2018). The sources of interest lie in the Galactic plane, which is represented by the dense region of the density sky map.

3.5 Discussion

Overall, the method for selecting precursors from the light curves of the likely progenitors was successful, with 21 unique sources being identified as slow rising transients. This selection does not necessarily imply that they are LRN precursors, as they could, for example, be a long period variable star or have a different physical explanation for their variability. Therefore, further information about the sources must be considered, such as their spectral energy distribution in UV, optical, and infra-red, the presence of high energy emission, and their spectra.

By considering the information of the sources provided in Table 3.4 along with the light curves presented in Figure 3.8 it was deduced that sources 2, 4, 5, 6, 8, 10, 12, 13, and 20 were highly unlikely to be LRN precursors due to their light curves exhibiting long term variability. Furthermore, sources 2, 5, and 8 are classified in the Simbad catalogue (Wenger et al., 2000) and the ATLAS variable star catalogue (Heinze et al., 2018) as types of variable stars, see Table 3.4. It was also argued that sources 7, 9, and 11 were unlikely to be LRN precursors as their light curves appear to be similar in nature to the pre-peak phase of sources 4 and 13.

For sources 15, 16, 17, 18, 19, and 21 it cannot be said if they are likely precursors or not as they do not show signs of a long term variability that would suggest they are not precursors. The light curves also do not have a long enough brightening phase, due to the brightening occurring at the end of the light curve, from which the sources can be determined to be likely precursors. Furthermore, there is no information on if these sources are Hα emitters, which is used as an indication for mass transfer happening in the precursor systems. Nevertheless, it could also be possible that the sources have only recently become Hα emitters, meaning that previous observational studies would have missed them. Given this, it is worth following up on sources 15, 16, 17, 18, 19, and 21 in the coming years as they could be possible LRN precursors.

Lastly, sources 1, 3, and 14 were determined to be precursor candidates based on the information in Table 3.4 and their light curves in Figure 3.8. Firstly, the light curves of sources 1 and 14 both exhibit a magnitude increases of $\sim$ 0.2 mag over $\sim$ 2 years, while source 3 shows an increase of $\sim$ 0.6 mag over $\sim$ 6 years. All three sources have an average brightening rate of $\sim$ 0.1 mag/year which is roughly half the average rate of V1309 Sco's precursor (0.18 mag/year) and about a third of the average rate of M101 OT2015-1's precursor (0.27 mag/year). Although the rate of brightening for these three sources is much less than the two observed LRN precursors, it could be the case that they are LRN precursors that happen to have lower brightening rates. Furthermore, both sources 3 and 14 have been identified as Hα emitters (Robertson & Jordan, 1989; Kohoutek, L. & Wehmeyer, R., 1999; González & González, 1956a) and are classified as emission-line stars in Simbad (Wenger et al., 2000). This information indicates that there is recombination of ionized gas in the systems possibly due to mass transfer and outflows from L2 and L3, supporting their identification as LRN precursors. To confirm the true nature of the likely precursor sources 1, 3, and 14, spectroscopic and higher cadence photometric follow up is necessary. Chapter 4 presents a preliminary follow up and literature research of source 3, which will be studied in future works.

The method presented here has successfully identified 21 sources as slow rising transients. From these, 3 are highly likely to be precursors of LRN. Next I discuss some limitations and improvements than can be made to the method.

The first limitation arises from the r-band light curve data obtained from the ZTF time domain survey. Some of these light curves exhibit high cadence observations which experience a sharp increase and decrease in magnitude, as is present at $MJD \sim 58450$ of the light curve in Figure 3.10. As is also shown in this figure, the rolling mean and hence the sum of differences are affected by these data points. In some cases, these peculiar data points can significantly affect the sum of differences value, which can lead to the method miss-identifying the sources as slow rising transients or not. Further investigation into these observations, found that they were taken as part of ZTF's private survey consisting of specifically requested observations, as opposed to the public survey which takes observations of the entire night sky every few days. Furthermore, the majority of the sources exhibiting these observations are classified in Simbad (Wenger et al., 2000) and the ATLAS variable star catalogue (Heinze et al., 2018) as eclipsing, and detached eclipsing binaries respectively, which explains the peculiar observations seen in the light curves.

Due to the sources being likely detached binaries, it is highly unlikely that any of them are LRN precursors, as detached binaries are not currently undergoing mass transfer. Therefore, false negatives, or in other words missed precursors, are unlikely to result from the light curves that exhibit a detached eclipsing binary behaviour such as that seen in Figure 3.10. However, false positives detections of these binaries remain an issue, with $\sim 9\%$ of the ATLAS classified slow rising transient detections having a classification of distant eclipsing binary, see Table 3.2. One way in which the number of false detections could be reduced would be to use a more advanced statistical method to find the general trend of the light curve. Currently the method uses a simple moving average, and so future work may wish to investigate the use of other methods such as locally weighted scatter plot smoothing (LOWESS) (Cleveland, 1979).

Some final limitations of the method include issues with light curves that contain very little

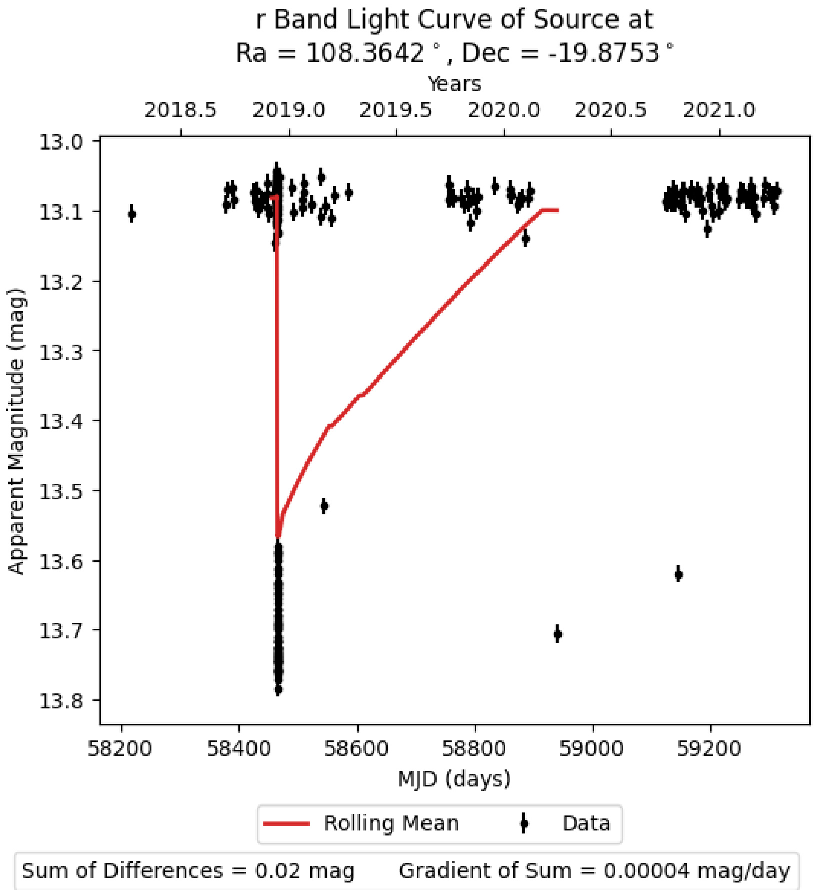


Figure 3.10: A ZTF r-band light curve that exhibits a peculiar sequence of observations at approximately MJD 58450. The vertical sequence of observations below a magnitude of 13.5 mag were taken during the same night as part of ZTF's private survey.

data, as well as issues with detecting precursors that start in the later portions of a light curve. Both of these limitations are discussed in detail in Section 3.3.2, however it is worth mentioning again the issue with the detection of precursors beginning in the later portions of a light curve. This particular issue is not a major problem with the light curve data used here as was discussed in Section 3.3.2, however, it can become a major issue if the method is applied to light curves with longer time spans. This is due to the fact that the sum of differences would be averaged over a longer time span and so the magnitude of the gradient for a precursor beginning in the later portions of the light curve would be smaller. With smaller magnitude gradients the precursors would be unlikely to meet the threshold gradient value of -2×10^{-4} mag/day and so would not be identified by the method as precursors. One way to solve this would be to increase the gradient threshold value, however, this could also increase the number of false positive detections, as is shown in Figure 3.5, and hence may not be the best solution. Another solution could be to take 3 year samples of longer time span light curves and then apply the sum of differences method, although this leaves the question of how to take the 3 year samples and if this could cause more false positives and negatives. It is possible the best solution to this problem may be to use an entirely different method to detect slow rising transients in longer time span light curves.

CHAPTER 4

V520 Mon

Presented in this chapter are the preliminary results of a more detailed study of source 3, here after referred to as V520 Mon, which is one of the sources identified as a LRN precursor candidate in Chapter 3. The investigation provided here consists of reviewing the existing literature and archival data of the source and searching for a periodic short term variability in its light curve.

4.1 Short Term Variability

V520 Mon's light curves, presented in Figure 3.8, show signs of a short term variability of about a few hundred days. To further investigate this, a short term period was searched for by using three different methods: Lomb-Scargle periodogram (Lomb, 1976; Scargle, 1982), box least squares periodogram (Kovcs et al., 2002), and conditional entropy periodogram (Graham et al., 2013). The aim of fitting a period to V520 Mon's light curves was to see if they exhibited a period that could be associated with an eclipsing binary, as this would support the idea of it being a LRN precursor. First the general increasing brightness trend was removed from the ZTF g and r band light curves, shown in Figure 3.8. This was achieved by fitting a third degree polynomial to the light curves and calculating the residuals, Next, the three period fitting methods were applied to the residuals using the the period range set by the Nyquist period and a third of the light curve time span. The Nyquist period was used as the lower limit as it represents the lowest theoretical period that can be obtained from the sampled data due to its sample rate. A third of the light curve time span was used as the upper period limit as I wanted to ensure that at least 3 periods were present in the available data. Presented in Figure 4.1 are the resulting periodograms, with the best fitting periods displayed in Table 4.1. The best fitting periods are those with the largest power in the Lomb-Scargle and box least squares periodograms, and those with the lowest conditional entropy in the conditional entropy periodogram. Produced from these periods were phase folded light curves which are shown in Figure 4.2.

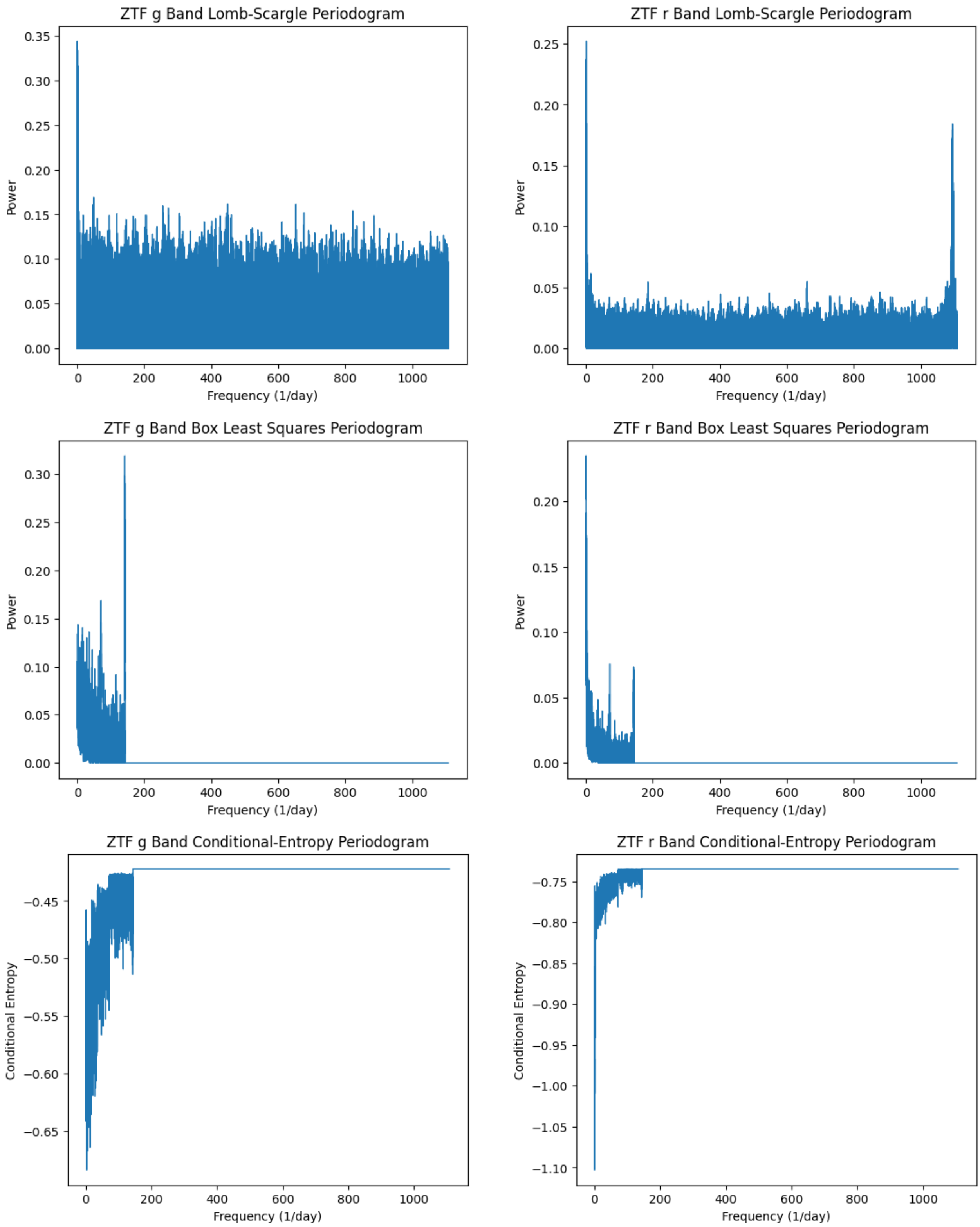


Figure 4.1: Periodograms of V520 Mon's ZTF g (left) and r (right) bands using the three different methods: Lomb-Scargle (top), box least Squares (middle), and conditional entropy (bottom). The frequency range of the periodograms is based on the Nyquist frequency and a third of the time span of the light curves. The best fitting frequencies are those with the highest power or lowest conditional entropy.

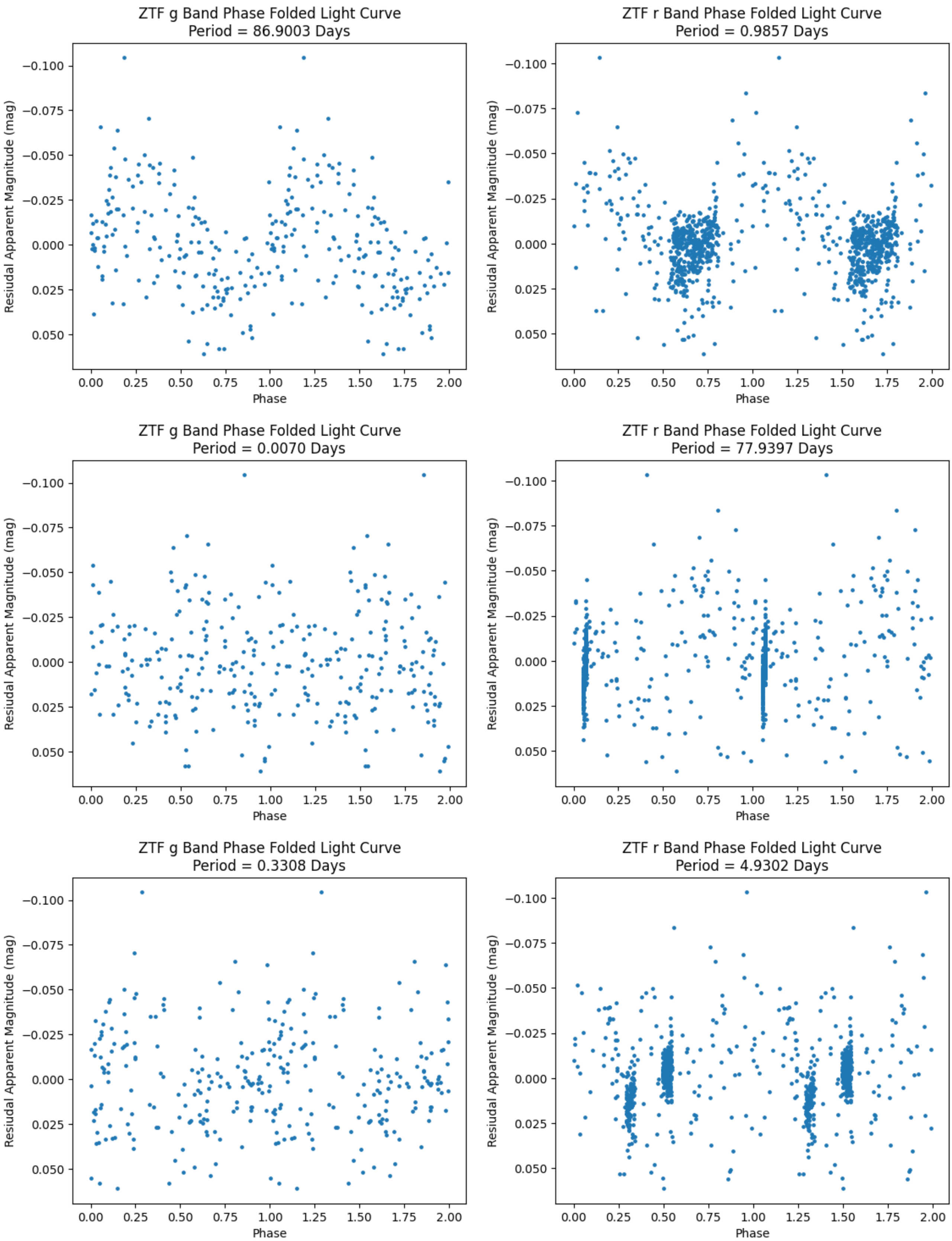


Figure 4.2: V520 Mon's ZTF g and r band light curves folded on the best fitting periods found using the Lomb-Scargle, box least squares, and conditional entropy periodograms.

	Period (days)	
	ZTF g-Band	ZTF r-Band
Lomb-Scargle	86.90	0.99
Box Least Square	0.007	77.94
Conditional Entropy	0.33	4.93

Table 4.1: Periods best fitting V520 Mon's ZTF g and r band light curves selected using the three different periodogram methods: Lomb-Scargle, box least squares, and conditional entropy.

From the periodograms shown in Figure 4.1 it can be seen that each one has a clear indication of which frequency best fits the light curves. However, the best fit frequencies do not agree with each other, as shown by the the best fit periods in Table 4.1 which vary between 0.007 days and 86.90 days. Furthermore, from the phase folded light curves presented in Figure 4.2 it is clear that none of the best fit periods provided a clean phase folded light curve, although the top two plots with periods of 86.90 and 0.99 days show the best agreement with a periodical signal. It is likely that the 86.90 day period is a harmonic of the much smaller true period, which is at most approximately 0.99 days. It is possible that the true short term period of V520 Mon is less than one day, however, due to the sample rate of the ZTF data, which is on average about twice per night at best, it is difficult to search for periods that are less than one day. Therefore, to be able to further investigate V520 Mon's apparent short term period there is the need for higher cadence observations.

From the analysis of V520 Mon's apparent short term period presented here, it cannot be said for certain that the system does exhibit a short term period which could be caused by an eclipsing binary. Despite not finding a short term period, it is possible that V520 Mon has a period that could not be found due to the limitation imposed by the cadence of the data.

4.2 Existing Literature

Provided in Table 3.4 is a summary of V520 Mon's (source 3) existing literature, which is further discussed in this section.

Firstly, V520 Mon has been previous identified as a Hα emitter by Robertson & Jordan (1989) who conducted an objective prism survey of fields along the celestial equator. Furthermore, the follow-up spectrum of V520 Mon obtained in October 2021 by Mogawana (2021) (see Figure 4.3), shows strong Hα emission. On top of this, the spectrum is complex and likely a composite of two different spectral types, meaning that V520 Mon is a likely binary system. The existence of Hα emission in the spectrum is a signature of the presence of recombining hydrogen gas in the system. Given that V520 Mon has been identified as a Hertzsprung gap star and is likely a binary system, the presence of hydrogen gas is likely to be the result of outflows from the system created by unstable mass transfer during RLOF. Depending on the mass-transfer rate, the formation and ejection of a CE could occur in a matter of years, which would support the idea that V520 Mon is a potential LRN precursor.

Another interesting observation of V520 Mon is that it has been observed to emit X-ray radiation in the energy range 1 - 12 keV (Evans et al., 2020; Pavlinsky et al., 2021). Given that V520 Mon is likely a binary and it produces X-rays, it is highly likely that the source of the X-rays is caused by the transfer of mass from a donor star onto a compact object such as a black hole, neutron star, or a white dwarf. The mass transfer could be caused by stellar wind

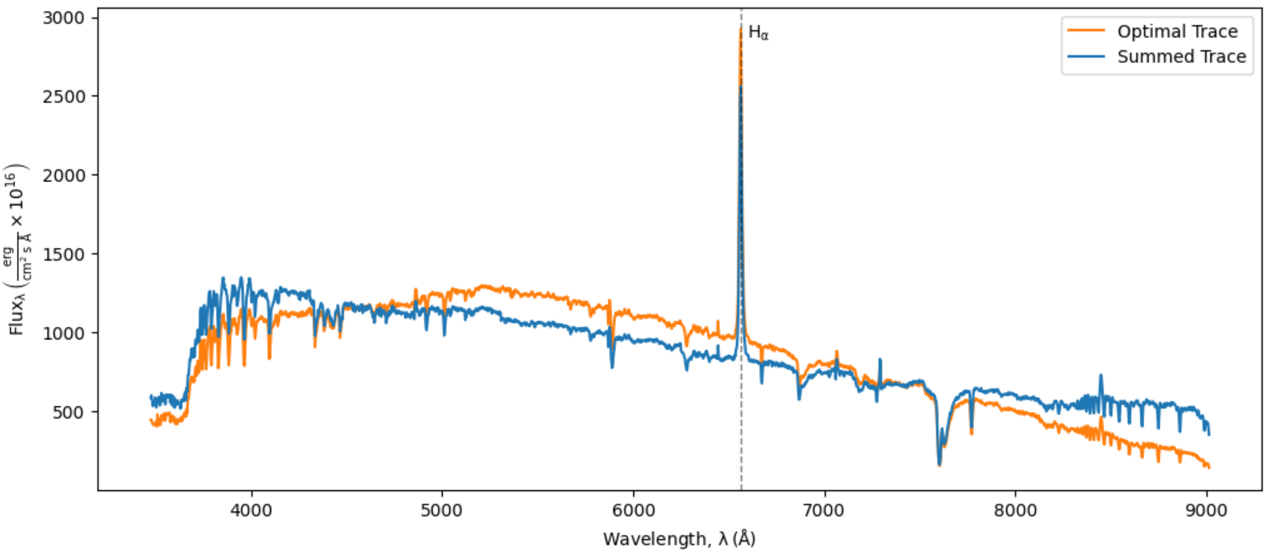


Figure 4.3: Flux calibrated spectrum of V520 Mon constructed from the data obtained in October 2021 by Mogawana (2021). There is the presence of a strong Balmer-alpha (Hα) emission.

or more likely RLOF, which is supported by the recombination of hydrogen gas in the system. Pavlinsky et al. (2021) briefly suggest that V520 Mon could be a high mass X-ray binary or a symbiotic binary.

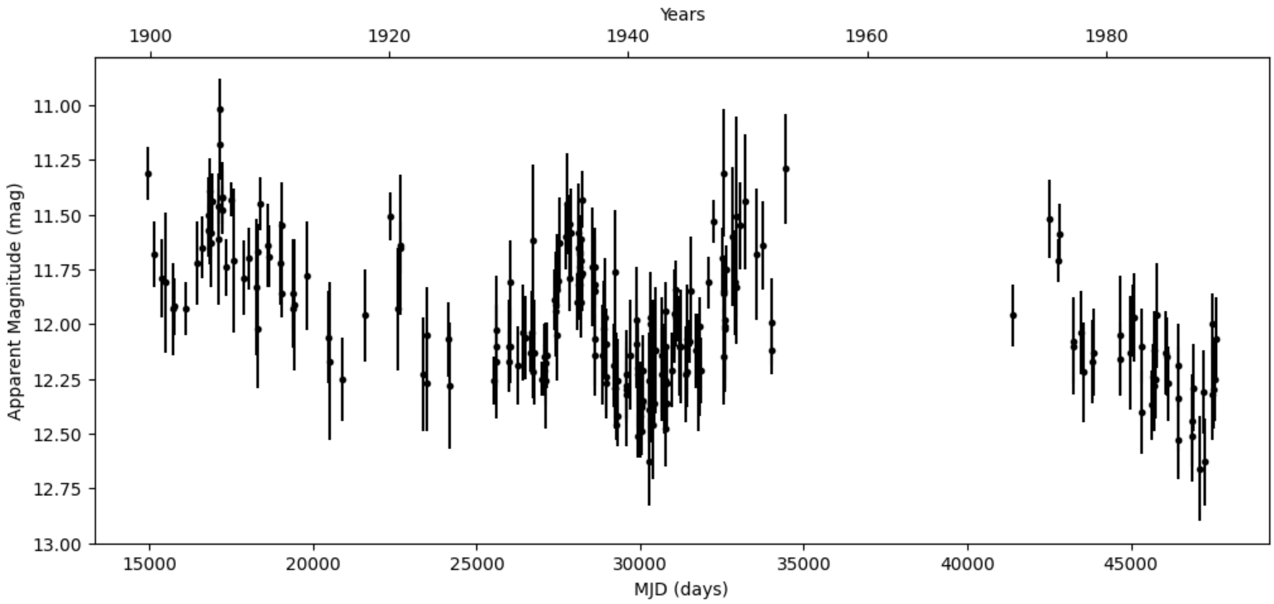


Figure 4.4: V520 Mon's historical light curve from the Digital Access to a Sky Century @ Harvard (Grindlay et al., 2011). Present in the light curve is a long term variability with a visual period estimation of 15 years.

Lastly, V520 Mon is classified as a long period variable in both Simbad (Wenger et al., 2000) and the ATLAS variable star catalogue (Heinze et al., 2018), the latter with a period of

761 days. However, based on the ZTF and ASAS-SN light curves shown in Figure 3.8 this period is unlikely to be the correct long term period of the source. Furthermore, the historical light curve of V520 Mon from the Digital Access to a Sky Century @ Harvard (DASCH) (Grindlay et al., 2011), shown in Figure 4.4, also shows that V520 Mon has a much longer period than 761 days, with a visual period estimation of about 15 years. Due to the long term variability of the source shown by the historical data, it is likely that the slow brightening seen in the ZTF light curves is not a LRN precursor but is instead a part of the source's variability. This does however leave the question of what is causing the long term variability.

4.3 Conclusion of V520 Mon

Through the analysis of V520 Mon light curves, current literature, and its spectrum, there is sufficient evidence to believe that it is a binary system. This evidence includes a complicated spectrum that likely consists of multiple sources, as well as the presence of Hα emission which is evidence of hydrogen gas in the system. This gas could be the result of mass transfer between binary components due to RLOF, which is also supported by the X-ray emission, as this could have also resulted from the transfer of mass onto a compact object such as a white dwarf neutron star, or a black hole.

In addition to V520 Mon being a likely binary system, the historical light curve spanning approximately 100 years shows signs of a period of approximately 15 years. Although the cause of this periodicity is unclear, it is possible that the slow rising brightness seen in the ZTF g and r band light curves is just part of this cyclical variation, and is thus not a LRN precursor. Further follow up observations and investigations into V520 Mon are needed in order to determine the true nature of this system.

CHAPTER 5

Conclusions

The method for identifying Galactic LRN precursors was presented in Chapters 2 and 3. The application of this method to Gaia EDR3 data allowed me to identify $\sim 1.6 \times 10^6$ likely YSG as possible progenitors of LRNe. From the ZTF light curves of these likely progenitor sources, 21 of them were found to exhibit a slow rising trend in brightness. On further investigation of the 21 sources, only three were found to be LRN progenitor candidates, requiring further follow-up studies. Although ruled out as LRN precursors, the nature of the variability in the remaining 18 sources is not entirely clear, and therefore it may also be of interest for future works. One of the three potential LRN precursor candidates, V520 Mon (source 3), was further investigated using archival data found in literature and additional follow-up observations. This preliminary analysis was consistent with the source being a binary undergoing RLOF. However, it was not clear if the RLOF will result in the formation of a CE on dynamical timescales, and so it cannot currently be said if V520 Mon would become a LRN. Additionally, the unknown nature of V520 Mon's approximate 15 year period seen in the historical light curve introduces additional complexity to its interpretation, and is left for future works to investigate.

The methods presented in Chapters 2 and 3 resulted in the identification of three potential LRN precursor candidates. Summarised here are the limitations and improvements to these methods which are further discussed in Sections 2.5 and 3.5.

The first limitation of the method is the number of false positive detections of slow rising transients that are caused by ZTF's private survey observations of detached eclipsing binaries, which account for $\sim 9\%$ of the identified slow rising transients. It is possible that the use of a moving average to find the general trend of the light curves may not be sufficient. Therefore, a possible improvement to the method may be to use a more advanced statistical method, such as LOWESS (Cleveland, 1979), to find the general trend of the light curves. This may potentially reduce the number of false positives, and is left for a future work to explore fully.

The second limitation of the method is that LRN precursors beginning in the later part of the light curve may not be detected. This is due to the method using the light curve's net change in magnitude averaged over its time span as the rate of brightening. Therefore, the magnitude of this value is less than the magnitude of the true rate of brightening of the source, with the reduction in the value increasing with the start time of the brightening. This means that sources that should be identified as slow rising transients may not meet the threshold value for being selected. This limitation of the method is a minor issue with the data used here, however, if the time span of the data is much longer than ~ 3 years then the limitation becomes more prevalent. One proposed solution to this would be to take 3 year samples of longer time span light curves, although this may be impractical. As such it is likely that a different method would be needed to search for slow rising transients in longer time span light curves.

Other improvements to the method are aimed at increasing the number of sources that are considered. The first of these improvements is to expand the search of LRN precursors to all declinations. In the work presented here only sources above a declination of $-30°$ were considered due to the declination range of the ZTF time domain survey (Dec $\gtrsim -30$) (Bellm et al., 2018). Hence, to be able to consider all declinations, another all sky time domain survey would be required in addition to ZTF. The second improvement to increase the number of sources considered is to reduce the parallax and flux errors applied to the fitting data set. This would increase the number of sources in the data set, although it would also decrease the uncertainty in the positions of these sources on the HRD, hence increasing the number of falsely identified Hertzsprung gap sources. However, this negative effect could be reduced by switching to the use of Gaia DR3, which is set to be released in 2022 (DPAC, 2021b). Gaia DR3 will contain extinction values for each source and so these could be used to obtain the intrinsic brightness and colour of the source. This would thus reduce the uncertainty in the positions of sources on the HRD and hence the number of falsely identified Hertzsprung gap stars.

To conclude, the presented method to identify LRN precursors has provided three high confidence candidates for LRN precursors that will become object of study in future work. Furthermore, the suggested improvements to the method could potentially enhance the number of candidates of potential LRN precursors. If these candidates are truly LRNe precursors then they will provide invaluable opportunities to investigate the astrophysical processes involved in the formation and ejection of the common envelope in binary systems. The detailed study of such systems will shed light on one of the least understood phases of binary stellar evolution (Izzard et al., 2011). Hence, the work presented here has potential to greatly advance our understanding of both LRNe and binary stellar evolution.

Bibliography

Alerts G. S., 2020, Gaia20bsc, Available at `http://gsaweb.ast.cam.ac.uk/alerts/alert/Gaia20bsc/` (03/11/21) 3.3.1

Bailer-Jones C. A. L., Rybizki J., Fouesneau M., Demleitner M., Andrae R., 2021, The Astronomical Journal, 161, 147 2.2

Bellm E. C., et al., 2018, Publications of the Astronomical Society of the Pacific, 131, 018002 (document), 1.3, 2.2, 2.3, 2.7, 2.5, 3.2, 3.3.1, 3.3, 3.3.1, 3.4, 3.5, 3.3.2, 3.4, 3.1, 3.4, 3.8, 3.9, 5

Blagorodnova N., et al., 2017, The Astrophysical Journal, 834, 107 1, 1.2, 1.2, 2.1, 2.5, 3.1

Blagorodnova N., et al., 2021, Astronomy & Astrophysics, 653, A134 1.2, 2.5

Brown A. G. A., et al., 2018, Astronomy & Astrophysics, 616, A1 (document), 1.1, 2.2, 2.1, 2.1, 2.2

Brown A. G. A., et al., 2021, Astronomy & Astrophysics, 650, C3 (document), 2.2, 2.2, 2.2, 2.3, 3.3

Cao Y., Kasliwal M. M., Chen G., Arcavi I., 2015, The Astronomer's Telegram, 7070, 1 1.2

Choi J., Dotter A., Conroy C., Cantiello M., Paxton B., Johnson B. D., 2016, ApJ, 823, 102 (document), 2.2, 2.1, 2.6, 2.3

Cleveland W. S., 1979, Journal of the American Statistical Association, 74, 829 3.5, 5

Connor R., Cardillo F. A., Moss R., Rabitti F., 2013, in Brisaboa N., Pedreira O., Zezula P., eds, Similarity Search and Applications. Springer Berlin Heidelberg, Berlin, Heidelberg, pp 163–168 2.3

Coyne G. V., MacConnell D. J., 1983, Vatican Observatory Publications, 1, 73 (document), 3.4

Coyne G. V., Wisniewski W., Otten L. B., 1978, Vatican Observatory Publications, 1, 257 (document), 3.4

DPAC 2021a, Gaia Archive, Available at `https://gea.esac.esa.int/archive/` (25/11/21) (document), 1.1, 2.2, 2.2

DPAC 2021b, Gaia Data release 3, Available at `https://www.cosmos.esa.int/web/gaia/data-release-3` (01/12/21) 2.5, 5

Darwin G. H., 1879, Proceedings of the Royal Society of London, 29, 168 1.1.2

Dotter A., 2016, ApJS, 222, 8 (document), 2.2, 2.1, 2.6, 2.3

Drout M. R., Massey P., Meynet G., Tokarz S., Caldwell N., 2009, The Astrophysical Journal, 703, 441460 2.1

Eggleton P., 2006, Rapid non-conservative processes. Cambridge University Press, p. 209230, doi:10.1017/CBO9780511536205.006 1.1.2

Evans P. A., et al., 2020, ApJS, 247, 54 (document), 3.4, 4.2

Gaia Collaboration et al., 2016, Astronomy & Astrophysics, 595, A1 (document), 1.1, 1.3, 2.2, 2.1

Gaia Collaboration et al., 2018, A&A, 616, A10 (document), 1.1

Gimnez ., Guinan E., Niarchos P., Rucinski S., 2006, Astrophysics and space science, 304, 3 1

González G., González G., 1956a, Boletin de los Observatorios Tonantzintla y Tacubaya, 2, 16 (document), 3.4, 3.5

González G., González G., 1956b, Boletin de los Observatorios Tonantzintla y Tacubaya, 2, 19 (document), 3.4

Goranskij V. P., et al., 2016, Astrophysical Bulletin, 71, 82 1.2, 1.2, 3.1

Graham M. J., Drake A. J., Djorgovski S. G., Mahabal A. A., Donalek C., 2013, Monthly Notices of the Royal Astronomical Society, 434, 2629 4.1

Grindlay J., Tang S., Los E., Servillat M., 2011, Proceedings of the International Astronomical Union, 7, 2934 (document), 4.4, 4.2

Heinze A. N., et al., 2018, The Astronomical Journal, 156, 241 (document), 3.1, 3.4, 3.4, 3.2, 3.4, 3.5, 4.2

Ivanova N., et al., 2013a, A&A Rev., 21, 59 1.1.2

Ivanova N., Justham S., Avendano Nandez J. L., Lombardi J. C., 2013b, Science, 339, 433 3.1

Izzard R. G., Hall P. D., Tauris T. M., Tout C. A., 2011, Proceedings of the International Astronomical Union, 7, 95102 1.1.2, 1.2, 5

Jayasinghe T., et al., 2019, MNRAS, 485, 961 (document), 3.4, 3.8

Karttunen H., Krger P., Oja H., Poutanen M., Donner K. J., 2017, Stellar Structure. Springer, Berlin, Heidelberg, doi:https://doi.org/10.1007/978-3-662-53045-0 1.1.1

Kippenhahn R., Weigert A., Weiss A., 2012, Evolution Through Helium Burning: Intermediate-Mass Stars. Springer Berlin Heidelberg, Berlin, Heidelberg, pp 367–384, doi:10.1007/978-3-642-30304-3_31, https://doi.org/10.1007/978-3-642-30304-3_31 1.1.1, 2.1

Kochanek C. S., Adams S. M., Belczynski K., 2014, MNRAS, 443, 1319 1.3

Kohoutek, L. Wehmeyer, R. 1999, Astron. Astrophys. Suppl. Ser., 134, 255 (document), 3.4, 3.5

Kostrzewa-Rutkowska Z., et al., 2018, Monthly Notices of the Royal Astronomical Society, 481, 307 (document), 3.3, 3.3.1, 3.3.1, 3.1, 3.3.1

Kovcs G., Zucker S., Mazeh T., 2002, Astronomy & Astrophysics, 391, 369377 4.1

Livio M., Soker N., 1988, ApJ, 329, 764 1.1.2

Lomb N. R., 1976, Ap&SS, 39, 447 4.1

Luo A.-L., et al., 2015, Research in Astronomy and Astrophysics, 15, 10951124 (document), 3.4

MacConnell D. J., Coyne G. V., 1983, Vatican Observatory Publications, 2, 63 (document), 3.4

MacLeod M., Macias P., Ramirez-Ruiz E., Grindlay J., Batta A., Montes G., 2017, The Astrophysical Journal, 835, 282 2.5

Maíz Apellániz, J. Weiler, M. 2018, A&A, 619, A180 2.2

Masci F. J., et al., 2018, Publications of the Astronomical Society of the Pacific, 131, 018003 (document), 1.3, 2.2, 2.3, 3.2, 3.3.1, 3.3, 3.3.1, 3.4, 3.5, 3.3.2, 3.4, 3.1, 3.4, 3.8

Mason E., Diaz M., Williams R. E., Preston G., Bensby T., 2010, Astronomy & Astrophysics, 516, A108 1.2

McLachlan G. J., Lee S. X., Rathnayake S. I., 2019, Annual Review of Statistics and Its Application, 6, 355 2.1

Moe M., Di Stefano R., 2017, ApJS, 230, 15 1

Mogawana O., 2021, V520 Mon Spectrum (document), 4.2, 4.3

Nakano S., et al., 2008, IAU Circ., 8972, 1 1.2, 1.2

Nelemans G., 2005, Astronomical Society of the Pacific Conference Series, 330, 27 1.1.2

OGLE 2021, The Optical Gravitational Lensing Experiment, Available at http://ogle.astrouw.edu.pl/ (06/07/21) (document), 1.4, 1.2, 3.3.1, 3.2

Paczynski B., 1976, in Eggleton P., Mitton S., Whelan J., eds, IAU Symposium Vol. 73, Structure and Evolution of Close Binary Systems. p. 75, https://ui.adsabs.harvard.edu/abs/1976IAUS...73...75P 1.1.2, 1.1.2

Pavlinsky M., et al., 2021, SRG/ART-XC all-sky X-ray survey: Catalog of sources detected during the first year (arXiv:2107.05879) (document), 3.4, 4.2

Paxton B., Bildsten L., Dotter A., Herwig F., Lesaffre P., Timmes F., 2011, ApJS, 192, 3 (document), 2.2, 2.1, 2.6, 2.3

Paxton B., et al., 2013, ApJS, 208, 4 (document), 2.2, 2.1, 2.6, 2.3

Paxton B., et al., 2015, ApJS, 220, 15 (document), 2.2, 2.1, 2.6, 2.3

Paxton B., et al., 2018, ApJS, 234, 34 (document), 2.2, 2.1, 2.6, 2.3

Pearson K., 1894, Philosophical Transactions of the Royal Society of London Series A, 185, 71 2.1

Pedregosa F., et al., 2011, Journal of Machine Learning Research, 12, 2825 2.3, 2.3

Pejcha O., 2014, ApJ, 788, 22 1.1.2, 1.2, 1.2

Pejcha O., Metzger B. D., Tomida K., 2016, MNRAS, 455, 4351 1.2

Postnov K. A., Yungelson L. R., 2014, Living Reviews in Relativity, 17 1, 1.1.2

Rich R. M., Mould J., Picard A., Frogel J. A., Davies R., 1989, ApJL, 341, L51 1.2

Rigault M., 2018, ztfquery, a python tool to access ZTF data, doi:10.5281/zenodo.1345222, https://doi.org/10.5281/zenodo.1345222 3.2

Robertson T. H., Jordan T. M., 1989, AJ, 98, 1354 (document), 3.4, 3.5, 4.2

Rousseeuw P. J., 1987, Journal of Computational and Applied Mathematics, 20, 53 2.3

Sana H., et al., 2012, Science, 337, 444 1

Scargle J. D., 1982, ApJ, 263, 835 4.1

Schlafly E. F., Finkbeiner D. P., 2011, ApJ, 737, 103 (document), 3.3

Schwarz G., 1978, The Annals of Statistics, 6, 461 2.3

Shappee B. J., et al., 2014, ApJ, 788, 48 (document), 3.4, 3.8

Stephenson C. B., Sanduleak N., 1977, ApJS, 33, 459 (document), 3.4

Tauris T. M., van den Heuvel E. P. J., 2006, Formation and evolution of compact stellar X-ray sources. pp 623–665 (document), 1.2, 1.3

Tylenda R., et al., 2011, Astronomy & Astrophysics, 528, A114 (document), 1, 1.2, 1.4, 1.2, 3.1, 3.3.1, 3.2

Udalski A., 2003, Acta Astronomica, 53, 291 (document), 3.3.1, 3.2

Udalski s., Soszynski I., Szymanski M., Kubiak M., Pietrzynski G., Wozniak P., Zebrun K., 2003, Acta Astronomica, 53 1.2

Weiler, Michael 2018, A&A, 617, A138 2.2

Wenger M., et al., 2000, A&AS, 143, 9 (document), 3.4, 3.4, 3.5, 4.2

Wevers T., et al., 2017, Monthly Notices of the Royal Astronomical Society, 473, 3854 3.3.1, 3.3.1

ZTF team 2021, Zwicky Transient Facility Public Data Release 7, Available at `https://www.ztf.caltech.edu/page/dr7#14` (02/11/21) 3.2, 3.4

von Neumann J., 1941, The Annals of Mathematical Statistics, 12, 367 3.3, 3.3.1

APPENDIX A

Literature Review

Luminous Red Novae:
Recent Observational and Theoretical Studies, and Future Pre-Outburst Detections

URN: 6526653[1]

Department of Physics, University of Surrey, Guildford GU2 7XH, United Kingdom

(Dated: 1 November 2021)

In this literature review I discuss the literature on recent luminous red novae (LRNe), which are transient events thought to be caused by common envelope ejections and stellar mergers. I explore both the observational studies and the theoretical modelling that has been conducted on recent LRNe. From the literature discussed, it is evident that there is much more to learn about LRNe and the processes that cause them. However, the literature also shows that the current method for studying LRNe precursors is limited by the archival, coincidental data used. Generally the archival data has a small number of observational epochs, as well as a small range in the available filter bands. Furthermore, the archival data consists of very few of the available techniques that could be used to investigate LRNe. One way in which we can avoid these limitations is by taking real time observations of the precursor. This would allow for a wide range of filter bands to be used, as well as all of the available techniques. However, to perform real time observations of the LRNe precursor, it first has to be detected, which there is no known method for currently. As such, a methodology that will be able to detect LRNe precursors is proposed.

I. INTRODUCTION

Luminous red novae (LRNe) are a class of intermediate luminosity transients, due to their peak brightness being between that of novae (–6 to –8 mag) and supernovae (< –14 mag)[1]. They are thought to be caused by stellar mergers[2–4] and common envelope ejections[1], which are ejections of mass, resulting in unique characteristics seen in observations. It is these unique characteristics that led to the new class of nova being termed in 2007[5].

The field of LRNe, although not known at the time, began with the detection of a nova-like event in 1988[6–8]. This nova-like event was an unexpected discovery made by Rich *et al.* [6], who were studying luminous stars in the bulge of the galaxy M31. In their study they compared images of M31's bulge that were taken several years apart, and in doing so they noticed that in the latter image a bright star had appeared[6]. The bright star was classified by Rich *et al.* [6] as a red variable. The same bright star was also independently discovered by Tomaney and Shafter [7], who were searching for classical novae in M31. The observations made by Rich *et al.* [6], Tomaney and Shafter [7], and Bryan and Royer [8] indicated that this event was nova-like, however, it was not like any known transient class such as supernovae or classical novae.

Six years after the nova-like event in M31, a second peculiar nova, Nova Sagittarii 1994 1, was discovered by Hayashi, Yamamoto, and Hirosawa [9]. The observations[9,10] showed that it was unlike any known class of novae, with Martini *et al.* [11] comparing it to the red variable seen in 1988[6–8]. Although Martini *et al.* [11] suggested that the closest match to Nova Sagittatrii 1994 1 was the red variable in M31, they also discussed the significant differences between the two events, one of which was the difference between the peak brightness of them.

The next major observations that brought the field of LRNe into existence didn't come until 2002 and 2006, when the peculiar outbursts V838 Mon[12] and M85OT2006-1[5] were discovered respectively. V838 Mon was described as having a mysterious nature by Munari *et al.* [12], who also concluded that the event was similar to the red variable in M31 and Nova Sagittarii 1994 1, stating that they could be a new class of astronomical objects. Furthermore, the outburst M85OT2006-1 provided yet another example of an event that was nova-like, however not of a know class[5]. Hence, Kulkarni *et al.* [5] coined the term luminous red novae to describe these peculiar nova-like events.

Since the coining of the term LRNe, the population of this class of astronomical objects has increased, and many theories behind the driving process of them have been suggested. However, the processes that cause LRNe are still not fully understood. The gaps in the knowledge are discussed in this review, which explores the observational and theoretical advancements made recently in the field of LRNe. More specifically, I review the current literature on a handful of recent LRNe, discussing the observations and the theoretical models that have been produced to explain them.

The text is organised as follows: in section II I provide information for the non experts, which includes the relevant background theory on stellar and binary evolution for LRNe. Then in section III I discuss the state of the art observations and theoretical work of recent LRNe, as well as highlighting the gaps within the current literature. I then provide details in section IV of my own research and how it will provide a solution to the gap in the current literature. Finally, I summarise the current literature and my research in section V.

II. BACKGROUND THEORY

In the following section I provide theory on stellar and binary evolution that is relevant for understanding the the later sections within this literature review.

A. Stellar Evolution

The evolution of a single star has many phases throughout its lifetime, starting from a cloud of dust and gas, to ending their lives in a variety of manners. Here I only provide information on the phases of stellar evolution that are relevant for understanding LRNe, which are the main sequence, and the post main sequence up until the red giant branch.

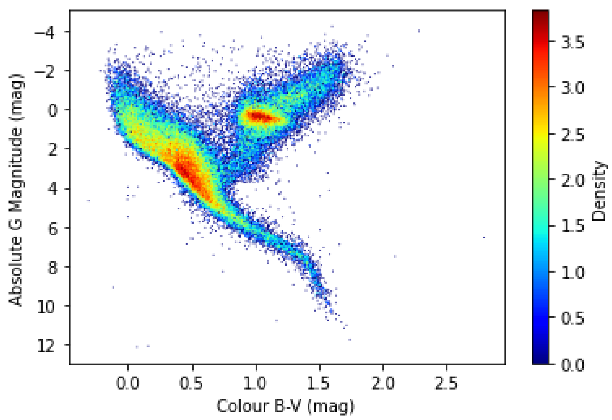


FIG. 1. Observed colour magnitude diagram constructed from non-variable, nearby sources in the Gaia DR1 and Hipparcos catalogues, as part of my project training. This colour magnitude diagram consists of the absolute G magnitudes calculated from the Gaia DR1 data, and the colour (B–V) from the Hipparcos catalogue. The two dense regions are the main sequence (upper left to bottom right) and the red giant branch (upper right).

The main sequence is the phase of stellar evolution where stars generally spend the majority of their time. The main sequence can be seen as the upper left to bottom right dense region in the colour magnitude diagram, also known as Hertzsprung Russell diagram, in figure 1. This figure is constructed from non-variable (colour and G magnitude errors <5%), nearby sources (parallax error < 20%) within the Gaia DR1 and Hipparcos catalogues, which contain data of stars within the Milky Way. During the main sequence phase stars burn hydrogen in their cores into helium through fusion processes. To stop the star from collapsing in on itself due to its own gravity, the gas pressure applies an opposing force. This balance of gas pressure and gravitational force is known as hydrostatic equilibrium, and is a self regulating process. It is self regulating because if the gas pressure decreases then the gravitational force will cause the star to contract. This contraction increases the temperature of the core which results in a higher fusion rate and thus a further increase in temperature. This in turn raises the gas pressure, causing the star to return to equilibrium.

When a main sequence star exhausts its supply of hydrogen in its core, it enters the post main sequence phase, which leads to the giant phase. Due to the pressure and temperature within the core not being high enough, helium burning within the core cannot start. This results in a loss of core temperature, gas pressure, and hydrostatic equilibrium, resulting in the star contracting. This contraction heats the shell (or envelope) of hydrogen surrounding the core, eventually causing hydrogen fusion to occur within the shell. The hydrogen shell burning produces more energy than hydrogen burning in the core, and so there is more gas pressure than previously, when the star was in the main sequence. This causes the star to establish a new equilibrium state by expanding to a larger radius than when it was on the main sequence. This evolution of the star results in it moving from the main sequence towards the red giant branch. In a Hertzsprung Russell diagram, as shown in figure 1, the star will move from the left dense region towards the right dense region. The evolution of the star can also be represented by evolution tracks, which are formed from theoretical models and show the position of a star of a given mass in a Hertzsprung Russell diagram at different stages.

B. Binary Evolution

The evolution of binary systems can result in various scenarios, such as neutron star mergers or peculiar types of supernovae, however here I will only present the relevant information needed to understand the literature on LRNe. This includes information about binary systems where both components of the system are main sequence stars, or where one of the components is a main sequence star and the other is an evolving main sequence star.

Binary systems start out as detached binaries, meaning that the two main sequence components are not in contact, and are therefore in hydrostatic equilibrium and within their Roche lobe. A Roche lobe is the region around a star in a binary system within which material is gravitationally bound to the star. The Roche lobe can be seen as the darker equipotential curve in figure 2, which shows the gravitational equipotential lines in the orbital plane of a binary system with masses M_1 and M_2. Also shown on this figure are the five Lagrange points (L_{1-5}), which are positions where the forces due to the two masses and the centrifugal force are all balanced. There are many ways in which the detached binary can evolve, although not all of them are relevant for the discussion on LRNe. The evolution path that is relevant is the formation of a common envelope, which is where the secondary component (smaller star) is engulfed by the primary's (larger star) envelope[14]. There

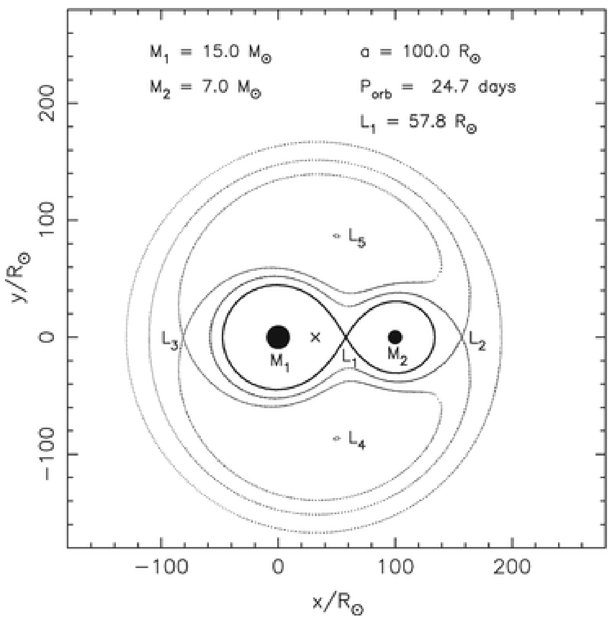


FIG. 2. Gravitational equipotential lines in the orbital plane of a binary system consisting of masses M_1 and M_2, as constructed by Tauris and van den Heuvel [13]. Indicated on this figure is the centre of mass by the cross between the two masses, as well as the Roche lobe of the binary, which is represented by the darker equipotnetial line. Also shown on this figure are the five Lagrange points denoted by L_{1-5}.

are two mechanisms that are are thought to lead to the formation of a common envelope: Darwin instability[15], and dynamically unstable mass transfer[3]. The formation of a common envelope has been summarised in figure 3, and further details of the two formation mechanisms are discussed next. The outcomes of the common envelope phase applicable to LRNe are left for discussion in section III.

The first mechanism for common envelope formation is Darwin instability, which is an instability of the binary system due to the spin angular momentum of the primary component and the angular momentum of the orbit[15]. This instability arises from the primary star evolving and consequently expanding, as discussed in section II A, which causes the star's rotational velocity to slow through conservation of angular momentum. However, the rotation of the star wants be tidally synchronised with the orbit. This causes angular momentum transfer between the orbit and the star, spinning up the primary star. Synchronisation is reached if the spin angular momentum of the primary (J_1) is less than a third of the orbital angular momentum, J_{orb}, ($J_1 < \frac{1}{3} J_{orb}$). If this is not the case then the extraction of angular momentum from the orbit causes a runaway angular momentum transfer scenario. This occurs as the decrease in the orbital angular momentum causes the orbit to shrink, which increases the orbital frequency. The primary star then needs more angular momentum to stay tidally synchronised with the orbit, and so more angular momentum is transferred from the orbit. This eventually leads to the orbit shrinking such that the two binary components form a common envelope, with the secondary being engulfed by the primary.

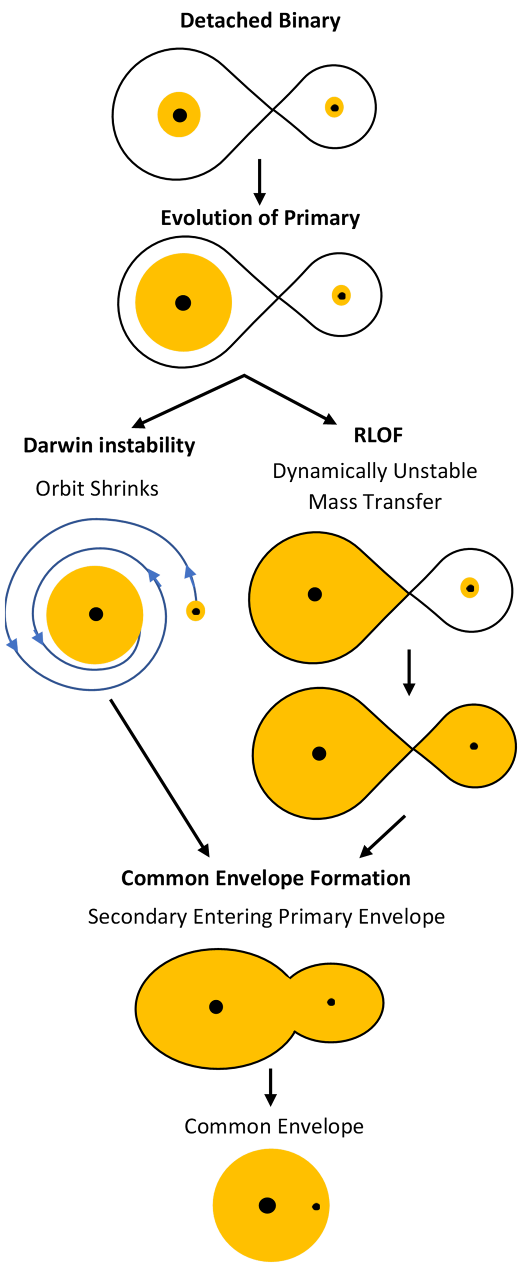


FIG. 3. Summary of the formation of a common envelope though the mechanisms of Darwin instability and Roche lobe overflow (RLOF), after Tauris and van den Heuvel [13]. Throughout the figure the cores of the two stars are represented by the black circles, while their envelopes are represented in yellow. The black tear drop shaped lines represent the Roche lobe of the binary system.

The second mechanism for common envelope formation is dynamically unstable mass transfer. This mechanisms starts with one of the main sequence components in the binary, usually the more massive primary, evolving as mentioned in section II A, which leads to it expanding and filling its Roche lobe. This results in mass being lost from the Roche lobe of the primary star, which is transferred to the secondary star through the L_1 point. This process is called Roche lobe overflow, and the consequences of it on the binary system depends on the rate at which mass is transferred. The mass transfer rate is highly dependent on the stability of the mass transfer, of which there are three scenarios: stable mass transfer, thermal-timescale mass transfer, and dynamically unstable mass transfer. For the case of common envelope formation, only the dynamically unstable mass transfer case is relevant[3].

Dynamically unstable mass transfer occurs when the primary star cannot adjust to the disturbance of its hydrostatic equilibrium due to the mass transfer, on a dynamical timescale. The dynamical timescale is defined as the timescale within which a star counteracts a perturbation made to its hydrostatic equilibrium. As the primary star can not adjust quickly enough to the disruption of its hydrostatic equilibrium, it continues to fill its Roche lobe, which leads to a runaway increase in the mass transfer rate. This runaway mass transfer then leads to the formation of a common envelope.

III. STATE OF THE ART

Here I discuss the current literature and advancements that have been made recently in the field of LRNe. More specifically, I explore the observational studies and theoretical models produced for the recent LRNe events: V1309 Sco, M31LRN 2015, and M101 OT2015-1. These three events have been been selected for discussion as the studies of them cover the different phases of the LRN events, and they are also some of the most recent events that currently have more than a single study.

A. LRN V1309 Sco

The first of the recent major LRNe events is LRN V1309 Sco, which was originally discovered in 2008 by Nakano *et al.* [16]. Mason *et al.* [17] identified that the transient event was not of a common nova class, suggesting that it could be a LRN. Here I discuss the observational studies that were conducted on V1309 Sco and its progenitor system. I also discuss theoretical models that have been developed to explain the observations.

1. Progenitor

One of the only observational studies into the nature of V1309 Sco's progenitor is that by Tylenda *et al.* [2], who used archival photometric data from the Optical Gravitational Lensing Experiment[18,19] (OGLE), specifically OGLE-III and OGLE-IV. The observational data of V1309 Sco from OGLE-III and OGLE-IV that Tylenda *et al.* [2] analysed, can be seen in figure 4. The figure shows the light curve of V1309 Sco in the Cousins I filter, with the I-band magnitude being displayed against both Julian dates (JD) (bottom axis) and years (top axis).

Tylenda *et al.* [2] analysed the progenitor phase of the light curve, which occurs before the start of the outburst at approximately 2454500 JD. One notable and interesting feature of the progenitor light curve is the ~ 0.5 magnitude scatter, which is due to a short term variability in the progenitor system[2]. Tylenda *et al.*[2] showed that the variability from 2002-2007 had a period of ~ 1.4 days, and that the period was exponentially decreasing with time[2]. Tylenda *et al.*[2] also noted that their period fitting method also highly suggested a period of ~ 0.7 days. By looking at the phase folded light curves from 2002-2006, seen in the upper plot of figure 5, it is clear as to why the periods of ~ 1.4 days and ~ 0.7 days are likely periods. The lower part of this figure shows the phase folded light curves from 2007, which have only a single maxima per phase, compared to the double maxima per phase of the 2002-2006 light curves.

Tylenda *et al.*[2] interpreted that the variability could be due to three possible causes: pulsation, rotation of a single star, and an eclipsing binary. Of these three likely causes, it was concluded that the progenitor of V1309 was an eclipsing binary system[2]. Furthermore, Tylenda *et al.*[2] also suggested that the decreasing period of the binary could be due to an instability in the system, which would cause the binary orbit to shrink, and thus lead to a merger event[2]. The instability was suggested to be either caused by Darwin instability or due to mass loss from the outer Lagrange point (L_2) and RLOF[2].

A study by Pejcha[3] investigated the instability that could have caused V1309 Sco, arguing that the timescale that the instability operated on must be shorter than the tidal friction timescale (timescale that tidal friction dissipates energy), based on the observed rate of change of the binary's period being shorter than this timescale[3]. As the instability must be shorter than the tidal frictional timescale, this excludes Darwin instability, as it operates on this timescale[3]. Pejcha[3] also argues that the timescale of the instability must be longer than the dynamical time scale, as it took V1309 Sco several hundred days to reach peak brightness. This leads to the possibility that the instability is caused by dynamically unstable mass loss[3], which Tylenda *et al.*[2] also suggested.

2. Outburst and Post-Outburst

Tylenda *et al.*[2] also investigated the outburst of V1309 Sco, again using the OGLE I band filter light curve seen in figure 4. The outburst of V1309 Sco starts at approximately JD 2454500. One of the noticeable features of the light curve in figure 4 is the ~ 1 magnitude dip before the outburst. Tylenda *et al.*[2] suggested that this decrease in magnitude was due to the formation of an optically thick excretion disc, which would have been caused by the mass loss through L_2. The dimming of the system would only be seen if the binary was viewed near the equatorial plane, which is the case for V1309 Sco, as is evident from the upper phase folded light curves in figure 5[2]. This theory is also supported by theoretical modelling by Pejcha, Metzger, and Tomida[4], where they model the mass loss from L_2. One result from the models produced by Pejcha, Metzger, and Tomida[4] is shown in figure 6. The figure shows the surface density of the L_2 outflow from the binary, as viewed from above the equatorial plane. The binary is located at the centre, where the potential of the L_2 point is highlighted in red. As the binary rotates the L_2 outflow expands radially, which forms a spiral pattern. From similar results as that shown in figure 6, Pejcha, Metzger, and Tomida[4] showed that the L_2 mass loss forms an optically thick outflow, which if viewed at the right angle (near edge on) could obscure the system.

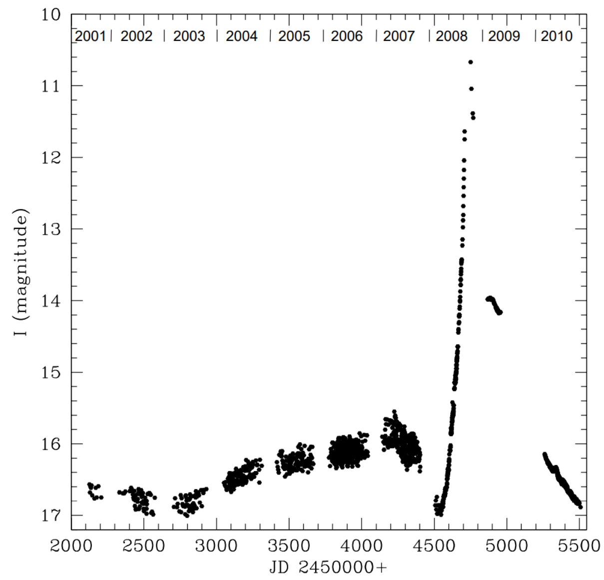


FIG. 4. Light curve of V1309 Sco in the I band filter as constructed by Tylenda *et al.*[2] from archival OGLE[19] data. The time of the observations is given in Julian Dates (JD) (bottom) and in years (top). Note the scatter in the light curve which suddenly disappears after approximately 2454500JD.

The next interesting feature of the light curve in figure 4 is the gradual brightening of the system which follows the disappearance of the signs of an eclipsing binary, which occurs in February 2008. Tylenda *et al.*[2] attributed the disappearance of the signs of an eclipsing binary to the formation of a common envelope, as this would mean that the individual components would no longer be visible. The formation of a common envelope was also used to explain the gradual brightening of the system[2]. This brightening was suggested to be caused by the transfer of angular momentum from the secondary's core to the common envelope[2]. Although this theory seems plausible, the mass loss models produced by Pejcha[3] and Pejcha, Metzger, and Tomida[4,20] suggest that this gradual brightening is due to the expanding L_2 mass loss outflow in the equatorial plane, shown in figure 6, which forms an expanding pseudo photosphere, a luminous envelope from which heat and light radiate. As already established, V1309 Sco was viewed edge on and supposedly obscured by the optically thick outflow. This would mean that that expanding photosphere would replace the binary system's luminosity, resulting in the observations of the system being of the expanding photosphere.

The gradual rise in brightness was followed by a brightness increase of a factor ~ 300 over ~ 10 days[2,3]. Tylenda *et al.*[2] related this increase in brightness to a possible sign of the final disruption of the secondary's core. Pejcha[3] also suggested that the brightening could be a sign of the binary merging, however, they also proposed a different mechanism

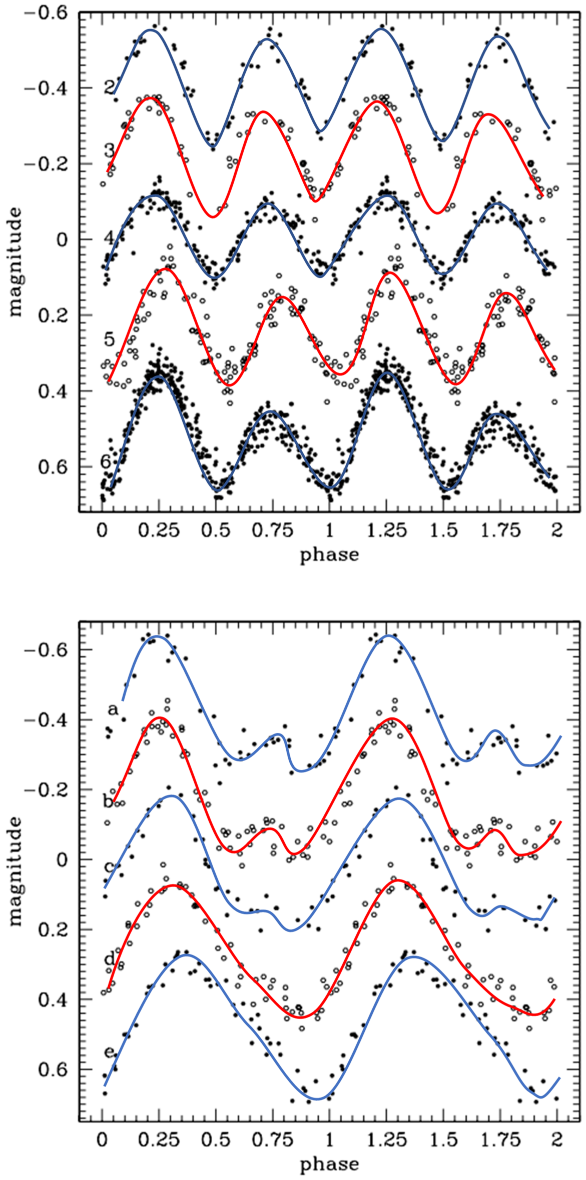


FIG. 5. Phase folded light curves of V1309 Sco from 2002 to 2006 (upper plot) and 2007 (lower plot) divided into five subsamples in each plot, after Tylenda *et al.*[2]. The zero point of the magnitude scale is arbitrary, and the light curves are instead ordered in terms of time, the top light curve being the earliest.

that could have caused the rapid increase. This mechanism involved significantly increasing the radiation efficiency, the efficiency of energy dissipation through radiation, of the L_2 outflow photosphere[3], which could be achieved by a passing shock wave. However, Pejcha[3] stated that there is no clear origin of a shock that could achieve the needed increase in the radiation efficiency.

As is seen in figure 4, the brightness of V1309 Sco reached its pre-outburst brightness approximately two years after peak brightness[2]. The temperature of the system was much cooler after the outburst compared to before[2,17]. This cooler temperature is suggested to be due to dust formation around the remnant of V1309 Sco from the L_2 outflow[2,21].

3. V1309 Sco Summary

The observational studies of V1309 Sco discussed here provide one of the first insights into the progenitor systems of LRNe. Despite there being only one observational study into the progenitor system, a substantial amount of knowledge was gained about the progenitors of LRNe. Furthermore, the models of L_2 mass outflows by Pejcha[3] and Pejcha, Metzger, and Tomida[4,20] provide theoretical arguments that convincingly explain the observations. However, there remains a debate about the causes of some of the features seen in the observations, notably the brightness increase to peak. Although V1309 Sco has provided insights into the causes of LRNe, it alone is not enough to convincingly say that all LRNe are caused by unstable mass transfer with L_2 outflows, and binary mergers.

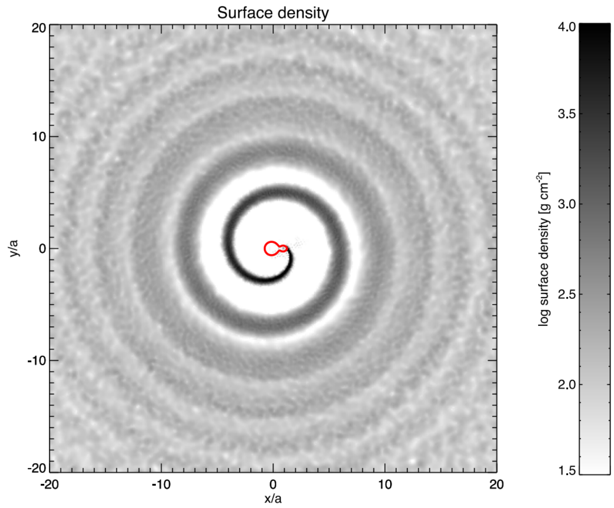


FIG. 6. Pejcha, Metzger, and Tomida[4] simulation results of the surface density of the outflow from L_2 as viewed from above the equatorial plane of the binary. The red line at the centre of the figure represents the potential surface containing L_2. The spiral represents the L_2 outflow, which leaves the L_2 point and radially expands.

B. M31LRN 2015

Another recent and important event, discovered by the transient detection system MASTER, occurred in the Andromeda galaxy (M31) in January 2015[22]. Initially Kurtenkov *et al.*[23] identified the transient event as a classical nova, however, Kurtenkov *et al.*[24] later classify the event as an LRN based on the observations of the transient's evolution. Here I discuss the observations that were made of M31LRN 2015, as well as various theoretical models that have been developed to explain it.

1. *Progenitor*

As was the case for V1309 Sco, there were no real time observations of the progenitor system of M31LRN 2015. Due to this, Williams *et al.*[25], MacLeod *et al.*[26] and Blagorodnova *et al.*[27] made use of the available archival data in order to analyse M31LRN 2015's progenitor.

The study by Williams *et al.*[25] made use of the Hubble Space Telescope's (HST) archival data to locate the likely progenitor of M31LRN 2015, as well as the Local Group Galaxies Survey (LGGS)[28] to analyse the progenitor. From the photometric observations provided in the LGGS, Williams *et al.*[25] found that the emission spectrum of the progenitor system exhibited a strong Hα emission excess. It was suggested that this could be due to a supernova remnant, however, this is unlikely to be the case as the system lacked an SII emission[25], which is usually present in supernova remnants. Other explanations of the source of the strong Hα excess are: directly from the progenitor, and from a period of mass loss from the progenitor[25].

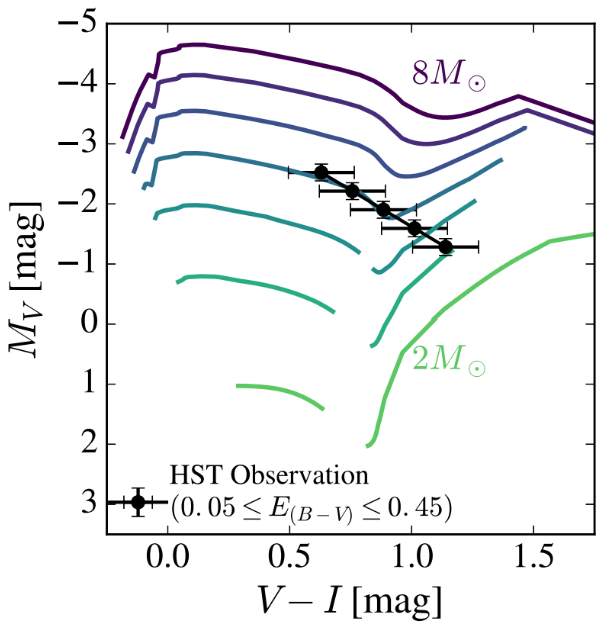


FIG. 7. Colour magnitude diagram, created by MacLeod *et al.*[26], comparing the HST observations presented by Williams *et al.*[25] and the post main sequence stellar evolution tracks computed in MESA. The plotted evolution tracks are of stars with integer masses from 2 to $8M_\odot$ and include only the portions where the radius of the star is increasing. This is due to the interest in binary interaction, which can only occur if the primary is expanding.

MacLeod *et al.*[26] further analysed M31LRN 2015's progenitor and precursor by comparing the observations presented by Williams *et al.*[25] to detailed stellar models, from which they inferred the properties of the progenitor source. To compare the observations of the progenitor with stellar models, MacLeod *et al.*[26] assumed that if the system was a binary then the observations were dominated by the primary component.

MacLeod *et al.*[26] inferred some interesting properties of the progenitor from figure 7, which shows a colour magnitude diagram that compares the HST observations presented by Williams *et al.*[25] and the post main sequence stellar models computed in MESA[29]. The stellar evolution tracks present in figure 7 are of stars with integer masses between 2 and $8M_\odot$. Due to the interest in binary interaction, only the portions of the evolution tracks with increasing stellar radius are plotted[26]. From figure 7 it can be seen that the HST observations of M31LRN 2015's progenitor are consistent with a post main sequence star with a mass between 3.3 and $5M_\odot$[26]. Furthermore, the observations correspond to portions of the stellar evolution tracks where the radius of the star is increasing. This implies that if the system is a binary, then the primary star would be expanding and engulfing the secondary[26], which would therefore result in the formation of a common envelope. The implied formation of a common envelope thus supports the theory of ejections and binary mergers causing LRNe. Lastly, figure 7 shows that the progenitor is in the giant phase, meaning that it would be the more luminous component of a binary, unless the companion was also a giant which is very unlikely[26]. This therefore justifies the assumption that the if the system is a binary, then the primary dominates the observations.

Blagorodnova *et al.*[27] studied the progenitor of M31LRN 2015 by analysing the archival photometric data from the Palomar Transient Facility (PTF)[30,31], and the Canada-France-Hawaii-Telescope (CFHT). The archival data available from these two facilities provided enough data to analyse the progenitor up to five years before the outburst.

The resulting light curves of M31LRN 2015 are very similar to those of V1309 Sco, shown in figure 4. Some of the similarities between the two events' light curves are: the gradual brightening several years prior to the outburst, and a dip in brightness before the outburst. MacLeod *et al.*[26] suggested that tidal heating of the progenitor, due to the forces of the two stars on one another, could be responsible for the observed light curve, however, Blagorodnova *et al.*[27] dismissed this idea due to the complex behaviour of the light curve. Instead Blagorodnova *et al.*[27] proposed that a possible explanation is mass loss from the L_2 point during RLOF, as is also suggested by Pejcha *et al.*[32], and was also suggested to be the case for V1309 Sco[3,4].

Although the observed light curve of M31LRN 2015's progenitor is very similar to V1309 Sco's, there are some differences. One of these is that M31LRN 2015's light curve has a plateau after the dip in the light curve and before reaching peak brightness, whereas V1309 Sco smoothly transitioned from the dip into the outburst. Blagorodnova *et al.*[27] suggest that this could be caused if the L_2 outflow was marginally bound to the binary system, and fell back towards the binary, interacting with it. Pejcha, Metzger, and Tomida[20] supported this with theoretical modelling of the marginally bound L_2 outflow.

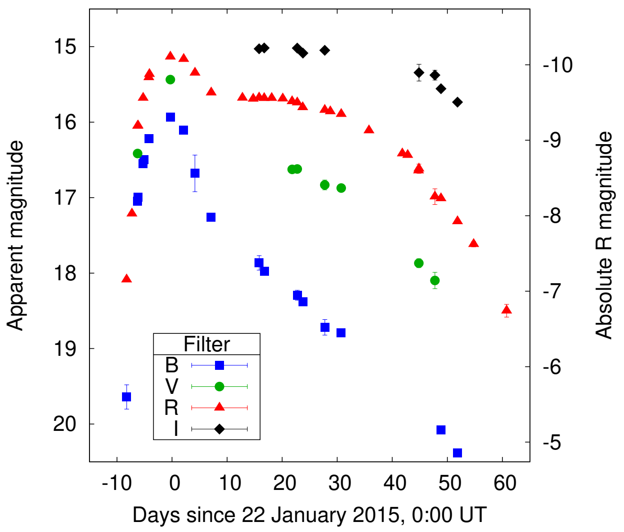


FIG. 8. Light curves of the outburst of M31LRN 2015, presented by Kurtenkov, A. *et al.* [33], in the filter bands: B, V, R, and I. The peak of the outburst occurred on 22 January 2015, with the peak magnitude reaching an apparent magnitude of R $\sim$ 15.1 mag, which is equivalent to an absolute R magnitude of $\sim --10.2$ mag. Both the apparent magnitude (left axis) and the absolute R magnitude (right axis) are present on this plot.

2. *Outburst and Post-Outburst*

The observations of the outburst of M31LRN 2015 presented by Kurtenkov, A. *et al.* [33] are shown in figure 8. This figure shows the light curves of M31LRN 2015 in the filter bands, B, V, R, and I. The observations were taken over a 70 day period starting 10 days before the peak of the outburst, which occurred on the 22 January 2015. As is shown by figure 8, the light curve peak is followed by a decrease in the blue (B) filter, while the redder filters (R and I) exhibit a plateau.

MacLeod *et al.* [26] explain the light curve, seen in figure 8, to be the result of the formation of a common envelope and subsequently a stellar merger. They begin by suggesting that the peak of the light curve is best explained by an ejection of a small amount of mass, approximately $10^{-2}M_\odot$, with velocities above the escape velocity of the system[26]. These ejecta are likely to be caused by the binary's secondary component entering the primary's envelope, which causes shocks through the outer envelope[26,34]. MacLeod *et al.* [26] also suggest that mass ejections are the cause of the plateau seen in M31LRN 2015's light curves. They argue that the inspiraling of the secondary in the common envelope causes a slower, more massive outflow that has a lower temperature, and thus a lower peak luminosity than the previous ejecta[26]. The estimates of the effective temperature during the plateau are low enough for hydrogen recombination to occur[26], which is where hydrogen nuclei are able to recapture their electron to form atomic hydrogen. MacLeod *et al.* [26] say that hydrogen recombination stabilised the temperature and estimate the mass of the ejecta, in this theory, to be approximately $\sim 0.3M_\odot$. Blagorodnova *et al.* [27] back this up with their independent estimation of an ejected mass of $0.21^{+0.53}_{--0.13}M_\odot$. However, these estimations of ejected mass are for the scenario where the light curve is powered by recombination, which causes tension with the shock powered scenario presented by Metzger and Pejcha [35]. To explain M31LRN 2015's light curve with the shock powered scenario, the models of Metzger and Pejcha [35] suggest that a maximum of ejected mass of $0.1M_\odot$ is required[27]. Williams *et al.* [25] also came to the conclusion that the mass ejection was of $0.1M_\odot$, based upon the theoretical models of Ivanova *et al.* [36]. However, Williams *et al.* [25] also suggested suggest that the entire envelope could have been ejected, which is inconsistent with the arguments presented by MacLeod *et al.* [26] and Blagorodnova *et al.* [27].

3. *M31LRN 2015 Summary*

As with V1309 Sco, the studies of M31LRN 2015 provide a useful insight into the progenitor systems of LRNe, as well as the processes behind LRNe. There are competing theories and models that provide explanations for M31LRN 2015's observed light curves, however, it is unclear from the current observations as to which one is most likely correct. Furthermore, the lack of in depth observations of M31LRN 2015's progenitor meant that it could not be fully understood, with inferences about the progenitor being made based upon the archival data available.

C. M101 OT2015-1

M101 OT2015-1 is another LRNe outburst that was observed in February 2015, although its progenitor was discovered two years earlier in 2013 when the intermediate Palomar Transient Factory survey identified it as a slow rising source[1,37]. As will be discussed, M101 OT2015-1 has a plethora of available archival and follow up observations, however there have only been two studies on M101 OT2015-1[1,38]. Here I will discuss the observations that presented in these studies.

1. *Progenitor*

To study the progenitor system of M101 OT2015-1, Blagorodnova *et al.* [1] and Goranskij *et al.* [38] used archival data from the Canada–France–Hawaii Telescope (CFHT)[39], Pan-STARRS-1/GPC1[40–42], Isaac Newton Telescope/Wide Field Camera, and the Sloan Digital Sky Survey data release 10[43]. Blagorodnova *et al.* [1] presented the archival observations of M101 OT2015-1 in the form of light curves, as seen in the left plot of figure 9.

From the archival data, Blagorodnova *et al.* [1] noted that there was a lack of a periodic variation in the progenitor system, unlike those seen in V1309 Sco. This led them to believe that the system could be a binary where the primary component outshone the secondary, resulting in no detectable variation in the observed light curves[1]. By assuming that the

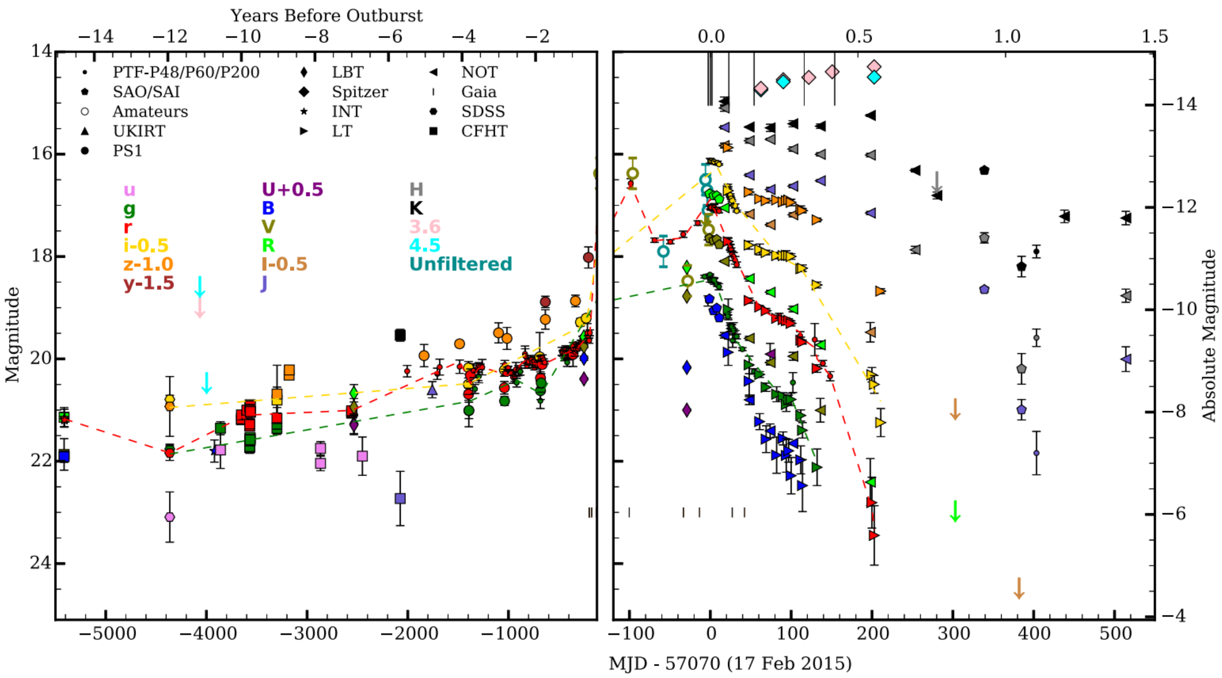


FIG. 9. Light curves of M101 OT2015-1 constructed from historical archival data and follow up observations. The filter and instrument used for the observations are noted by their colour and shape respectively. The light curve magnitudes are presented in both absolute and apparent magnitudes. The light curve epochs are displayed in modified Julian date (MJD) and years, relative to the secondary peak of the outburst. Note the time scale difference between the left and right plots.

progenitor observations represented a single star, Blagorodnova *et al.* [1] estimated that the radius of the progenitor source was $220 \pm 13 R_\odot$. They also compared the progenitor observations to the BPASS[44] stellar model[1], which can be seen in figure 10. In this figure the first three observations of M101 OT2015-1's progenitor are shown along with the BPASS stellar evolution tracks of stars with initial masses of 17-20 $M_\odot$. The stellar evolution tracks shown are of the main sequence, which is to the left of the "hook" located around a temperature of 25,000K, and the post main sequence, which is to the right of the "hook". From the comparison of the observations and stellar evolution tracks presented in figure 10, the mass of the progenitor was estimated to be between 18 and 19 $M_\odot$[1]. Also from the figure, it is clear that the progenitor star is in the post main sequence phase, which is also suggested by the observations presented by Goranskij *et al.* [38]. Based on this Blagorodnova *et al.* [1] suggested that the progenitor star is a yellow super giant that is evolving from the main sequence towards the red giant branch. Due to this suggested evolution of the star, if it was a part of a binary, then the formation of a common envelope would occur[1].

2. *Outburst and Post-Outburst*

The observations of the outburst and post-outburst of M101 OT2015-1, presented by Blagorodnova *et al.* [1], are shown in the right plot in figure 9. The lack of data points during the first peak, around 100 days before the second peak, is due to the sun blocking the event[1]. The light curves of M101 OT2015-1 in this plot exhibit characteristic features, which Goranskij *et al.* [38] compared to previous LRNe. One of the features that Goranskij *et al.* [38] investigated, is the gradual increase in brightness leading to the outburst. The brightness increase seen is 2.2 mag and occurs over 5.5 years. Goranskij *et al.* [38] suggested that the gradual increase could be caused by the formation of a common envelope, as was seen with V1309 Sco[38]. The proposition that a common envelope formed is also supported by the estimates of the radii of the photosphere over the course of the event. Goranskij *et al.* [38] estimated that the radius of the photosphere 240 days before the outburst was $400 R_\odot$, increasing to 3300 $R_\odot$ at the second peak, and then further increasing to 4700 $R_\odot$ at 120 days after the event. Blagorodnova *et al.* [1] also estimated the radii of the photosphere, estimating it to be 230 ± 13 $R_\odot$ six years before the outburst, and increasing to 6500 $R_\odot$ during the second outburst maxima. Although these estimations do not agree with one another, they both show signs of a significant increase in the photospheric radius, which could be caused by the formation and ejection of a common envelope as Goranskij *et al.* [38] suggests.

The common envelope theory for M101 OT2015-1 is further supported by the quick colour evolution in the blue filters, which can be seen in figure 9 by the quick decrease in brightness in the bluer bands (B, g, V). The quick colour evolution provides evidence that there is optically thick dust forming in the system[1]. The spectroscopy of M101 OT2015-1 also confirms the presence of this medium, as well as suggesting that it is newly formed. This newly formed dust could have been formed from an ejected common envelope.

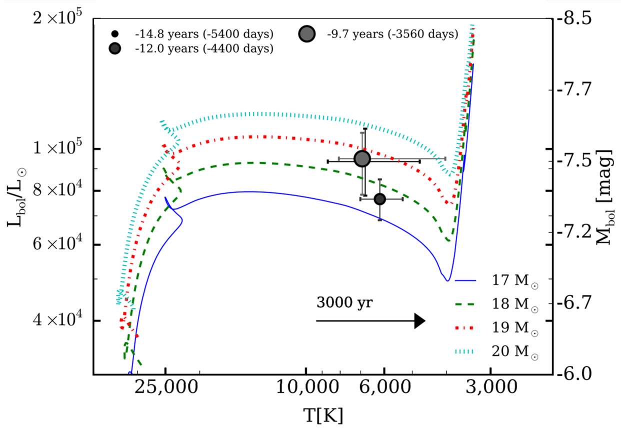


FIG. 10. Hertzsprung Russell diagram showing the stellar evolution tracks[45] for stars with a mass range 17-20$M_{\odot}$, and the first three observations of M101 OT2015-1[1], which are at least 2500 days before the outburst. This Hertzsprung Russell diagram represents the parater space of temperature (T) in Kelvin, and bolometric luminosity (L_{bol}) in terms of solar luminosity, as well as bolometric absolute magnitude (M_{bol}). The stellar evolution tracks are of the main sequence phase, which is the track located to the left of the "hook" seen at temperatures of approximately 25,000K, as well as the post main sequence, to the right of the "hook". Also on this figure is a visualisation of much the temperature of a 18 M_{bol} star changes over 3000 years.

3. *M101 OT2015-1 Summary*

The two studies of M101 OT2015-1, by Blagorodnova *et al.* [1] and Goranskij *et al.* [38], present valuable insights into LRNe. They use the available archival data to analyse the progenitor system and the lead up to the outburst. These observations showed that the progenitor system was a yellow super giant that, if in a binary system, outshone its companion star. Further investigations of the event showed that the radius of the photosphere had significantly increased from a few hundred solar radii, to a few thousand solar radii. Lastly there was evidence of newly formed, optically thick dust in the system. The results presented by Blagorodnova *et al.* [1] and Goranskij *et al.* [38] provide reasonable evidence of a common envelope ejection as the cause of the M101 OT2015-1. Although a common envelope ejection seems to be the likely cause of M101 OT2015-1, there is little investigation into other plausible causes. Meaning that the true causes of M101 OT2015-1 remain unclear.

D. Summary

Throughout the last two decades many theories have been developed to explain LRNe, with the more prominent theories being common envelope ejections, mass losses from L_2, and stellar mergers. The observational and theoretical studies, presented in previous sections, provide equal support for these three different explanations of the processes behind LRNe. This leaves some uncertainty in the true nature of the processes that occur during LRNe events, which is not helped by the fact that archival data is used to study their progenitors and precursors. The use of archival data to study LRNe has provided many useful observations that have provided us with knowledge of the progenitor systems. Some of the key findings are that the progenitors of LRNe are binary systems that undergo an orbital period decrease, one of the binary components is likely to be an evolving star such as a yellow super giant, and there is the onset of Roche lobe overflow which likely causes the formation of a common envelope and or L_2 mass loss. Although a lot has been learnt over the recent years, there is still a lot to be understood about LRNe. The main limitation is that archival data is used and ideally we would want to identify a LRN before its peaks.

In general the archival data available for the progenitor and precursor of an LRN is limited, in both the number of observations and the range of filters bands. For example V1309 Sco has an excellent number of pre-outburst observations, see figure 4, however, they were only in a single band. The archival data for M101 OT2015-1 is the best set of observations, shown in figure 9, for a LRNe progenitor and precursor so far, although there is a lack of observations at the time of the first outburst. This lack in range and number of observations limits the analysis of the LRNe progenitors and precursors, and so limits our understanding of the processes that occur.

IV. RESEARCH

The main gap in the current literature is that archival data is used to study the precursors and progenitors of LRNe. This data is limited in both the number of observations and the variety of bands due to the observations being coincidental. Furthermore, there is a large range of different techniques that are not utilised, such as: infrared spectroscopy, adaptive optics assisted imaging, and optical interferometry.

The use of just the archival data therefore restricts our understanding of LRNe precursors and the processes that occur, and so to further our understanding of them we need to improve the data available for the precursor. One way to do this would be to take real time observations of the LRNe precursors, as they could make use of the full range of filter bands and techniques available in facilities around the world. However, to do this we would first have to detect the LRNe precursor. As of current there is no known methodology for detecting LRNe precursors, and so my research project will provide the first methodology for detecting Galactic LRNe precursors. I will now discuss the planned details of how this will be achieved, and the expected outcomes.

A. Project Details

The first planned step in my methodology for detecting Galactic LRNe precursors is to identify likely LRNe progenitor systems from a data set of known Galactic sources. As was discussed in previous sections, the most likely cause of LRNe are common envelope ejections and stellar mergers. One formation path for a common envelope is through the evolution of one of the components, which has to experience a fast expansion. Such stars that experience this rapid expansion are yellow super giants, and so systems containing them are likely LRNe progenitors. To be able to identify YSGs from a Galactic population I will be using the fact that the YSG phase lasts around 1% of the MS phase, which implies that the population of them is small[46]. Therefore, the YSGs will reside in a thinly populated region of the Hertzsprung Russell diagram, which is known as the Hertzsprung gap.

To utilise this knowledge, I will define the Hertzsprung gap by modelling the density of a Hertzsprung Russell diagram, constructed from well behaved sources within the Gaia data release 2 (DR2) catalogue[47]. There are a range of suitable modelling methods that I will be exploring, such as the Gaussian mixture modelling method and the K means clustering method. When I have explored the benefits and disadvantages of the different modelling methods and have decided on which is best for my purpose, I will then use the model to select likely progenitor systems. The resulting model will represent the dense regions of the Hertzsprung Russell diagram, which are the main sequence and red giant branch. Therefore, I will compare stars from the Gaia DR2 catalogue with the model, and those that do not fit it will be selected as likely LRN progenitor systems.

With the identified likely progenitor systems, possible LRNe precursors will then be selected from them. To select the precursors, the light curves of the likely progenitors will be analysed using already available transient detection methods. One of the transient detection methods that I will explore is the use of the Von Neumann statistic coupled with skewness of the light curve, as was used previously by Kostrzewa-Rutkowska *et al.* [48] to detect transients from the Gaia data. I will explore the suitability of this method for detecting LRNe precursors, as the features of their light curves differ from those of other, more common transients.

I will also be further analysing the light curves of the identified precursors by searching for periodicity within them. The purpose of this is to identify any signs of a binary system, as was seen with V1309 Sco's progenitor[2]. This will provide further evidence that the system is a LRNe precursor, as the progenitor systems of LRNe are binaries. To search for peridoicity in light curves, a range of different methods are available, however, they have different scenarios where they are best suited. As such I will investigate the uses of the different techniques and if they are suitable to find periodicities that might be found within the light curves of LRNe precursors.

B. Expected Outcomes

The final expectation of the methodology that I am developing, is that it should be able to identify at least one true LRNe precursor within the Gaia DR2 catalogue of Galactic sources. This is based on the observed rate of the initiation of the common envelope phase that are brighter than an apparent magnitude of –3 mag. This rate is approximately 0.5 per year[49], and as the LRNe precursor phase occurs over three to five years, an approximate estimation for the current observable number of Galactic LRNe precursors is about two. Therefore to find at least one true precursor would be an achievement.

V. CONCLUSION

The field of LRNe has progressed significantly over recent years, with observational studies of a handful of events providing us with evidence that their progenitors are binary systems that likely undergo a common envelope phase. Along with these observations different theories have been developed to explain them. The most prominent of these theories are common envelope ejections, L_2 mass loss, and stellar mergers. Although there have been major developments in the field, there is still much debate around the true nature of LRNe. Currently our understanding of the precursor phase of LRNe has been drawn from archival, coincidental observations. These observations are the current limiting factor in fully understanding the nature of LRNe, as these archival observations are usually only in a few different bands, contain a small number of observational epochs, and are from a small range of techniques. One way in which we could expand upon our knowledge of the precursor phase would be to obtain real time observations. To be able to take real time observations, the precursor must first be detected. Currently there is no such method to detect LRNe precursors, and so I will produce a methodology that is capable of detecting Galactic LRNe progenitors. The expectation of this methodology is that it is able to identify at least one true LRN precursor, which has been based off of the observed Galactic rate of common envelope initiations.

ACKNOWLEDGMENTS

I would like to thank my supervisors for the helpful discussions of the work presented here. I would also like to thank my visiting tutor and anonymous readers for comments on an early version of this work.

[1] N. Blagorodnova, R. Kotak, J. Polshaw, M. M. Kasliwal, Y. Cao, A. M. Cody, G. B. Doran, N. Elias-Rosa, M. Fraser, C. Fremling, and et al., The Astrophysical Journal **834**, 107 (2017).
[2] R. Tylenda, M. Hajduk, T. Kamiński, A. Udalski, I. Soszyński, M. K. Szymański, M. Kubiak, G. Pietrzyński, R. Poleski, Ł. Wyrzykowski, and K. Ulaczyk, Astronomy & Astrophysics **528**, A114 (2011), arXiv:1012.0163 [astro-ph.SR].
[3] O. Pejcha, ApJ **788**, 22 (2014), arXiv:1307.4088 [astro-ph.SR].
[4] O. Pejcha, B. D. Metzger, and K. Tomida, MNRAS **455**, 4351 (2016), arXiv:1509.02531 [astro-ph.SR].
[5] S. R. Kulkarni, E. O. Ofek, A. Rau, S. B. Cenko, A. M. Soderberg, D. B. Fox, A. Gal-Yam, P. L. Capak, D. S. Moon, W. Li, A. V. Filippenko, E. Egami, J. Kartaltepe, and D. B. Sanders, Nature **447**, 458 (2007), arXiv:0705.3668 [astro-ph].
[6] R. M. Rich, J. Mould, A. Picard, J. A. Frogel, and R. Davies, ApJL **341**, L51 (1989).
[7] A. B. Tomaney and A. W. Shafter, ApJS **81**, 683 (1992).
[8] J. Bryan and R. E. Royer, PASP **104**, 179 (1992).
[9] S. S. Hayashi, M. Yamamoto, and K. Hirosawa, IAUC **5942**, 1 (1994).

[10]R. M. Wagner, R. Bertram, P. M. Kilmartin, B. A. Skiff, A. C. Gilmore, and F. Farrell, IAUC **5944**, 1 (1994).
[11]P. Martini, R. M. Wagner, A. Tomaney, R. M. Rich, M. della Valle, and P. H. Hauschildt, AJ **118**, 1034 (1999), arXiv:astro-ph/9905016 [astro-ph].
[12]U. Munari, A. Henden, S. Kiyota, D. Laney, F. Marang, T. Zwitter, R. L. M. Corradi, S. Desidera, P. M. Marrese, E. Giro, F. Boschi, and M. B. Schwartz, A&A **389**, L51 (2002), arXiv:astro-ph/0205288 [astro-ph].
[13]T. M. Tauris and E. P. J. van den Heuvel, "Formation and evolution of compact stellar X-ray sources," in *Compact stellar X-ray sources*, Vol. 39 (2006) pp. 623–665.
[14]B. Paczynski, in *Structure and Evolution of Close Binary Systems*, Vol. 73, edited by P. Eggleton, S. Mitton, and J. Whelan (1976) p. 75.
[15]G. H. Darwin, Proceedings of the Royal Society of London **29**, 168 (1879).
[16]S. Nakano, K. Nishiyama, F. Kabashima, Y. Sakurai, C. Jacques, E. Pimentel, D. Chekhovich, S. Korotkiy, T. Kryachko, and N. N. Samus, IAU Circ. **8972**, 1 (2008).
[17]E. Mason, M. Diaz, R. E. Williams, G. Preston, and T. Bensby, A&A **516**, A108 (2010), arXiv:1004.3600 [astro-ph.SR].
[18]s. Udalski, I. Soszynski, M. Szymanski, M. Kubiak, G. Pietrzynski, P. Wozniak, and K. Zebrun, Acta Astronomica **53** (2003).
[19]"The optical gravitational lensing experiment," Available at `http://ogle.astrouw.edu.pl/` (06/07/21) (2021).
[20]O. Pejcha, B. D. Metzger, and K. Tomida, MNRAS **461**, 2527 (2016), arXiv:1604.07414 [astro-ph.SR].
[21]R. Tylenda and T. Kamiński, A&A **592**, A134 (2016), arXiv:1606.09426 [astro-ph.SR].
[22]V. Shumkov, V. Lipunov, E. Gorbovskoy, N. Tiurina, P. Balanutsa, A. Kuznetsov, M. Pruzhinskaya, D. Denisenko, A. Rufanov, V. Vladimirov, K. Ivanov, S. Yazev, N. Budnev, O. Gress, V. Poleshchuk, E. K. A. Parkhomenko, A. Tlatov, D. Dormidontov, V. S. V. Yurkov, Y. Sergienko, D. Varda, E. Sinyakov, V. Gabovich, V. Krushinsky, I. Zalozhnih, A. Popov, A. Bourdanov, D. Buckley, S. Potter, A. Kniazev, M. Kotze, V. Shumkov, S. S. V. Vladimirov, V. Lipunov, E. Gorbovskoy, N. Tiurina, P. Balanutsa, A. Kuznetsov, M. Pruzhinskaya, D. Denisenko, V. Rufanov, K. Ivanov, S. Yazev, N. Budnev, O. Gress, V. Poleshchuk, E. K. A. Parkhomenko, A. Tlatov, D. Dormidontov, V. S. V. Yurkov, Y. Sergienko, D. Varda, E. Sinyakov, V. Gabovich, V. Krushinsky, I. Zalozhnih, A. Popov, A. Bourdanov, D. Buckley, S. Potter, A. Kniazev, M. Kotze, and S. Shurpakov, The Astronomer's Telegram **6911**, 1 (2015).
[23]A. Kurtenkov, E. Ovcharov, P. Nedialkov, A. Kostov, R. Bachev, R. V. M. Dimitrova, V. Popov, and A. Valcheva, The Astronomer's Telegram **6941**, 1 (2015).
[24]A. Kurtenkov, T. Tomov, S. Fabrika, E. A. Barsukova, A. F. Valeev, P. Pessev, K. Vida, L. Molnar, K. Sarneczky, V. P. Goranskij, K. Hornoch, M. Henze, A. W. Shafter, E. Ovcharov, P. Nedialkov, A. Kostov, S. Valenti, and M. Stritzinger, The Astronomer's Telegram **7150**, 1 (2015).
[25]S. C. Williams, M. J. Darnley, M. F. Bode, and I. A. Steele, ApJ **805**, L18 (2015), arXiv:1504.07747 [astro-ph.SR].
[26]M. MacLeod, P. Macias, E. Ramirez-Ruiz, J. Grindlay, A. Batta, and G. Montes, The Astrophysical Journal **835**, 282 (2017).
[27]N. Blagorodnova, V. Karambelkar, S. M. Adams, M. M. Kasliwal, C. S. Kochanek, S. Dong, H. Campbell, S. Hodgkin, J. E. Jencson, J. Johansson, S. Kozłowski, R. R. Laher, F. Masci, P. Nugent, and U. Rebbapragada, MNRAS **496**, 5503 (2020), arXiv:2004.04757 [astro-ph.SR].
[28]P. Massey, K. A. G. Olsen, P. W. Hodge, S. B. Strong, G. H. Jacoby, W. Schlingman, and R. C. Smith, AJ **131**, 2478 (2006), arXiv:astro-ph/0602128 [astro-ph].
[29]B. Paxton, L. Bildsten, A. Dotter, F. Herwig, P. Lesaffre, and F. Timmes, The Astrophysical Journal Supplement Series **192**, 3 (2010).
[30]N. M. Law, S. R. Kulkarni, R. G. Dekany, E. O. Ofek, R. M. Quimby, P. E. Nugent, J. Surace, C. C. Grillmair, J. S. Bloom, M. M. Kasliwal, L. Bildsten, T. Brown, S. B. Cenko, D. Ciardi, E. Croner, S. G. Djorgovski, J. van Eyken, A. V. Filippenko, D. B. Fox, A. Gal-Yam, D. Hale, N. Hamam, G. Helou, J. Henning, D. A. Howell, J. Jacobsen, R. Laher, S. Mattingly, D. McKenna, A. Pickles, D. Poznanski, G. Rahmer, A. Rau, W. Rosing, M. Shara, R. Smith, D. Starr, M. Sullivan, V. Velur, R. Walters, and J. Zolkower, Publications of the Astronomical Society of the Pacific **121**, 1395 (2009).
[31]A. Rau, S. R. Kulkarni, N. M. Law, J. S. Bloom, D. Ciardi, G. S. Djorgovski, D. B. Fox, A. Gal-Yam, C. C. Grillmair, M. M. Kasliwal, P. E. Nugent, E. O. Ofek, R. M. Quimby, W. T. Reach, M. Shara, L. Bildsten, S. B. Cenko, A. J. Drake, A. V. Filippenko, D. J. Helfand, G. Helou, D. A. Howell, D. Poznanski, and M. Sullivan, Publications of the Astronomical Society of the Pacific **121**, 1334 (2009).
[32]O. Pejcha, B. D. Metzger, J. G. Tyles, and K. Tomida, ApJ **850**, 59 (2017), arXiv:1710.02533 [astro-ph.SR].
[33]Kurtenkov, A., Pessev, P., Tomov, T., Barsukova, E. A., Fabrika, S., Vida, K., Hornoch, K., Ovcharov, E. P., Goranskij, V. P., Valeev, A. F., Molnár, L., Sárneczky, K., Kostov, A., Nedialkov, P., Valenti, S., Geier, S., Wiersema, K., Henze, M., Shafter, A. W., Muñoz Dimitrova, R. V., Popov, V. N., and Stritzinger, M., A&A **578**, L10 (2015).
[34]N. Soker and R. Tylenda, Monthly Notices of the Royal Astronomical Society **373**, 733 (2006), https://academic.oup.com/mnras/article-pdf/373/2/733/4104626/mnras0373-0733.pdf.
[35]B. D. Metzger and O. Pejcha, Monthly Notices of the Royal Astronomical Society **471**, 3200 (2017), https://academic.oup.com/mnras/article-pdf/471/3/3200/19515512/stx1768.pdf.
[36]N. Ivanova, S. Justham, J. L. Avendano Nandez, and J. C. Lombardi, Science **339**, 433 (2013), arXiv:1301.5897 [astro-ph.HE].
[37]Y. Cao, M. M. Kasliwal, G. Chen, and I. Arcavi, The Astronomer's Telegram **7070**, 1 (2015).
[38]V. P. Goranskij, E. A. Barsukova, O. I. Spiridonova, A. F. Valeev, T. A. Fatkhullin, A. S. Moskvitin, O. V. Vozyakova, D. V. Cheryasov, B. S. Safonov, A. V. Zharova, and T. Hancock, Astrophysical Bulletin **71**, 82 (2016), arXiv:1611.04936 [astro-ph.SR].
[39]S. D. J. Gwyn, PASP **120**, 212 (2008), arXiv:0710.0370 [astro-ph].
[40]E. F. Schlafly, D. P. Finkbeiner, M. Jurić, E. A. Magnier, W. S. Burgett, K. C. Chambers, T. Grav, K. W. Hodapp, N. Kaiser, R. P. Kudritzki, N. F. Martin, J. S. Morgan, P. A. Price, H. W. Rix, C. W. Stubbs, J. L. Tonry, and R. J. Wainscoat, ApJ **756**, 158 (2012), arXiv:1201.2208 [astro-ph.IM].
[41]J. L. Tonry, C. W. Stubbs, K. R. Lykke, P. Doherty, I. S. Shivvers, W. S. Burgett, K. C. Chambers, K. W. Hodapp, N. Kaiser, R. P. Kudritzki, E. A. Magnier, J. S. Morgan, P. A. Price, and R. J. Wainscoat, ApJ **750**, 99 (2012), arXiv:1203.0297 [astro-ph.IM].
[42]E. A. Magnier, E. Schlafly, D. Finkbeiner, M. Juric, J. L. Tonry, W. S. Burgett, K. C. Chambers, H. A. Flewelling, N. Kaiser, R. P. Kudritzki, J. S. Morgan, P. A. Price, W. E. Sweeney, and C. W. Stubbs, ApJS **205**, 20 (2013), arXiv:1303.3634 [astro-ph.IM].
[43]C. P. Ahn, R. Alexandroff, C. Allende Prieto, F. Anders, S. F. Anderson, T. Anderton, B. H. Andrews, É. Aubourg, S. Bailey, and F. A. Bastien, ApJS **211**, 17 (2014), arXiv:1307.7735 [astro-ph.IM].
[44]"Binary population and spectral synthesis," Available at `https://bpass.auckland.ac.nz/` (20/07/21) (2018).
[45]J. J. Eldridge and C. A. Tout, Monthly Notices of the Royal Astronomical Society **353**, 87 (2004), https://academic.oup.com/mnras/article-pdf/353/1/87/18661978/353-1-87.pdf.
[46]M. R. Drout, P. Massey, G. Meynet, S. Tokarz, and N. Caldwell, The Astrophysical Journal **703**, 441–460 (2009).
[47]Gaia Collaboration, T. Prusti, J. H. J. de Bruijne, A. G. A. Brown, A. Vallenari, C. Babusiaux, C. A. L. Bailer-Jones, U. Bastian, M. Biermann, D. W. Evans, and L. Eyer, A&A **595**, A1 (2016), arXiv:1609.04153 [astro-ph.IM].
[48]Z. Kostrzewa-Rutkowska, P. G. Jonker, S. T. Hodgkin, L. Wyrzykowski, M. Fraser, D. L. Harrison, G. Rixon, A. Yoldas, F. van Leeuwen, A. Delgado, M. vanLeeuwen, and S. E. Koposov, Monthly Notices of the Royal Astronomical Society **481**, 307 (2018), https://academic.oup.com/mnras/article-pdf/481/1/307/25683940/sty2221.pdf.
[49]C. S. Kochanek, S. M. Adams, and K. Belczynski, MNRAS **443**, 1319 (2014), arXiv:1405.1042 [astro-ph.SR].

APPENDIX B

Simulated Light Curves Sum of Differences Results

Full results of the calculated sum of differences and sum of differences gradient from the simulated light curves with varying ratios of amplitude and period as was discussed in section 3.3.2.

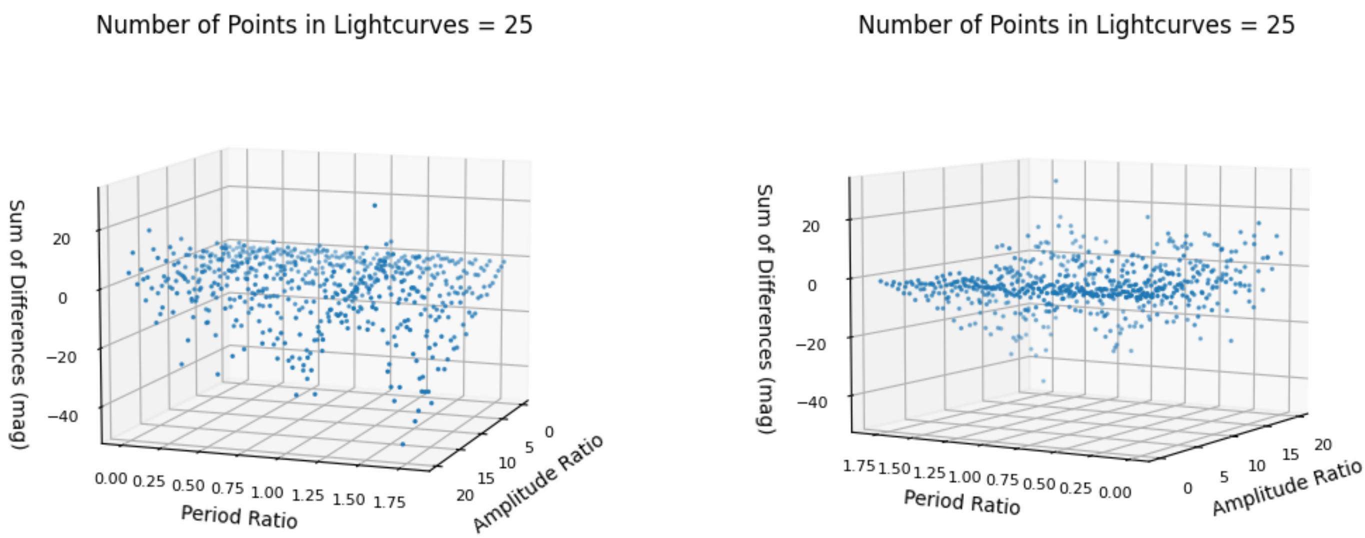


Figure B.1: Calculated sum of differences values for the simulated light curves with 25 data points, and varying period and amplitude ratios. The two plots show the results from different angles.

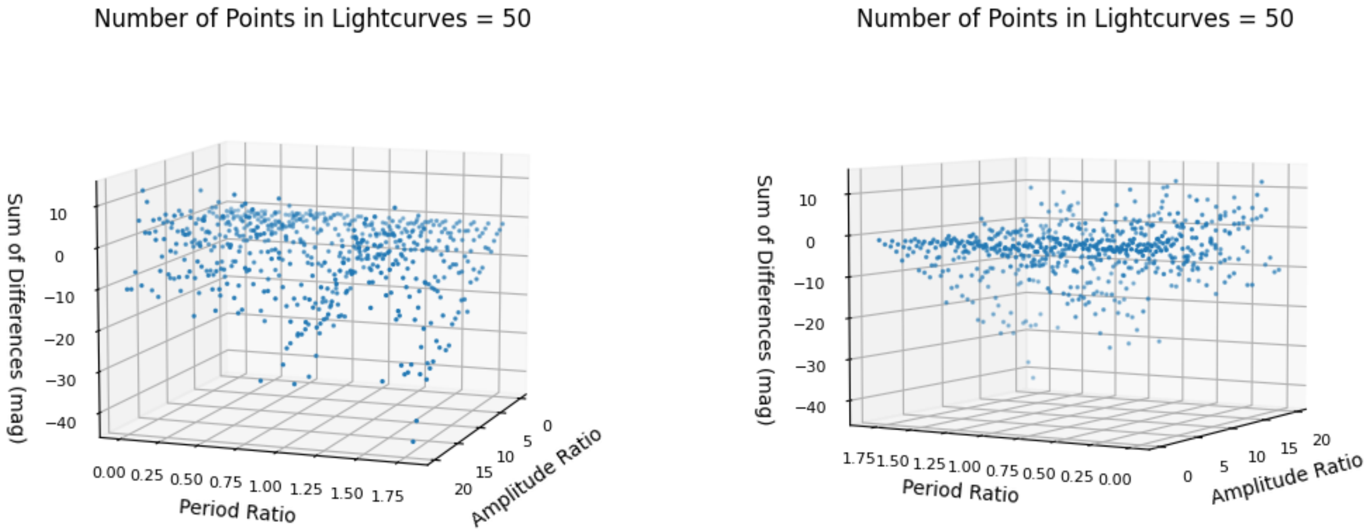


Figure B.2: Calculated sum of differences values for the simulated light curves with 50 data points, and varying period and amplitude ratios. The two plots show the results from different angles.

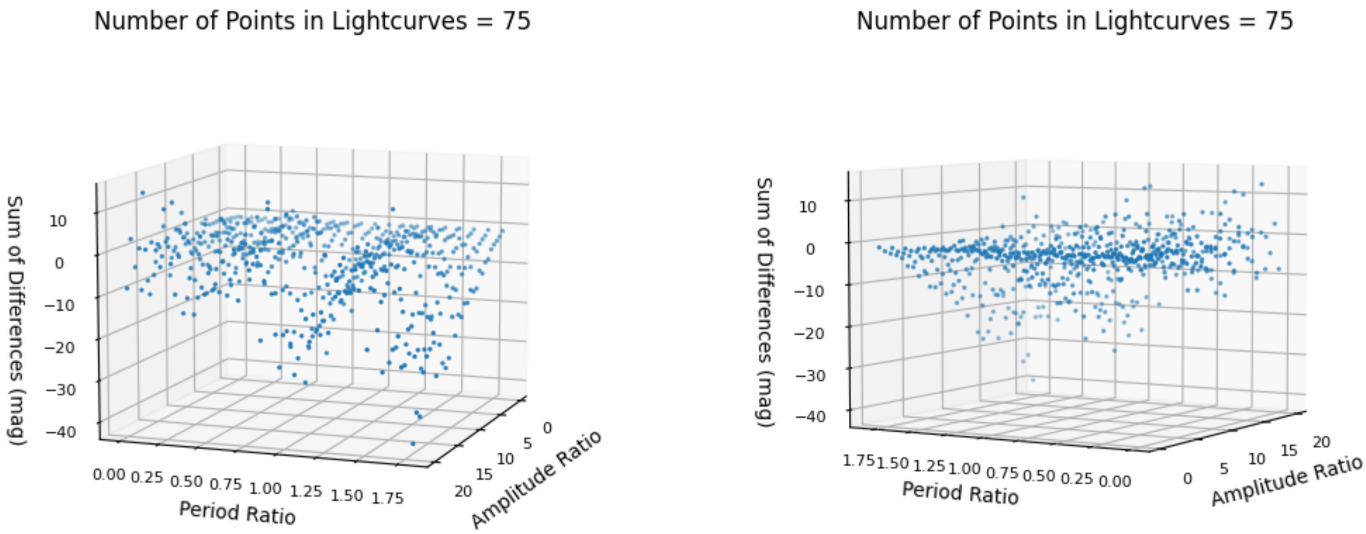


Figure B.3: Calculated sum of differences values for the simulated light curves with 75 data points, and varying period and amplitude ratios. The two plots show the results from different angles.

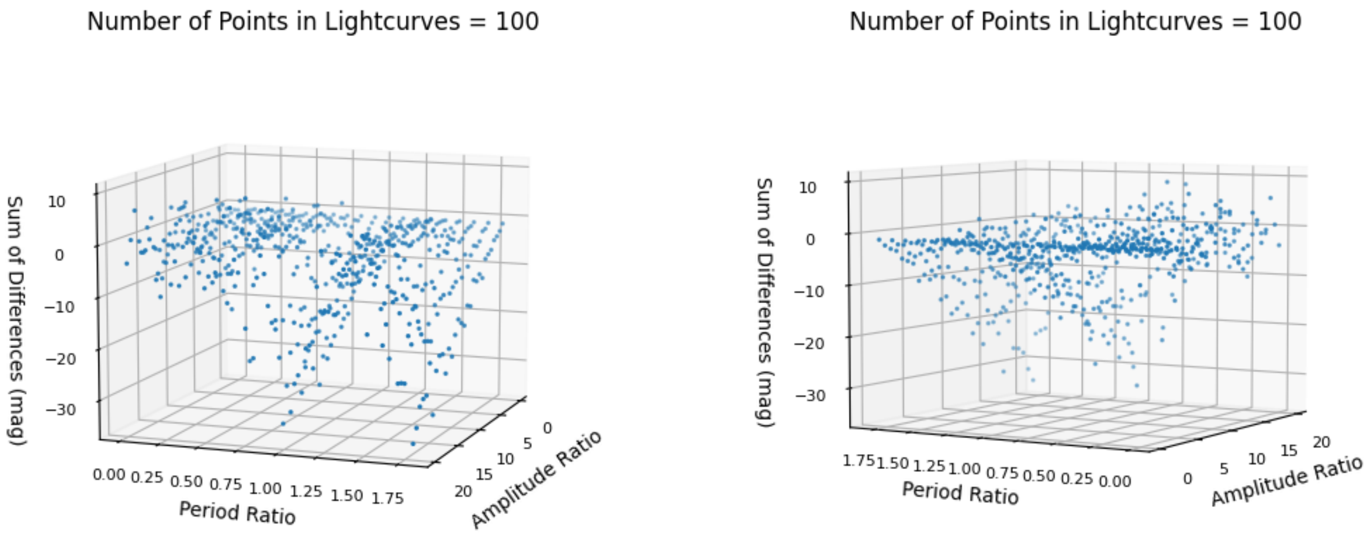


Figure B.4: Calculated sum of differences values for the simulated light curves with 100 data points, and varying period and amplitude ratios. The two plots show the results from different angles.

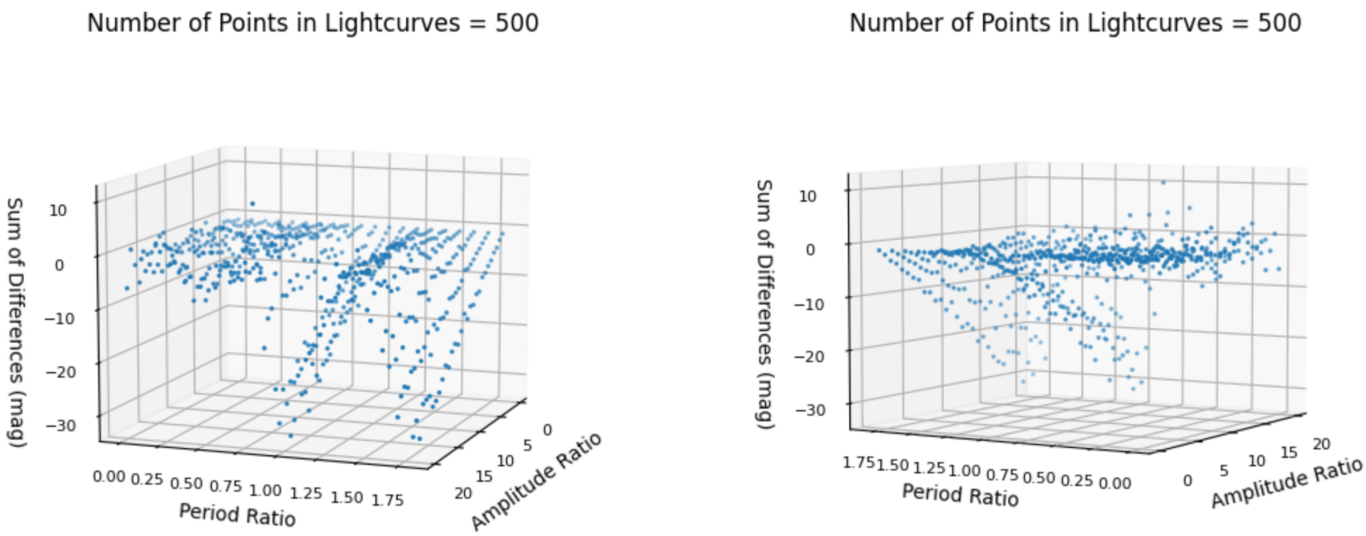


Figure B.5: Calculated sum of differences values for the simulated light curves with 500 data points, and varying period and amplitude ratios. The two plots show the results from different angles.

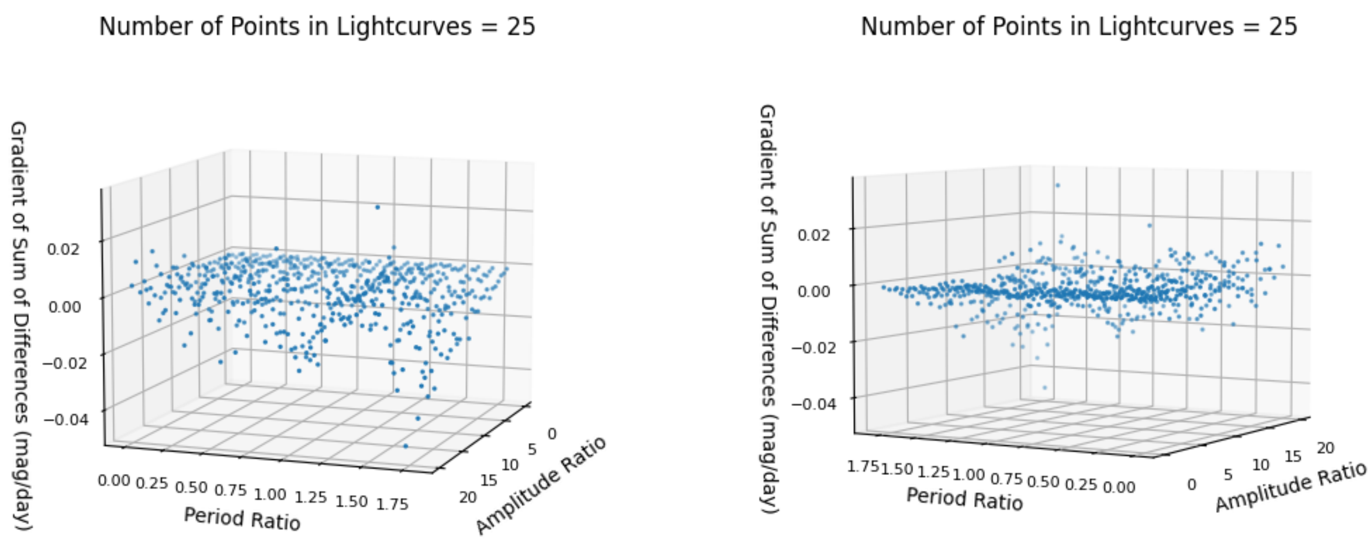


Figure B.6: Calculated gradient of the sum of differences values for the simulated light curves with 25 data points, and varying period and amplitude ratios. The two plots show the results from different angles.

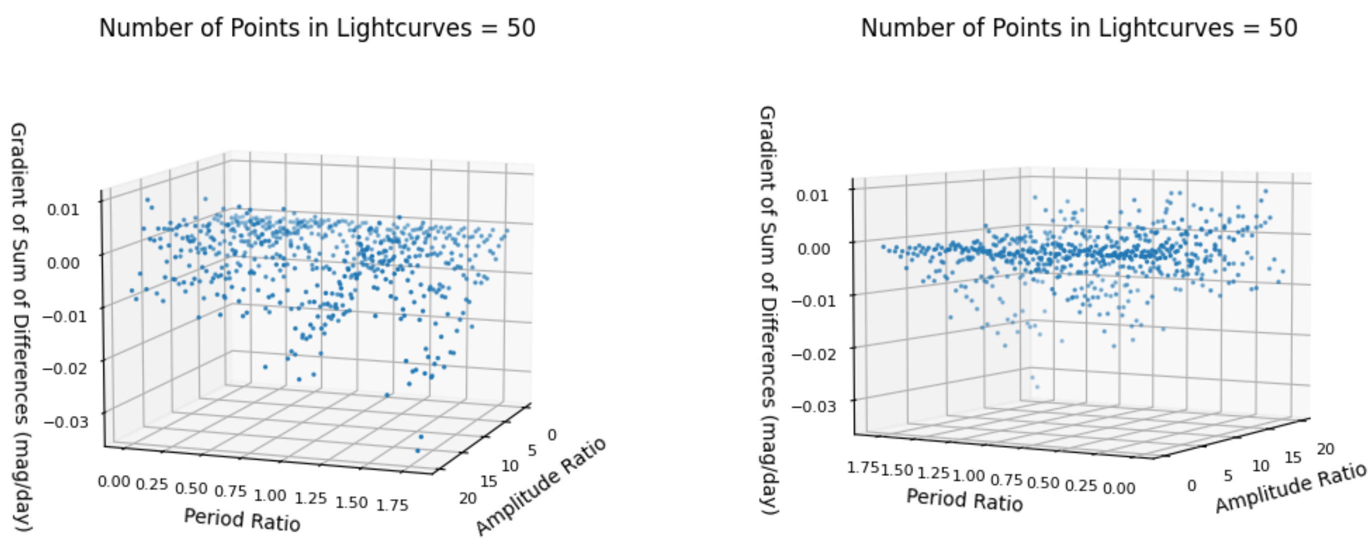


Figure B.7: Calculated gradient of the sum of differences values for the simulated light curves with 50 data points, and varying period and amplitude ratios. The two plots show the results from different angles.

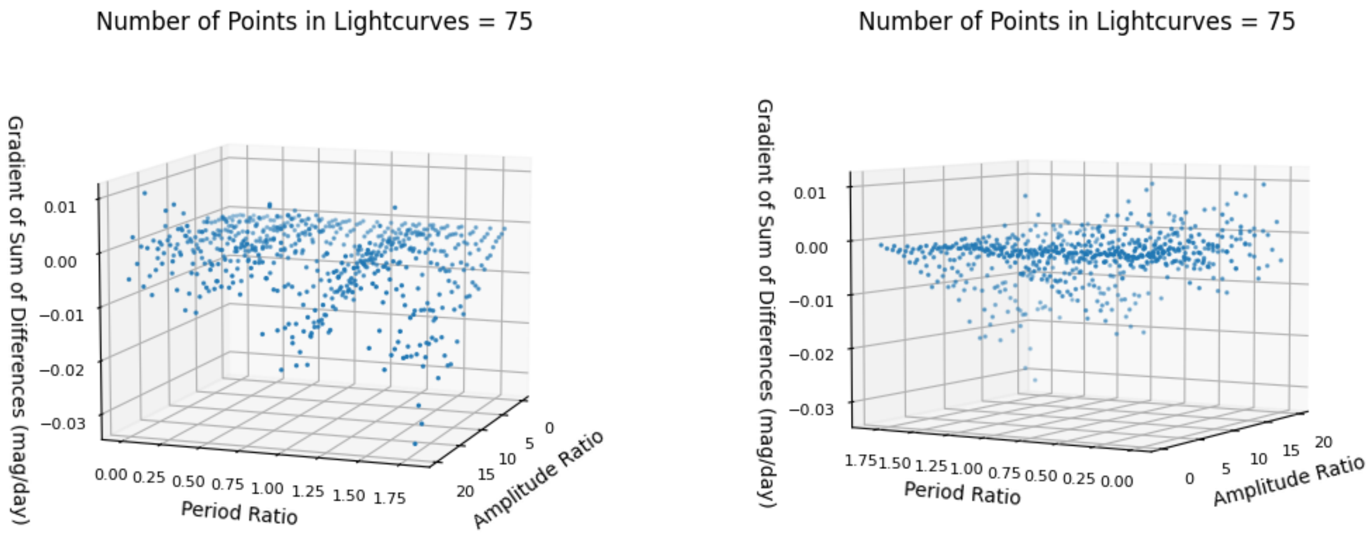


Figure B.8: Calculated gradient of the sum of differences values for the simulated light curves with 75 data points, and varying period and amplitude ratios. The two plots show the results from different angles.

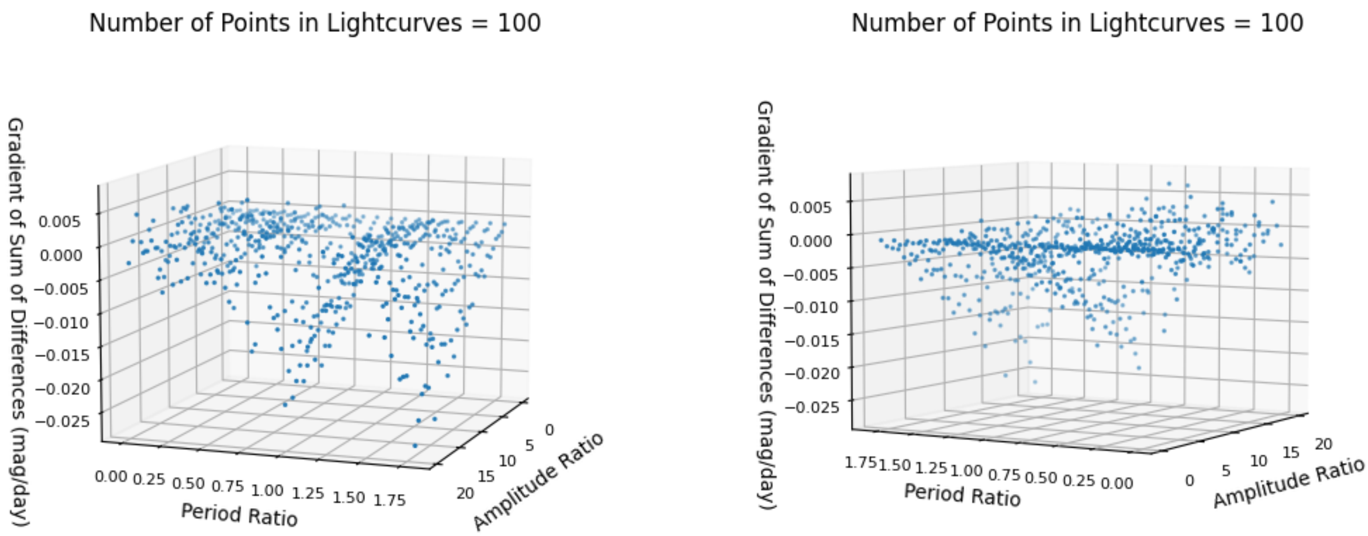


Figure B.9: Calculated gradient of the sum of differences values for the simulated light curves with 100 data points, and varying period and amplitude ratios. The two plots show the results from different angles.

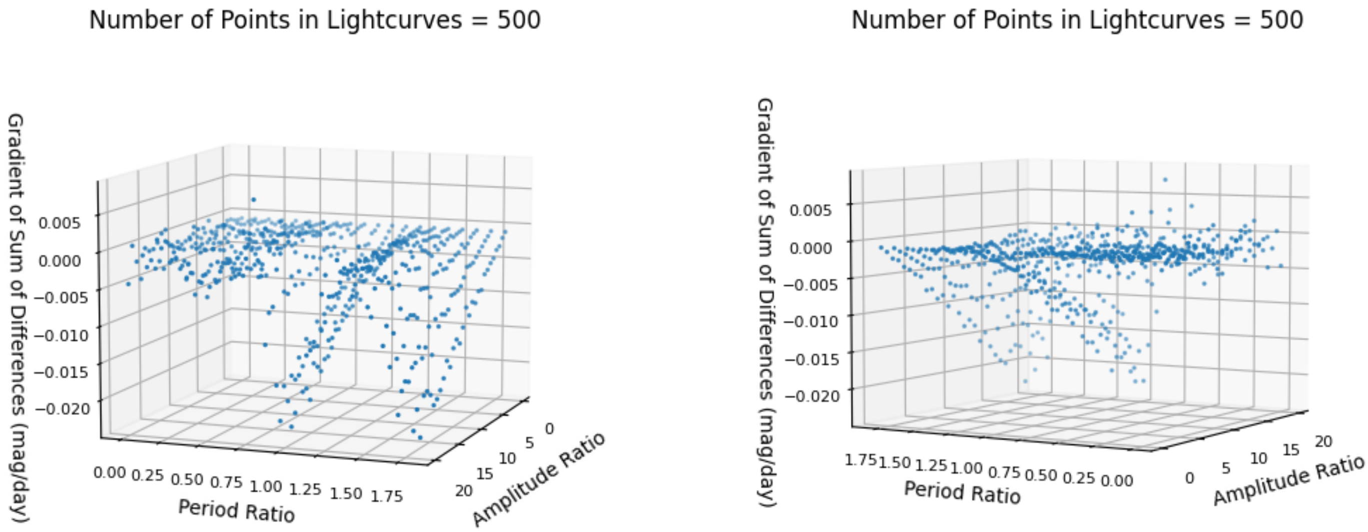


Figure B.10: Calculated gradient of the sum of differences values for the simulated light curves with 500 data points, and varying period and amplitude ratios. The two plots show the results from different angles.